Abstract Machines

Steppers, retained worlds and semantic connections in Mettapedia

Mettapedia Working Notes

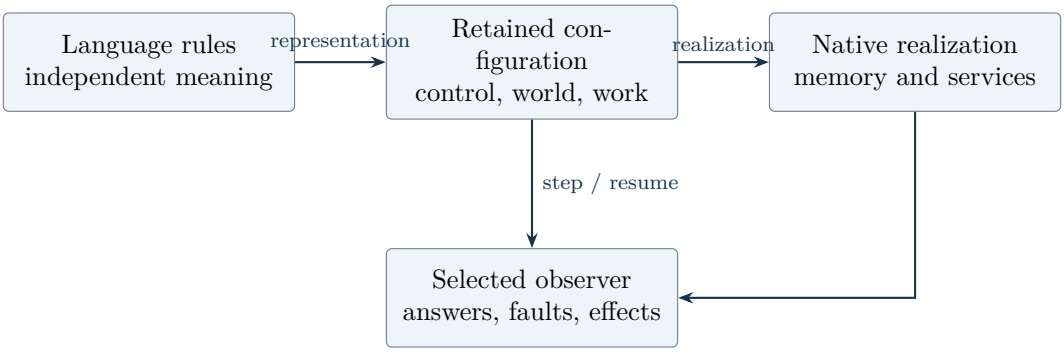
19 October 2026

Abstract

An abstract machine records unfinished work, its binding environment or world, and the observations already committed. This overview distinguishes three connections: independent language judgments, their retained operational machines, and particular implementation realizations. Common finite-block simulation and resumption laws support substantial CAM, fixed-C, HE and PeTTa instances without identifying their languages or strategies. The principal source correspondences, selected realization laws and their remaining gaps are stated separately. Later sections discuss regional execution and proof machines as applications and research directions; they do not enlarge the finite abstract-machine acceptance boundary.

1 One map, several machines

The language rules say what a computation means. The machine says how its unfinished work is represented. A VM additionally supplies a code image, loader and instruction-admission contract. A native implementation must realize those representations and transitions under its memory and service assumptions. Each connection needs its own theorem or qualification evidence. Section 13 develops the corresponding proof-checking machine: what follows from what, under which conditions, and how its evidence composes.



Similar outputs on examples test a connection. They do not replace a general correspondence theorem between independently defined semantics.

Family	Distinguishing retained information
CEK / Krivine	Value environments and argument frames / unevaluated closures.
Categorical CAM	Register, compiled closures, remaining code and value stack.
HE	Typed/minimal work, scoped bindings, alternatives and return/collection boundaries.
PeTTa	Ordered control, bindings, saved callers, alternatives and collection scopes.
Branch-local Need	Cell origins, memo states, branch worlds and reuse identity.
Refocusing	Legal context stack, focused redex and ordered candidate attempts.
GSLT / rho	Scoped search jobs, unmet obligations and occurrence-bearing receipts.
C-language core	Byte memory, continuations, pending expression occurrences.

The historical *Categorical Abstract Machine* and the machine for the *C language* are different instances. Neither supplies a universal fastest strategy. The broader observer and logical-world foundation is developed in `papers/metatheory.tex`.

The HE and PeTTa developments are further language instances, with their own demand, return, search and publication rules. Changing a storage representation changes its realization map. Changing a language rule changes its local semantic obligations and can change program observations. The surrounding execution and composition laws remain reusable where their hypotheses hold; a shared interface does not identify the languages or their strategies.

The realization boundary is a binding store, retained control, occurrence selection, world/services and publication. Their realization obligations are local loading, step-block, readout and resumption laws on formal component models. Literal C fragments and field inventories below are downstream worked examples; their theorems do not by themselves connect a current source revision or a compiled binary. No helper function or storage layout supplies the language semantics.

Principal correspondences and explicit profiles

The central distinction is

$$\text{independent source judgments} \longleftrightarrow \text{retained contextual machine} \longleftrightarrow \text{particular realization}.$$

The two arrows have different invariants and different coverage. For HE, `RecursiveExecution.Contextual.source_iff_observed` relates independent completed source judgments to actual machine paths under arbitrary retained callers and the declared lookup, dispatch and policy. Its public completion observer is a bag of joint caller/value observations. The stronger comparison with the plain-Atom worker in `FirstSuccessScopedAdequacy` has additional entry, coverage and materialization obligations. Those obligations describe that realization; they do not restrict the source language or replace the already constructed source/contextual correspondence. Their general recursive block composition remains open.

The HE `Completed.Policy` record selects reentry, marking, rebuilding and matching. `unmarked`, `ready`, `unchangedReady` and `finiteTypes` name local profiles, not whole-engine conformance claims. Type enumeration is a separate explicit parameter: `TypeOccurrencesRel` specifies the multiset of source occurrences, while `TypeEnumeration` provides one complete, unique enumeration for each query. A first-success consumer may observe that choice. Changing it therefore needs its local semantic law; a bag observer does not make all first-success choices equal.

PeTTa's saturated-list evaluator and its scoped destination machine likewise have separate source profiles. The latter retains open constraints, saved callers, ordered alternatives, collection scopes and commit delimiters. Its `Scoped.source_run_iff_derivation` compares actual finite execution with independent source rules. Neither the historical `Cetta*` module names nor those finite source theorems establish conformance of a native CeTTa or SWI binary. The CAM and fixed-C machines remain independent peer languages.

A small semantic change is absorbed locally when the revised component still satisfies the laws used by the surrounding machine. Reentry, marking, complete selection and return policy are explicit variation points. The tests are substantive: held-value reuse and reinterpretation have different derivations; reordered first-success casts can change a binding; a source reader can change anonymous-variable allocation without changing continuation machinery. A new observable operation or control form can require a new local correspondence. The interface does not promise that every language extension is free.

Boundary. The required principal rows are the common theory, historical CAM, fixed C, faithful HE/PeTTa profiles, shared-fragment comparison, retained interaction, component realization and current joined qualification. The final section records their remaining joins. Regional probability, universal machine and new logical-kernel proposals are research directions, not extra closing conditions for those rows.

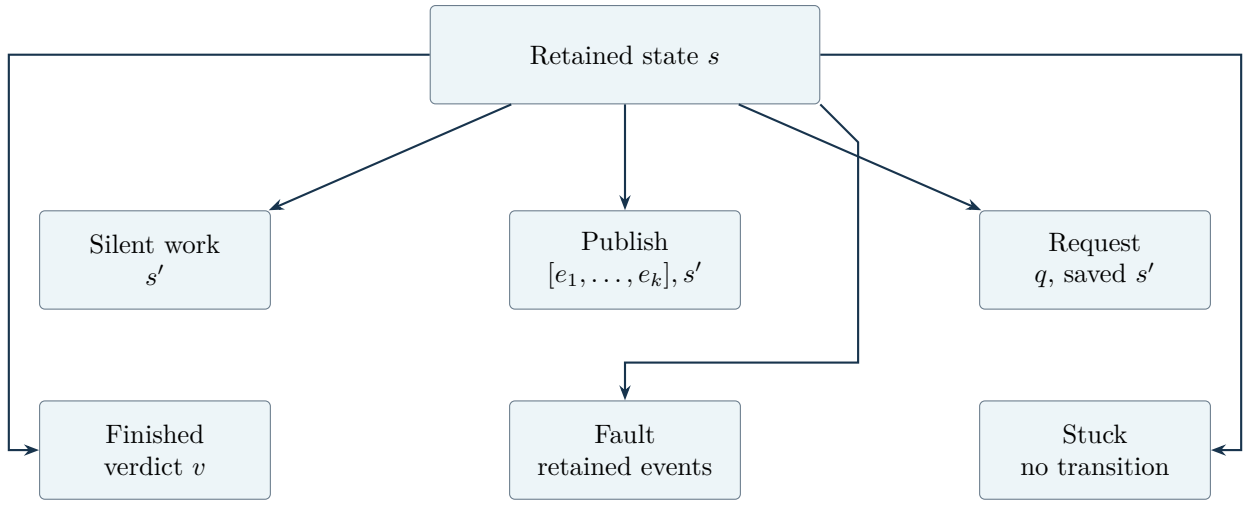
2 What the common stepper must retain

An individual run records an execution sequence; MeTTa need not promise one global answer enumeration order. Cross-machine observations may retain the multiset of full answer occurrences, including multiplicity, bindings and provenance, while forgetting enumeration. Stronger ordered theorems qualify a selected worker policy. Within each worker, resumption keeps its captured publication and remaining work without replay or omission. Bounded selection can choose different valid occurrences under different strategies. Observable effects and causal dependencies require their own preservation laws.

The simplest core is a state type, a loader and a partial step function. A missing step alone does not tell us whether a program answered or got stuck. The productive event interface therefore makes successful completion explicit:

$$\mathsf{step} : S \longrightarrow \mathsf{Option}(\mathsf{Transition}(S, E, V, Q)).$$

Here E is an event, V a verdict and Q a request. A transition can perform silent administration, publish events, finish, fail with events, or call out while saving its continuation. An absent transition is reported as stuck.



The arrows show possible transition or outcome kinds, not a sequence that visits every box. Exhausting fuel retains a residual; it proves neither completion nor divergence.

A finite runner returns an ordered event list and an outcome. After fuel exhaustion, running longer uses the saved residual and appends only the newly produced events. A suspended request instead needs explicit response delivery; extra fuel does not supply its answer. Finishing, faults, suspension, stuckness and exhaustion remain distinct. Separate price functions account for instructions, allocations or other declared resources. An administrative rank excludes an infinite silent stretch when its hypotheses hold; it does not assume the whole program terminates.

VM admission has two separate boundaries. A decoder admits a source compiler image and retains its captured environment. The historical CAM, its quotation/conditional extension and the graph CAM each prove the decoding, loading and admitted-execution correspondence. At runtime, an independently stated instruction table licenses the fetched operation in its current state. Admitting the code for an open variable does not supply its missing environment; an admitted image can therefore be stuck. Conversely, executable quotation code can lie outside the historical compiler image. The graph loader admits compiled recursion while rejecting arbitrary raw update instructions. Need's source-term loader constructs its required source heap. Source-atom and C-program entries retain their own language admission and realization obligations.

Three different choices in a stepper

Semantic alternatives, redex selection within a branch, and scheduling between branches have different contracts. For a pure snapshot, expanding one branch b can replace its frontier occurrence by the complete bag of its successors:

$$F' = (F - \{b\}) + \mathsf{Succ}(b).$$

Choosing which b to expand need not discard any semantic alternative. Choosing one successor and forgetting the others changes the computation. Occurrences remain distinct even when their printed terms agree. Fairness is a separate obligation for an infinite search; a finite pause establishes no exhaustive answer collection.

Redex selection inside a branch must still follow its evaluation policy. Call-time choice illustrates why unrestricted term rewriting is insufficient: copies of one shared choice must not become independent choices. Let-rewriting provides an explicit reduction presentation with soundness and completeness against its call-time-choice semantics [3]. Our branch-local Need instance has its own independently stated judgments and adequacy laws; that precedent does not establish native MeTTa adequacy automatically.

A source-level step also need not be one implementation instruction. Clements, Flatt and Felleisen separately prove preservation by instrumentation and reconstruction of the source reduction sequence [2]. The corresponding obligation here is a constructed source/configuration map and a productive block simulation. A debugger may expose that semantic view without making all optimizer frames ordinary program observations.

Scheduling shared effects requires further evidence: commuting actions, isolation, or an explicit serialization policy. The answer observer must also be declared. Alpha-renaming of internal bindings cannot erase a string that an ordinary operation has already produced from a variable's displayed name. These distinctions prevent a scheduler-independence claim from silently changing sharing, effects, or the language's observations.

The strongest useful form of correspondence

A related source/native pair should reproduce the same selected occurrences, with related residuals, at every relevant observation boundary. Native administration may take several steps for one source step. A ranked block simulation licenses that difference without erasing observable failures. The backward direction establishes no invention; the forward direction establishes that source behavior is realized. Cost bounds need an explicit resource model in addition to behavioral correspondence.

A logical machine with replaceable backends

The stable MeTTa machine is specified at the level of scoped evaluation operations. Its configuration retains the term or task, binding constraints, continuation, answer occurrences, observable world and required authority. The selected language profile determines argument evaluation, rule freshening, choice sharing, returned-value treatment and effects. A frame's address, hash-table layout or symbol number belongs to a backend realization. Logical object identity, freshness and ownership remain obligations wherever the declared operations can observe them; changing addresses does not license merging identities.

The representation boundary is a relation $R(b, s)$ between a backend state b and a logical configuration s. Several backend states may represent the same configuration. At the declared observation boundaries, a logical step $s \xrightarrow{e} s'$ must be realized by a finite backend block $b \overset{e}{\Longrightarrow} b'$ with $R(b', s')$, and backend behavior must satisfy the applicable reflection law. The block may contain internal

administration; productivity and endpoint laws prevent that administration from concealing divergence, stuckness or faults. This is a behavioral obligation. A uniform bound on the number or physical cost of backend instructions requires a separate argument.

The existing event-machine interface now proves this finite-block connection without a uniform cost premise. `Machine.run_related_of_one` derives whole-prefix correspondence from local implementations of individual inspections. Its resumption corollary retains the matched checkpoint and appends only the suffix. Local implementations in both directions give equal completed observations. Their outcomes retain faults, stuckness, requests and related saved states; a separate corollary reflects unbounded execution within the current interaction region. The local contracts compose through the actual middle state. Costed stages retain their stronger resource law.

A checked control makes the difference concrete: each logical publication has size-dependent internal work, with no single instruction bound for all admitted states. Both directions preserve its duplicate publications and completed observations; resumption appends the saved suffix, while restarting repeats the prefix. Existing forward-only controls still demonstrate that preservation alone cannot establish completion reflection. These generic laws do not supply the missing local CeTTa, Rust or SWI implementation instances.

The local obligations concern loading, scoped query and binding operations, invocation and return, host effects, retained control and observation. Their constructed laws are inputs to the existing run-composition theorems. Assuming the complete run correspondence as an implementation field would not discharge those obligations. A new storage representation can replace one backend instance while the language judgments and logical machine remain unchanged. Representation differences can be hidden only when the declared operations and future continuations respect the proposed relation; this includes the identities correlating several uses of one unresolved variable.

The finite execution judgment also accepts relational local rules. `DerivationSimulation` matches each independently justified source inspection by a finite target derivation. Its `ExecutionReadout` maps private event, verdict and request carriers to declared public observations; the state relation retains the actual saved configurations. An event contributes zero or more public records. Internal HE answer publication can therefore remain private while ordered effects stay in the public trace and the completed answer bag stays in the final verdict. `transport` composes the local blocks along an arbitrary finite source derivation, and `trans` composes realization stages through their shared intermediate rules and readout. `resume` starts the suffix at the already reached target residual and appends only its new events. `completed_iff` requires a simulation in both directions. The control `RelationalBlocks.no_backward` exhibits a forward realization whose target additionally drops one duplicate answer on a separate branch. Relabeling can hide private spellings while preserving both occurrences; it does not justify coalescing them. `CostSimulation.toDerivationSimulation` connects the deterministic theory to this relational theory on the machines' step graphs. It requires the separately constructed absent-transition law `MatchesStuck`. The CAM graph/tree refinement and the independent fixed-C machine/codec instantiate this adapter in both directions. It changes neither machine's source language nor its observation contract. These finite transport laws alone imply neither administrative productivity nor validation against a changed world.

The same relational laws now reach actual scheduled executions. `EventCursor.advance_simulation` transports the shared cursor's result to a finite target derivation, retaining its public history and the exact source account. `resume_simulation` constructs the target prefix and suffix at one reached intermediate configuration; it does not restart either prefix. When both rules are executable, `advance_machines` compares actual cursor results on both sides, with independently priced allowances. The CAM graph/tree and fixed-C machine/data comparisons instantiate this law in both directions. Their representation relations retain the caller stack or the C memory, activations and service status. These scheduled comparisons do not assert equality of physical costs or add a scheduling choice to the language semantics. A branching-target control retains two equal public occurrences through distinct private labels; no source scheduling allowance can realize the target's extra completed branch that discards one occurrence.

Implementation evidence has its own scope. A formal proof about a memory model, a source-fragment correspondence under ABI and service assumptions, and an executable conformance test support different claims. In particular, a model theorem conditional on a backend contract is not an unconditional correctness theorem for the compiled evaluator. Physical helper proofs strengthen an instance; they do not become definitions of MeTTa evaluation. The independent fixed C-core development remains a separate substantial instance of the common theory.

The same comparison framework also guides machine redesign. Explicit continuations can replace hidden evaluator calls; one invocation dispatcher can replace duplicated classification; a retained binding representation can replace repeated reconstruction. These are machine refinements when the specified observations are preserved. They earn efficiency claims only through an appropriate cost comparison and measurements of the actual implementation. A small change to the evaluation policy may instead yield a more coherent language. Such a change is an explicitly compared semantic candidate, with programs distinguishing the two policies and laws explaining the gain. It is not evidence for equivalence under the old observation. The relevant choices include fault versus ordinary failure, repeated evaluation versus ready values, and who may inspect, advance or publish a retained computation. A licensed transition and possession of authority to perform it remain separate judgments. Execution determinism, answer cardinality and probabilistic meaning are separate dimensions (Section 14). Focused graph holes and changes of resolution follow in Sections 15 and 16.1.

Existing owners. `Mettapedia/Machines/MachineSubstrate.lean`, `Mettapedia/Machines/ProductiveCoreAdapters.lean`, `Mettapedia/GSLT/Distinction/ProductiveBlocks.lean`, `Mettapedia/GSLT/Distinction/SpanTransport.lean`, `Mettapedia/GSLT/Distinction/SimulationComposition.lean`, `Mettapedia/GSLT/Distinction/OrderedAdministration.lean`.

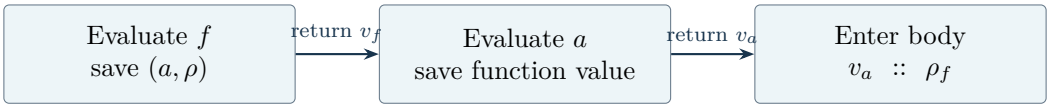
> **Boundary.** These are common execution interfaces. They do not make the selected logical kernels equal, or automatically prove a native evaluator refines them.

3 CEK and Krivine: one syntax, different demands

Both examples use de Bruijn lambda terms with natural-number literals. The CEK-shaped machine implements left-to-right call by value. Its environment contains values; its continuation first remembers an argument, then the evaluated function. The Krivine-shaped machine uses call by name: its argument stack contains closures whose evaluation is deferred.

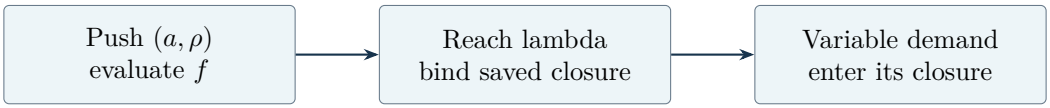
CEK stepper

For an application, evaluate the operator, evaluate the argument, and enter the function's captured environment extended by the argument value.



Krivine stepper

Push the argument closure, enter the operator, and bind the saved closure when a lambda is reached. A variable lookup resumes its closure.



The existing `CEK.eval_to_machine` and `Krivine.eval_to_machine` theorems construct machine paths from successful independent evaluator results, uniformly in the surrounding continuation. The common adapters expose checked answer boundaries, finite traces, resumption and computational accounts. This endpoint development does not mechanize the entire CPS/defunctionalization derivation.

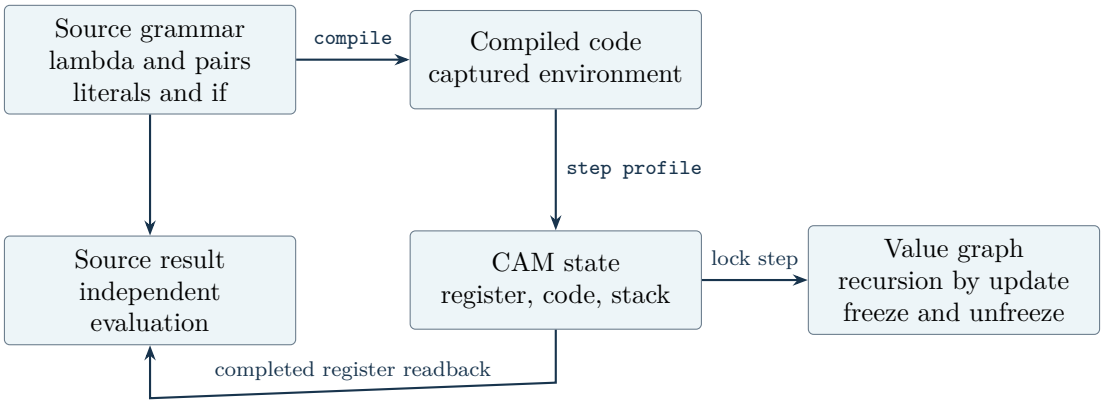
A discriminating term is $(\lambda x.\lambda y.x)\ 7\ \Omega$. Call by value evaluates the discarded divergent argument, so its evaluator returns no answer at every finite fuel. Call by name reaches 7 without demanding that argument. This is a proved strategy difference, not an implementation defect.

Existing owners. `Mettapedia/Machines/FunctionalCorrespondence.lean`, `Mettapedia/Machines/MachineSubstrate.lean`, `Mettapedia/Machines/ProductiveFunctionalMachines.lean`.

> **Boundary.** Call by name does not memoize. Repeated demand may repeat work; branch-local call by need requires the additional cell and cache laws below.

4 The historical Categorical Abstract Machine

The CAM follows Cousineau–Curien–Mauny's source strategy and instruction tables [1]. One indexed source grammar contains variables, lambda, application, pairs and projections, with literals and Boolean conditionals enabled by its extension index. Source closures retain source bodies; machine closures retain compiled instruction lists.



The tree machine and the value-graph machine share one instruction table. On the tree machine's instructions they move in lock step; recursion and delay run only on the graph.

Application compiles to saving the environment, evaluating the operator, restoring the environment, evaluating the argument, constructing their pair, and applying. The actual state retains the register, remaining instructions and stack throughout this work. The executable `step` agrees in both directions with the independent `Transition` rules.

The shared compiler and code/value decoder have proved inverses on their admitted images, including captured closure environments. `evaluation_iff_execution` connects independent evaluation to completed neutral-profile execution at both indices; completion requires empty code and stack. A conditional evaluates its condition, restores the saved environment and runs only the selected arm. A non-Boolean condition can be stuck. `Lambda.proposition_3_2` connects historical scoped weak source reduction to execution and readback at either syntax index, including literal and conditional source terms and captured scoped environments. The strategy does not reduce under a lambda. Typed source formation constructs actual Cartesian closed morphisms; `Typed.evaluation_iff_strict_execution` and the meaning theorems connect the active instruction table to that interpretation.

The historical constant-free compiler remains the disabled-index specialization; its no-quotation/no-branch laws and typed Cartesian-closed results keep that scope. The enabled source uses the existing native quotation/branch instructions.

Recursion and delayed evaluation follow the paper's Sections 4 and 5 on a value graph. Heap cells are couples, closures and frozen computations. Allocation appends a cell; `wind` updates the second component of an existing couple, as LISP's `rplacd` does. Recursion compiles as in the paper's Table 6 and leaves the cyclic closure $c = C{:}(s, c)$ in the register. `freeze` builds a frozen computation and `unfreeze` forces it at strict positions (Table 7); forcing is by name. The module records two corrections to the published listing: `unfa` and `unfb` rebuild the couple with exchanged components, and the `letrec` code of Section 5 has a stray `swap`.

An independent big-step judgment covers recursion, frozen values and strict primitives. `evaluation_iff_execution` relates it to completed execution without assuming termination; executions keep the caller's code and stack and never change existing cells. Heap words represent source values by finite derivations. A cycle represents both a recursive value and its unfolding (`Represents.not_functional`), so source syntax is not observable on the graph; represented words agree with

the depth-indexed unfoldings of their values, and bisimilarity is agreement at every depth (`Represents.bisimilar_iff`). `refinement` and `abstraction` are cost-one simulations between the tree machine and the graph machine on the tree's instructions. Allocation keeps the heap topologically ordered; `wind` breaks the order, and its cycle has no tree unfolding (`windHeap_no_unfolding`). Controls run the paper's factorial and lazy stream, a divergent loop, and stuck cases.

A memoizing variant uses the existing branch-local demand store. Each frozen cell has its own allocation identity and immutable origin. Starting evaluation records an owner; nested update frames retain the caller until the result can be committed. Later forces read the completed cache without restarting its producer. Separate allocations of equal source terms remain separate demands.

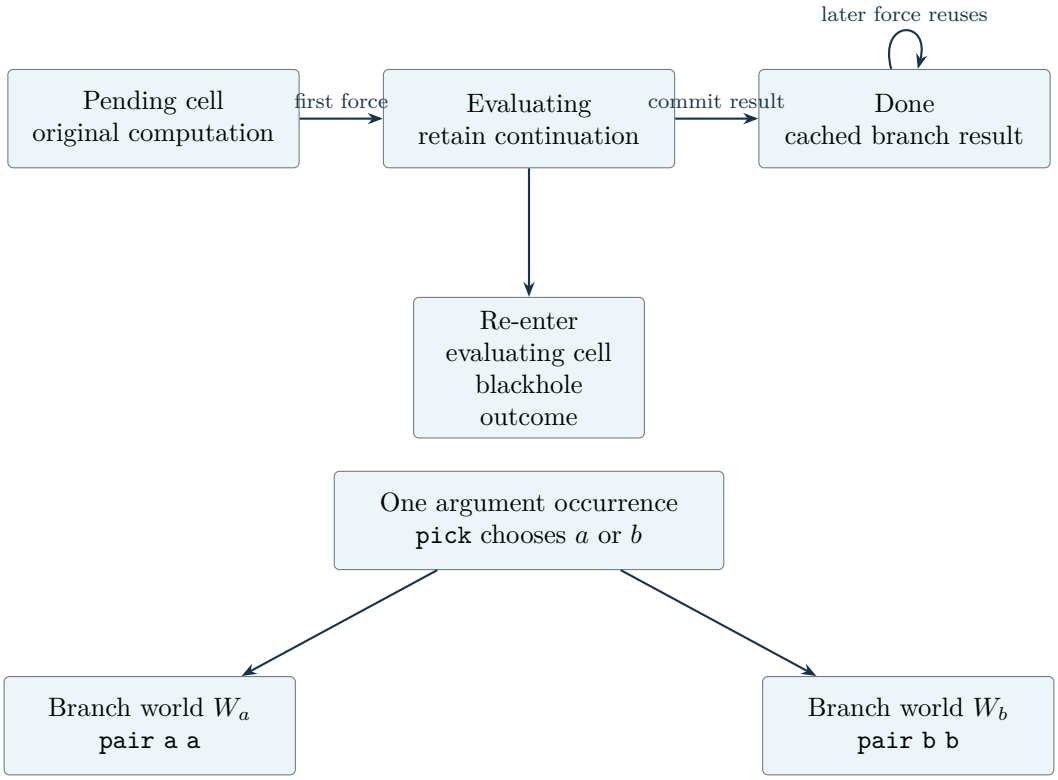
The independent source judgments `Need.Evaluation`, `Need.Invocation` and `Need.Force` use source constructors, an arena and a demand store. They have no machine control or transition premise. `Need.evaluation_iff_execution` proves both directions for complete values and the entire retained world, including recursion, conditionals, cached hits and nested updates. Its source-arena condition requires stored closure and suspension bodies to be compiled source terms; the empty loader establishes that condition directly. Active ownership blocks recursive demand. This is explicit memoizing semantics, distinct from the historical by-name strategy.

Existing owners. `Mettapedia/Machines/CategoricalAbstractMachine.lean`, `Mettapedia/Machines/CategoricalAbstractMachineGraph.lean`, `Mettapedia/Machines/CategoricalAbstractMachineControls.lean`.

> **Boundary.** Cartesian closure supplies typed meaning, not recursion or update; the graph machine gives those their own operational laws. The extended readback and memoizing correspondence are constructed. Logical instruction counts do not by themselves establish physical performance or a native-code compiler.

5 Branch-local Need: choose once, reuse in that branch

The existing call-time-choice core separates value expressions from computations. The distinction follows the call-time-choice separation between shared demand and unrestricted rewriting [3]. Its runtime distinguishes a memoized implicit suspension, a resampled implicit thunk, and explicitly quoted code. A cell retains its original code and environment while its cache changes state.



Both uses share one chosen value in their own branch. A cache written in W_a does not become the cache of W_b. Separate argument occurrences can still make independent choices.

The natural semantics `Derives` is independently defined. The `adequacy` and `observes_iff_halts` results connect it to the unchanged branch-local Need machine in both directions, keeping the ordered list of outcomes and duplicate occurrences. Failure and blackhole outcomes are explicit. The strict-position theorem ensures primitives receive normal forms under the modeled binding policies.

The finite weighted fragment is connected to these same machine runs. `needChain_machine_adequacy` covers a finite ground producer, an arbitrary finite chain of dependent producers and an arbitrary admitted final body. Each intermediate producer may force its immediate predecessor repeatedly. For sufficiently large frontier allowance, the actual frontier is completely halted and each ground answer occurs exactly as often as the independently defined weighted denotation requires. The relation also retains branch-local heaps, cached choices and ordered effects. No conversion of cell identities to ground values is used to obtain the counts.

Two machine-level controls distinguish the contracts. Forcing one duplicated producer twice gives two shared occurrences; resampling it gives four. Ignoring a failing producer succeeds under Need and fails under eager execution. The allowance here bounds frontier unfolding, not total work. The fragment does not admit arbitrary access to all earlier cells, and these laws do not supply mutable-world validation or execution authority.

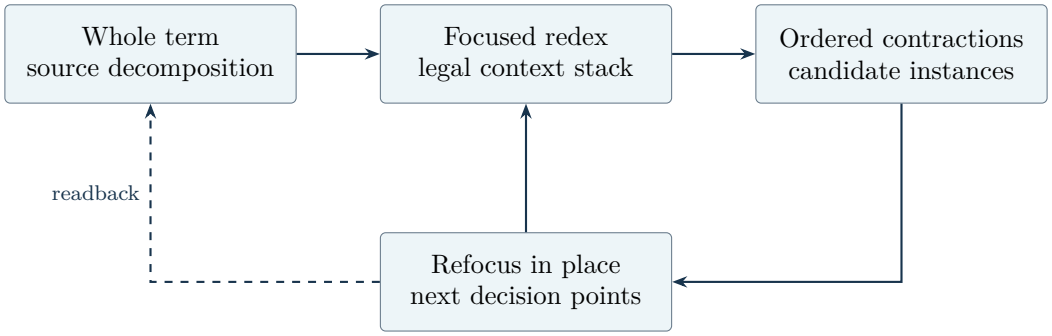
Quoted code is a value until explicit activation. Structural observation may materialize an implicit pending cell under its original world and force rights; it must not infer execution permission from a callable-looking written subtree.

Existing owners. `Mettapedia/Machines/BranchLocalNeed/CallTimeChoiceSpec.lean`, `Mettapedia/Machines/BranchLocalNeed/CallTimeChoiceBackward.lean`, `Mettapedia/Machines/BranchLocalNeed/CallTimeChoiceStrict.lean`, `Mettapedia/Machines/BranchLocalNeed/Weighted/StressAndCommitments.lean`, `Mettapedia/Machines/BranchLocalNeed/SharedContinuationInstance.lean`.

Boundary. This is source/model adequacy. Full native heap, producer/world identity, resource retention and evaluator correspondence require their own joins.

6 Refocusing: retain the route to the next redex

A reduction evaluator repeatedly decomposes a term, contracts a redex, and plugs the result back into its context. Refocusing keeps the context and searches from the contractum, avoiding unnecessary reconstruction when the strategy's laws justify it [4].



Frames are supplied by the evaluation strategy; their kinded stacks reuse `Quiver.Path`. A structural syntax position alone does not authorize evaluation. Traversal supplies local inspection and frame-resumption equations. The branching strategy supplies candidates in order and ordered instantiations for each candidate.

Inspection and rejected candidate attempts are administrative work, ranked by `SearchCredits`. A focused computational step remains visible even if its source and target terms are equal. A rejected candidate can be an empty block; an observable fault or unmatched redex cannot be discarded on that basis.

`refocusingComplete` constructs reverse tracing from local unique-decomposition laws. `blockSimulation`, `frontiers` and `outcomes` preserve ordered alternatives, multiplicity and pending work. `observable_stable` and `erasure_licensed` record the fault boundary. `Authored.ruleScan_simulation` supplies an actual GSLT rule-scan instance.

Where generalization is still needed

Several legal redex positions require an explicit position-selection policy. The present unique-decomposition instance must not be silently treated as a theorem about every parallel or nondeterministic traversal. Effectful attempts also need a history model: erasing a rejected attempt is licensed only if its observable commitments are accounted for.

Existing owners. `Mettapedia/Machines/Refocusing.lean`, `Mettapedia/Machines/RefocusingControls.lean`, `Mettapedia/OSLF/MeTTaIL/DerivedContexts.lean`.

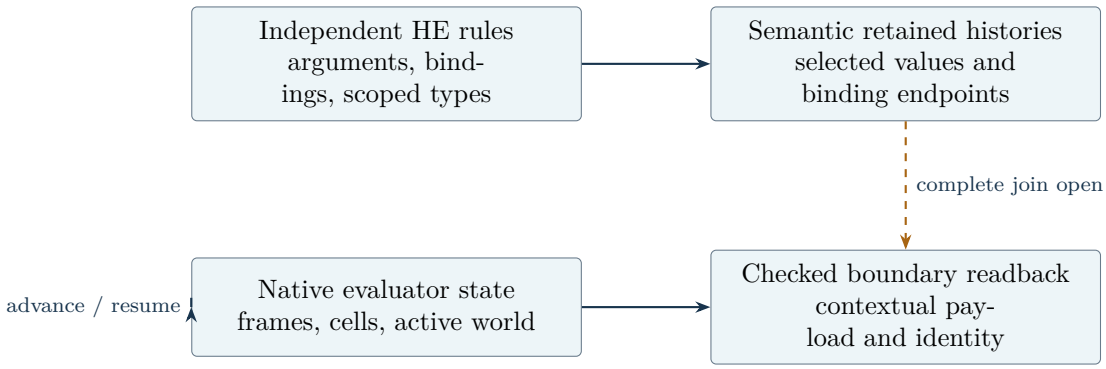
> **Boundary.** The authored rule scan is a real anchor. It does not yet establish the full native evaluator's contextual traversal or administrative rank.

7 HE and Prime: the retained representation boundary

HE, PeTTa and declared Prime profiles are distinct language selections. A common machine substrate can retain the selected package and its own state without making those dialects interchangeable. Changing the selected interpretation requires a justified crossing, not merely a new label.

The implementation boundary matters. LeaTTa is an auxiliary interpreter; its correctness and conformance to HE are not acceptance conditions for the CeTTa machine correspondence. A theorem about LeaTTa's instruction worker establishes a property of that worker, even when its source judgment is independently specified. It supplies no CeTTa transition map by itself. HE semantic claims answer to the independent judgments and published language contract. Rust HE and other engine runs supply separately labelled conformance evidence; disagreements remain explicit. The realization target is the retained driver's semantic component interfaces: selected occurrences, binding readout, caller/alternative control, world operations and publication. Concrete frame layouts and instruction phases belong to their downstream representation examples.

The HE source and abstract-machine dependency graph excludes LeaTTa implementation studies, including studies whose filenames do not name that implementation. Binding models, finite matcher completeness, freshening and valuation transport have their own mathematical owners. Neutral facts use `Spec.Match.SolutionTheory`, `BindingComposition`, `MatchSizeTheory` and `Spec.Bindings.ScopeObservation.ModelTransport`; they are not exported through an interpreter-comparison namespace. The shared `MeTTailCore.MeTTaSyntax.Atom` module owns the structural carrier in an independent package. Both the HE mathematics and implementation clients depend on that module. Every LeaTTa package module, including its former `Core.Atom` owner, is excluded from the HE source and machine graph. The shared leaf has no imports; the boundary check enforces this direction. The independent HE freshener realizes the alpha/freshness contract; comparisons with another implementation's freshener remain downstream instances.



The independent argument judgment constructs an ordered history of selected values and exact intermediate bindings. `RetainedArgumentSpecification.arguments_history_iff` gives both directions at that semantic boundary. Leaf and nonleaf service packages connect further parts of the independent evaluation.

Live scope is part of the evaluation judgment. A recursive equation call freshens its private variables away from the arguments, current bindings and saved caller context. This matters even for a single rule: an inner call to `inc` must not reuse the outer call's private variable. The executable evaluator, declarative relations and contextual step relation carry this scope explicitly. Context replacement consequently requires the appropriate agreement of live variable inventories.

Resource exhaustion is a separate observation from completed failure. The checked recursive machine retains the pending task and requests further resources when its evaluation depth or a supporting query is insufficient. Its depth-coverage checks follow the actual recursive dependencies; a successful check proves stability under every additional depth allowance. A completed checked run also certifies the whole evaluator result at every greater evaluation depth, for each of its six entry

points and internal list traversals. The proof increases the depths in every saved caller frame, not only the currently selected task. Equal answers at two shallow depths would not suffice: both can be empty while a deeper run returns a value. For all six entry points this completed execution constructs the existing evaluation certificate, including its eventual-stability witness and retained live scope. It is still a theorem about this evaluator profile; its matching and typing leaves do not disappear from the independent-language boundary.

Binding materialization now has a structural sufficient depth. A resolver marks fresh assigned keys, and syntax traversal decreases the current atom's height. The finite remaining-key set and the maximum stored value height give a decreasing rank for both ordinary and equality-aware resolution, including compound values and detected cycles. Control forms use this proved bound instead of a fixed numerical cutoff. Cycle detection is a different operation: the existing finite dependency analysis checks the complete graph of assigned values and their equality classes. Its result is equivalent to the independent absence-of-cycles judgment and can retain an explicit cycle witness. Every stored assignment matters, including a value hidden by a later lookup entry. Satisfiability together with nonvariable assignment payloads proves this admission condition by strict descent of interpreted term size. Nonvariable lookup results alone do not: a shadowed variable assignment is a counterexample. The older finite list evaluator is still an approximation: an empty list from that evaluator alone does not certify completed failure.

The service adapter uses this checked machine. Replenishment preserves its selected alternatives, caller stack, world and already emitted results. Completed checked executions refine the corresponding finite evaluator run; the resource request is propagated through the C service caller. Thus a shallow budget suspends a recursive service, while a sufficiently provisioned three-call example returns the successive values 3, 7 and 12.

The same checked execution has an actual data representation. Each advance reads its supplied configuration and preserves every saved frame, published occurrence and pending resource request. Reading after encoding is an identity; malformed input is refused. The shared partial-reader adapter has identity and composition laws and preserves the exact finite run, its suspended residual, stuckness and unbounded exhaustion. Its logical step account excludes the physical cost of encoding and decoding. In particular, an encoded exhausted or suspended run cannot become an accepted empty result.

Application selection also retains history. A selected branch starts after the preceding branches have advanced the state; its seed and entry state are therefore part of the correspondence. The shared argument-fold laws retain this chronological origin. A theorem about normalization from empty bindings cannot replace a theorem about that reached environment. General conversion soundness must supply normalization and binding invariants at the retained endpoint; the concrete nested identity and constructor examples establish those conditions directly. General normalization adequacy remains a separate obligation.

The equation-query observation retains a physical row together with the same raw rule, freshened pair, match, ambient merge and emitted work item. The explicit-scope query is sound at this boundary and observationally complete for any legal private freshening of that same row. A constructed permutation fixes the query, incoming bindings and retained caller names; valuation reindexing preserves the full output solution theory. Private binding records need not have identical spellings, and this existence theorem does not count semantic derivations. Separate laws for the actual HE continuation retain the changed binding frame of each instantiated body beneath any saved caller stack. Their finite realizations inherit completion, representation and exact saved-residual resumption from one shared proof. The service adapter carries each full value/binding occurrence into both its event log and endpoint reply; structured and represented execution share that result. The typed application continuation also retains those full results, including the actual success-priority filtering. A typed duplicate-equation control returns both changed frames to a saved caller through the real callback transitions. The argument continuation now resumes each head's remaining arguments under that head's selected frame. A second argument following two equation results keeps both

changed endpoints; restoring the common input frame preserves the packet values and fails the full-result observation. The same control law covers changed error and Empty results: it publishes that occurrence without evaluating the pending argument suffix. Its independent history records either the completed packet or the first failure and the unexecuted suffix. A saved head-return frame needs only the residual children; it does not replay the completed head. Duplicate rejected casts remain separate occurrences, and the public success-priority judgment excludes successful alternatives only when total rejection has independently been established.

The following query-item and materialized-entry laws additionally concern an auxiliary minimal interpreter. They describe useful representation boundaries, but do not identify that interpreter with Rust HE or construct a CeTTa step map. Completing its instruction coverage is not a prerequisite for the native correspondence.

The scoped-query completeness law additionally constructs the interpreter's actual `queryOpItemsOf Rule` work item and its `evalResult` continuation, preserving the entire saved minimal stack. These work items are already merged; the older HE equation task expects pre-merge candidates. Bounded HE application now has a constructed observation law at every fuel. The existing canonical native model promotes it to equality after actual materialization, and to invocation congruence because `evalOp` substitutes before dispatch. This does not license raw shape inspection of unresolved syntax, or establish transport through every later typed RHS. The actual query-item theorem uses the same materialization boundary to identify its emitted continuation with the independently freshened source result after substitution.

An independent normal finite presentation of the binding equations now supplies the materialized source entry. Its one-pass result and the actual query value agree up to renaming residual variables, by the existing principality theorem. Exact function arity, the broader function-head test and embedded-instruction classification are preserved separately. A raw head guard on this normal source value constructs both the independent raw-completion judgment and the actual finished query item. Function entries instead supply alpha-related body witnesses. This applies at the existing query materialization boundary; it does not authorize evaluating held operands or inserting substitution into raw structural instructions. Transporting arbitrary unresolved source syntax or executing every related body remains a further obligation.

Function completion preserves fuel, native state, pending alternatives, saved callers and earlier results. Open `cons-atom`/`decons-atom` steps and their `chain`/`return` bodies are constructed on both sides. The finished chain handler materializes its raw result under the selected output bindings; no fixed-result premise is needed by the general bridge. A bound head becomes a symbol while two correlated free occurrences stay shared. The actual complete driver suffix retains its pending queue and earlier results. Materialized entry is a separate operation: raw primitives do not substitute. A bound tail alias distinguishes those entries, and empty deconstruction exposes an uncovered source fault.

Unify has sound and complete finite endpoints, including shared and reflexive variables, success and refusal. Source and native merge orientations need only agree on their complete solution theories. The actual candidate list retains its occurrences; fallback syntax stays raw until materialization.

A caller-indexed terminal observer agrees with the actual function handler after materialization, and its prefix law preserves faults and publication. The nearest saved caller stays raw as the `NoReturn` subject; without a caller, the terminal is blamed. The existing source judgments now retain that caller and protect entered chain/function syntax in the query scope. This is the full-syntax frame policy; Rust's placeholder frames require an adapter. Malformed forms and the difference between core HE and extended LeaTTa instruction classifiers delimit those auxiliary results. These source judgments now materialize every finished function-body result before classifying it as terminal or scheduling an exposed instruction. The independent relation is generated by finite binding-equation presentation, exists exactly for satisfiable frames, and agrees with the actual resolver up to residual variable renaming. Controls expose both a return marker and a further instruction through a bound

variable. This corrects raw-syntax terminal classification; the general body/typed-continuation correspondence induction remains open.

Equation producers now earn this materialization from their actual matched output frame and retain the saved continuation. Native producers declare a separate raw-or-materialized binding mode, independently of whether they have completed interpretation. Raw delivery keeps the authored atom literally; materialized delivery must supply its independent presentation. These policies do not identify raw LeaTTa callbacks with Rust's post-merge materialization.

An invocation instance belongs to its saved configuration

The independent direct type-lookup fragment collects declarations over a multiset space, using `TypeOccurrencesRel`. A complete enumeration is supplied by `TypeEnumeration`; its soundness, totality and uniqueness laws keep every stored copy and choose one coherent enumeration per query. `CompleteTypeQueryRel` separately describes admissible individual queries; it is not an evaluator policy pooling all enumerations into one run. `StorageTypesRel` and `storageTypeService` explicitly name the older list-storage instance. The source does not prescribe that order. `enumeratedTypeService` and `scopedEnumeratedTypeService` supply alternative complete policies to the existing evaluation judgments. `TypeCastBatchRel.complete_space_perm` preserves the full set of admitted casts when storage is permuted. It does not assert that two fixed policies choose the same successful type: `first_success_can_change_binding` constructs actual source cast derivations with different type bindings. `dropped_duplicate_not_complete` rejects a traversal that loses one copy. Thus selection order is addressed before publication, rather than hidden by quotienting only the final answer list.

A schematic declaration such as `(-> $t $t)` shares one variable across its argument and return positions. Independent invocations receive distinct instances. A caller that already binds `$t` to `Number` must still be able to invoke the scheme at `Bool` through a fresh private variable. `scopedTypes` protects the names of the complete retained configuration, including the continuation and published prefix; declaration inspection remains literal. `EquationFreshening.getAtomTypesScoped_scopedTypeLookup` connects the actual query list to the independent scoped type service.

The applicability, coverage and first-success proofs now accept the actual type query. `fresh_polymorphic_scoped_selection` constructs the same changed binding in the worker and independent source scan. `fresh_polymorphic_scoped_step` executes that checked transition while preserving the saved stack and emitted prefix. The operator's selected signature is explicit; this local theorem does not yet prove the complete polymorphic application and return. `scoped_polymorphic_annotation_prefix` additionally derives that selected instance from the stored annotation and the actual expression checkpoint. Its two checked transitions retain the fresh variable, changed binding and saved context, with exact subsequent resumption. The shared equality-graph laws establish that linking fresh private instances leaves every unrelated ordered alias class unchanged, including its representative and value order.

The retained binding interface has a representation-independent constructor: `ReadoutCertificate.of_normal_theory` turns a normal finite table with exactly the source record's solution theory into one certified readout for all future operands. A finite tuple of the complete binding support and stored keys earns one model-fixed residual permutation; it is then extended to all variables and atoms. `ReadoutCertificate.fresh_assignment` extends this certificate by a fresh equation, including open values that avoid its new name. The native `scoped_type_assignment_frame` preserves older alias classes and constructs exact covered reads and pending-tuple absorption under that extended table. `scoped_polymorphic_annotation_frame` joins this construction to the actual two-transition annotation prefix, with its saved continuation and emitted answers. This is a reached-frame and prefix correspondence; the later function, argument and return correspondence remains separate.

The subsequent `scoped_polymorphic_function_entry_configuration_square` advances a unary repeated-variable arrow scheme through the actual selection boundary. It relates three checked worker transitions to one contextual function-entry step, constructs the changed certified frame, and retains arbitrary related caller stacks and prior occurrences. Its argument has one declared symbol type, and entry readiness is explicit. The executed caller-alias control reserves names in the saved caller and earlier publication, keeps the caller's `Number` binding, and gives the fresh instance `Bool`. A held-value reuse control separates that source policy from raw-worker reentry. These are earned selection and activation laws; they do not establish recursive application for arbitrary schematic signatures.

The quantifiers matter. A result table proved identical for every saved stack under literal lookup cannot be reused unchanged when the stack affects fresh names. The remaining scoped correspondence must follow the actual configuration and its chosen instantiation witness. Other legal source alpha spellings require a joint observation relation, rather than literal completeness for the executable's deterministic name choice. Likewise, the per-result `EvalRel` does not specify a completed answer bag: collection, multiplicity and success-priority errors need independent occurrence and completion evidence.

Changing a constraint representation also changes possible alias representatives. `BindingSourceAgreement.typeMatch_finite` constructs a finite source match after changing the input store, preserving its complete solution theory and variable support. The latter also preserves the complete ordered family of mutually fresh type instances. `typeCastBatch_scoped_finite` consequently keeps the first successful candidate and every duplicate failure occurrence. These laws construct the new source match and merge; they do not assume an evaluator correspondence. `TypeMatchRel.output_agreement` also compares any two actual source matches over equivalent incoming stores: their reached theories and support agree. For a fixed functional type query, `TypeCastBatchRel.output_agreement` preserves the first success and each failure occurrence, and `joint_observation_eq` preserves the complete ordered caller/value observation. The literal `StorageTypesRel` profile discharges that query condition. Scoped queries permit different legal alpha instances, so their witness alignment remains a separate obligation; they are not assumed functional. Ordinary future unification has a separate representation law. `ReadoutCertificate.support_union_apply` preserves the incoming support plus the variables of each operand. A source candidate succeeds exactly when its new equation and the incoming constraints have a common finite model. `UnifyCandidateRel.transport_certified_reads` therefore constructs the counterpart operation on an equivalent certified store, without assuming a second selected output. Both success and no-match are preserved. `certifies_paired_pending` earns the two later readout certificates, absorbs every earlier pending read and preserves the complete correlated packet observation. Opposite alias representatives satisfy this law; a normal but uncertified table can change success into failure. These are ordinary constraint-operation laws, with annotated readiness and host services still requiring their own correspondence. The worker's `scoped_worker_symbol_cast_enabling_bridge` now consumes that finite input-store cast and constructs the independent source batch, contextual frames and local machine step under arbitrary saved callers and prior publications. Its output relation retains the joint caller/value observer on every public scope. It no longer assumes a source-side output batch or supplied output-frame comparison. The primitive fold now derives that batch from the executable cast and its complete candidate coverage. `scoped_worker_symbol_cast_of_covered_selected_models` supplies this instance for nonvariable input assignments when each selected raw output has a finite model. A second constructor earns those models from coherent input classes and loop-free selected outputs. The admission is substantive: a covered raw merge can create a cycle even from a consistent, loop-free input. Conversely, an exhausted covered candidate has no finite source match. The compound-arrow control rejects its first candidate, selects the second, and preserves both correlated caller aliases. These laws construct source batches and joint output observations. The configuration-level fresh-symbol and repeated-arrow squares additionally construct the annotated output frame and retain its readout certificate, caller frames and earlier publications. A continuous control rejects an

earlier arrow, selects a fresh polymorphic instance, performs the next argument cast, and returns through the actual caller to publication and completion. Its published expression is held, as the selected marking policy requires. A second continuous suffix reaches a distinct fresh instance without resetting the first frame. These controls do not establish general schematic application adequacy at arbitrary arity or recursion depth.

The complementary `headsStopped_block` relates two worker administrative transitions to one contextual stopped-head rule. It retains the stopped occurrence and pending siblings without demanding arguments. The executed discriminator separates `Empty` from the ordinary symbol `Error`; an error expression has its own stopping rule. The continuing head-to-argument and argument-to-caller laws compose with these retained frames. General recursive composition must still earn the next readout and entry-readiness premises rather than infer them from erased output equality.

Literal covered-read equality is stronger. On unifying `$a` with `$b`, the maintained alias store reads `$a`, whereas the contextual left-binding unifier reads `$b`. The formal control `variable_match_has_no_literal_frame_transport` excludes that literal relation for every output of this contextual operation. Nevertheless, `variable_match_joint_public_readout` proves agreement of the actual joint caller/return packet. The general retained comparison must preserve this joint meaning, readiness and future operations; agreement of the constraint theory alone is not that comparison. The negative `wildcard_without_model_is_not_finite` also keeps wildcard acceptance separate from consistency of the caller.

The cast correspondence therefore has two distinct conclusions. `ScopedConfigurationRel.cast_selected_step` uses a certified canonical presentation match for subsequent control; its payload and type-query policy are parameters. `ScopedJointFrameRel.lift_finite_cast` instead lifts any finite-model source cast to certified frames and preserves joint public answer bags, including opposite open-alias representatives. It does not claim literal control alignment. Re-presenting a constraint record retains its environment and readout, so `Frame.represent` cannot create a missing canonical match. For a fixed syntactic query, `scopedEnumerationTypeLookup_congr` preserves lookup under equal protected-name sets and equivalent binding records. This does not identify different subjects produced by subsequent materialization.

Frame-read stability and evaluation status are separate obligations. `FrameUnassignedOn` permits open operands whose variables are absent from the retained substitution's domain. Empty frames satisfy it for every query. Certified materialization earns it again, so `Frame.read_idempotent` preserves the entire annotated value, including nested readiness marks. A pending sibling retains this property across a cast only when the cast's new assignment keys avoid that sibling. The `private_type_cast` law records the required write footprint explicitly; fresh candidate names alone do not establish that footprint. For a private type cast, `Frame.CastRenewRel.pending_reads` derives preservation from the existing source freshness theorem: the incoming presentation and both matched operands avoid the pending scope in their keys and value variables. Annotated unification inherits this theorem through its proved erasure. `InvocationSelectionRel.selected_fresh` earns whole-arrow and domain freshness from the actual scoped selection. This is stronger than the key-only condition needed for idempotent materialization; the two invariants are not interchangeable.

The existing contextual machine therefore remains parameterized by reentry policy. Reinterpretation admits a materialized value regardless of its status; reuse retains the active-entry condition and may deliver a held expression directly. The plain-atom worker realizes the reinterpreting profile. `ScopedResultRel.materialized_entry_ready` constructs its next entry obligation from an actual covered read, including open results. Erasing marks does not justify identifying the worker with a reusing profile. These local laws do not yet prove the complete recursive invariant across all selected casts and saved callers.

Nested success priority is an observation boundary

The location of publication is observable even in a closed, untyped example:

```
(= (mixed) Good)
(= (mixed) (Error Early Failed))
(= (later) (Error Later Failed))
!(Box (mixed) (later))
```

The mixed child publishes `Good` and discards its error before its caller resumes. The later child then fails. The checked retained machine, under both invocation-type policies, finishes with only `(Error Later Failed)`, agreeing with the tested Rust HE and CeTTa profiles. Filtering only after the whole tuple can instead resurrect the discarded early error. Priority therefore does not commute with resumption.

The existing raw mutual judgments propagate unfiltered child results. `NestedPriorityControls.raw_parent_early` constructs such a derivation; `outer_only_source_cannot_cover_checked_run` proves that their outer-only `EvalRel` cannot characterize even the atom support of this completed run. This diagnoses the formal source boundary, rather than a disagreement between the engines. The upstream tuple and argument rules call the published, filtered `metta` judgment recursively.

The independent `OrderedProtocol.Completion.SuccessPriorityRel` states publication of a supplied finite batch. Its laws preserve duplicates, exact support and representation changes, and descend through batch permutation. The stateful traversal keeps effects already produced by discarded branches. The theorem `successPriority_source_iff` identifies the actual worker filter's entire occurrence list, and `completed_priority_retained_filter` consumes its saved filter frame without changing the reached space, caller stack or earlier publications. These are local laws: they do not establish that a supplied batch exhausts a recursive source search. Source/machine adequacy additionally requires completed child phases at every licensed evaluation boundary. A prefix containing only errors remains incomplete.

The traversal law `OrderedStateFlatMapRel.split` exposes the actual middle world at any cut of the source occurrences. Its converse is the existing append law. `PublishedTraversalRel.simulate` lifts local child refinements to ordered, completed publication under a relation between worlds and an error-class-preserving result map. Neither the world relation nor the result map prescribes a storage representation. Mapping equal results does not merge their separate occurrences, and discarding a result does not restore an earlier world.

The positive mutual family `Completed.AtomRel` supplies recursively published finite child batches over contextual values and certified value frames. Entry reads the actual reached environment before classifying delivery, casting or interpretation. Tuple traversal keeps the child values, and each selected physical equation completes its own body after contextual unification. The complete caller batch is read before success-priority classification; marking follows that classification.

Reentry, marking, rebuilding and type-match admissibility are explicit local policies. The unmarked intermediate profile reinterprets expressions and marks none. The recursive rules also support reuse of held values, retaining nested statuses and the selected environments. Cast renewal either retains the whole frame or follows the selected source type comparison. It does not replay the output binding record to infer value provenance. This finite-frame contract is stronger than unrestricted raw type matching. For example, an incoming constraint $x = (y)$ can be individually consistent, and a structural type match of y against (x) can independently produce $y = (x)$. The bare merge admits their union, which has no finite-term model. `cyclic_type_match` constructs this raw success, whereas `cyclic_cast_has_no_contextual_lift` excludes any certified finite activation for its result. This is a counterexample about the raw source relation, not a claim about scoped engine execution.

`CastResultRel.lift_iff_models` characterizes the missing condition exactly: a completed raw

cast has a contextual lift if and only if each output constraint record has a finite model. Its constructive direction lifts the selected structural match through contextual unification, preserving the original payload and each diagnostic occurrence. The finite-model guard rejects every representation of the cyclic success; ordinary logical unification therefore selects its failure branch. A positive wildcard cast retains the same nonempty activation. A finite-type profile can enforce this condition, or a scoped admission theorem can establish it for the eligible uses. Neither repair follows from representative renaming alone. Rational trees would require a different value/readout contract; no raw default or runtime is silently changed here. The completed cast relation now takes its matching policy independently of the lookup policy. `TypeCastBatchRel.change_matching` and `matching_iff` transport a complete cast when the matching relations agree on the actual query. In particular, `TypeMatchingPolicy.raw_finite_cast_iff` preserves the raw cast on the domain where its candidates have finite models, including the first-success decision and failure occurrences.

The explicit candidate `Policy.finiteTypes` changes only match admissibility from the ready-value profile. The guard is tested during candidate selection, not after committing to the first raw success. `CastResultRel.lift_finite_batch` constructs every contextual frame without a separate caller-supplied model premise, and `AtomRel.finite_cast` constructs the corresponding full source phase. `finite_cast_step` supplies its actual machine transition beneath any saved continuation and prior publication prefix. `TypeCastBatchRel.total` constructs a source cast from any complete finite type query under an arbitrary matching policy. `finite_cast_progress` combines this existence law with finite-frame admission to construct a machine successor, without assuming a completed cast or termination of the enclosing recursive evaluation. With the candidate list $[(x), \%Undefined\%]$, incoming constraint $x = (y)$ and expected type y, the raw profile chooses the cyclic first candidate. The finite profile rejects it and selects the later wildcard, retaining the incoming frame. `finite_cast_phase` and `finite_fallback_run` construct this source completion and its completed execution through the same machine. The general adequacy theorem remains quantified over the policy; no new continuation forms or control machine are introduced. This does not establish conformance of the candidate to scoped HE or to any runtime. `held_variable_phase` completes a held value obtained through a nonempty certified environment under arbitrary demand. The control `held_call_profile_separation` changes only reentry: the same held call is delivered under reuse and reduced by the shared equations under reinterpretation. Under reuse, held function-shaped data stays reusable; function syntax selected for interpretation requires instruction execution.

The general laws exclude mixed published successes and errors, split caller traversal without losing the suffix, and preserve equal error occurrences with distinct contextual frames. Expression entry remains the read-only untyped tuple/ordinary-equation profile. Declared typed-argument traversal is covered below; typed-function selection and application, minimal bodies, host effects and collection require their further constructors and correspondence. `NestedPriorityControls.completed_scoped_source_matches_run` constructs an independent contextual source batch for the example above and supplies a contextual small-step run with the same erased atom list as the retained scoped worker. The general source/control result below applies to the declared core; this control does not establish general refinement of the scoped worker.

A contextual small-step machine

The existing recursive-control owners also contain the contextual phase machine. Its configuration is

$$\langle S,\ C,\ K,\ O\rangle,$$

where S is the read-only space, C is pending work, a returned batch or a saved child context, K is a stack of first-order continuations, and O is the ordered published prefix. The eighteen work modes are atom, tuple, tuple branches, typed arguments, argument branches, function, function branches, invocation traversal, caller traversal, call, equation bodies, return, minimal execution, function-body iteration, raw invocation, raw equation bodies, one boundary and boundary traversal.

Their operands retain contextual values, certified activations and the source's explicit live/protected scopes. Continuations close a phase, form a tuple, resume tuple branches or callers, and append or prepend the retained occurrence batches. Typed argument frames retain source values and declared demands and distinguish successful argument packets from propagated failures.

The local transition relation has seventy-seven rules. Entry classification, complete casting, static signature selection and equation queries, and contextual unification are primitive premises. Recursive evaluation is work to be executed beneath a continuation. Closing a phase reads its complete caller batch, applies success priority and marks the selected values. Only a returned batch with an empty continuation can be drained into O, one occurrence per step. The final observer accepts only an exhausted return queue and empty continuation; an accepted observation has no outgoing transition (`observed_final`). This observer retains complete logical values and activations; public atom and scoped-binding views are further declared projections.

`local_source_back` reconstructs independent source work across every local transition. The mutual completeness proof follows the eighteen source judgments under an arbitrary surrounding continuation. Together they give `source_iff_finished` and `source_iff_observed` for every retained configuration, including its already published prefix. In particular, `loaded_source_iff_observed` proves

$$\mathsf{Completed.AtomRel}(S, a, T, M) \quad \Longleftrightarrow \quad \exists c.\ \mathsf{load}(S, a, T) \longrightarrow^{*} c \ \wedge\ \mathsf{observe}(c) = \mathsf{some}(M),$$

with fresh input syntax, the empty certified activation and fixed declared dispatch, lookup and policy parameters. No machine execution is used to define the source judgment.

`RecursiveExecution.Contextual.Protocol.Rule` reads those actual machine steps through the common finite execution judgment. Internal control steps, one-occurrence publication and the final observer have separate rules; stuckness is not successful empty completion. `steps_exhausted` and `exhausted_steps` preserve both the reached configuration and the exact new publication history. `source_iff_derivation` proves that independent source completion agrees with protocol completion. The binding-changing function/chain control executes through this adapter under arbitrary saved callers and earlier publications. The generic finite-block composition can therefore use the relational HE machine without determinizing its primitive fresh-name choices. Its remaining worker and concrete realization instances still have to establish their local laws. Resource grants and live-world service protocols remain separate adapters.

The existing world machine also has a constructed common execution reading, `RecursiveExecution.Contextual.Stateful.Protocol.Rule`. It retains the actual world, context, continuation and publication history at each cut. Its source-to-protocol correspondence reuses the independent stateful source judgment, and its completed verdict includes the reached world. The declared public observer pairs a world observation with the joint caller/value answer bag. The world observation can retain the source's effect history; equal answer bags alone do not establish world agreement.

The previously proved `RestoreRepresentation.step` supplies the local law of an actual `DerivationSimulation` instance. Its common transport theorem executes the nonidentity alias representation through two restored occurrences, a prior publication and a live caller. Completion agrees in both directions, with duplicate answer observations retained. The effectful control additionally composes two native writes at the actual first-write cut. A reset world is distinguished even with the same final answer bag. These results connect locally completed host-operation contracts; executing and authorizing an external service protocol remains a separate obligation.

One concrete service connection is supplied by `RecursiveService.AuthorizedAppend.operation_refinement_square`. The declared indexed-space append operation uses the existing store action's safety and update judgments. The source and logical machine retain arbitrary caller frames and prior publications, while the existing permission-checked provider performs the append. Its first provider request reaches the updated store and retains a return continuation. Resumption completes that return with one charged write; restarting the primitive appends a second copy. `source_compl`

etion_iff also inverts the independent source instruction, fixing the permission, complete reply and entire reached world. Revocation before the operation instead retains the store and produces a service fault; no completed source result, including an empty batch, is derivable. This is the existing nonconsuming permission profile, not a new authority registry or a native commit/arbitration proof. It supplies one local operation instance rather than correctness of every host operation.

The retained service boundary is constructed in the same owner as AuthorizedAppend.Retained.operation_refinement. Its recognizer follows only the selected operational context chain. The returned site carries a proof that the native instruction occurs at that licensed focus; an arbitrary subtree containing the same syntax does not become executable. The inner context identity selects the writer, while all enclosing context labels, inner caller stacks, the root caller stack and earlier publications remain retained. A local source-authorized world step corresponds to a service call, one provider request, its reply, and reinstatement of the reached evaluator configuration. These are four cursor/callback inspections, not a constant bound on the provider's internal work.

The intermediate cut is an actual handled cursor packet. Its provider world already contains the write, while the saved caller still contains the input world. Resumption reads the packet's world and publishes only the reply; it cannot issue the write again. Restarting the instruction in that world instead produces two stored copies, as an executed negative control shows. A refused operation returns a fault paired with its unchanged world and has no completed source instruction derivation. A bare cancellation contains no world, so the adapter does not infer one from the stale caller. The existing callback composition and stuckness laws have constructed instances with a three-inspection bound per closed callback transition. This is one declared operation boundary under arbitrary retained operational contexts, not a complete native dispatcher or an interleaving commit/arbitration theorem.

In the same namespace, validated_resume_reached_cut joins dependency validation to an actual relational AM prefix. The saved store is the store reached at the pause, including earlier own writes. An explicit external update chain supplies the current store; validating the selected reader's reached stamp proves its current source agrees. The operation's right is checked in the current authority world. Successful rebasing changes only that world, and the licensed suffix performs the one pending write with the existing service protocol. Both caller stacks, activation frame, context labels and all prior occurrences survive. A stale dependency or revoked right instead returns the exact saved configuration before any provider operation. Controls distinguish an unrelated edit, a changed dependency and revocation; they also reject the original pre-write stamp. This is a direct Contextual Stateful AM connection for the declared append service, independent of the plain-atom worker and its type-annotation admission. External rebasing is not an evaluator step, and the theorem does not transport arbitrary evaluator actions across arbitrary changes of world.

retained_prefix keeps the exact earlier values and activations at every later cut. held_entry_exact prevents a completed reuse from replacing the held result; completed_atom_no_mixed_errors excludes mixed final successes and errors. A nonempty-environment control runs below any saved continuation without changing earlier publications, and its negative control rejects a newly computed replacement. The reentry discriminator now supplies both source completions and both corresponding machine paths.

The machine inhabits the common reduction-span interface with its actual transition endpoints. Primitive fresh-instance choices remain relational; their concrete selection and implementation comparison require realization laws. This is a finite-completion theorem for the declared read-only core and its typed invocation and minimal phases. Finite completion does not determine when successes may become observable during an unfinished search. Grounded host execution, captured binding collections, effects, answer-prefix productivity, full source progress/divergence, answer-origin readout and the general scoped-worker/native comparison remain separate coverage obligations.

Typed arguments retain their declared demands

`ArgumentsRel` pairs each source value with its declared expected type. Each child runs through `AtomRel`, with the remaining source values and demands kept live. `ArgumentBranchesRel` processes each returned occurrence under its actual reached frame before proceeding to its siblings. A changed error or `Empty` stops that occurrence's remaining argument requests; an unchanged literal error is argument data. These are different evaluation contexts from ordinary tuple construction.

Success data must be distinguished from a propagated failure. The declared HE encoding is `(Ok (arguments ...))`. The contextual value codec retains whole children and its erasure agrees with the existing atom codec (`unpackArguments_erases`). The protocol invariant `ArgumentsRel.packet_or_error` admits only successful packets or propagated failures, including when the argument payload begins with `Error`. A concrete runtime may realize this logical boundary with its own result representation; the theorem does not prescribe a C layout.

The two argument continuations resume the remaining occurrence traversal or prepend a selected whole value to each completed tail packet. They reuse the machine's existing append/prepend control. `arguments_complete` and `argument_branches_complete`, together with the extended `local_source_back`, join both source families to the general retained configuration correspondence. Recursive child evaluation is explicit pending work, not an argument-evaluator callback.

`raw_arguments_exact` preserves arbitrary open values and nested statuses under raw `Atom` demands in the empty activation; `raw_arguments_cannot_reinterpret` rejects a changed completed packet. Nonempty-frame controls propagate an error read through a bound variable without evaluating that occurrence's later requests. The binding-change control starts with $x = y$, casts the first argument to obtain $y = T$, and reads the second argument x as T. Its reached frame certificate is earned by actual contextual unification. The complete run and its variant beneath arbitrary saved continuations preserve the frame and earlier publications, for every declared policy. These laws are consumed by typed function assembly.

The scoped worker comparison separately earns the read at publication. `ScopedResultRel.materialized_readResult` relates the native value $b.\mathsf{applyFull}(v, d)$ to the contextual annotated read, using the covered native read and both materialization laws. It does not assume that the read is the original syntax. Its stable specialization and complete-list version feed the actual filter/publication square without dropping repeated occurrences. None of these erased-read equalities establishes active status.

Argument entry also retains the variables in every remaining value and demand. The contextual request inventory and the worker's grouped values and types need not be the same list; their protected-name support agrees. `argsCons_step` consumes that agreement at arbitrary arity. The stopped changed-error branch is a finite block of two worker steps and one contextual step, retaining the skipped requests' surrounding caller and prior output.

A symbolic execution family composes these operations. From a returned head, it casts a first symbol against a fresh binding variable, evaluates an arbitrary finite tail of admitted raw symbols, assembles the application and publishes it to the saved caller. A tail of length n requires $8n + 16$ silent worker steps. The reached frame and the contextual held publication remain related beneath arbitrary related saved callers and prior output. This is the explicit `finiteTypes` candidate: reuse, successful marking and preserved rebuilding remain unchanged. An executed three-argument instance reaches completion; a separate two-occurrence publication step retains both equal inputs. A bound-held-operand control earns the materialized erasure on both sides but refutes active-entry admission solely by its held status.

> **Boundary.** The family starts at prepared head return and its tail consists of raw symbols. It does not establish general recursive argument evaluation, typed return, minimal-body adequacy or whole scoped-worker refinement.

Typed invocation separates selection from recursive work

`InvocationScanRel` describes a complete ordered signature scan. For each arrow, `ApplicationChecksRel` checks every argument-type occurrence under every incoming logical frame, then checks the return demand. A type occurrence either supplies the primitive contextual constraint renewal or has a negative match witness. Repeated declarations remain repeated occurrences. A nonempty applicability batch selects that arrow and discards earlier diagnostics; exhaustion retains all failed blocks and whether tuple fallback is eligible. The scope-aware lookup is a parameter of this finite operation. It contains no call to recursive language evaluation.

`ApplicationChecksRel.source_member` projects every accepted frame to the existing independent `ApplicationSuccessRel`. The contextual constraint renewal is constructed from a selected finite type match (`Frame.TypeRenewRel.of_modelled_match`); finite matching makes each single comparison total. This is not termination of recursive evaluation or an equivalence with every presentation of the historical raw constraint model. The latter's finite-model obstruction and the explicitly distinct matching policies remain as stated below.

The selected batch enters `InvocationsRel`. Each occurrence evaluates its head at the selected arrow, evaluates its arguments at their declared demands, and passes the assembled application through the selected return gate before advancing the next occurrence. An `Expression` return changes the call demand to `\%Undefined\%`, following the declared interpretation rule. Function head failures skip their pending argument requests. Tuple fallback appends its saved diagnostic block before the ordinary completed publication boundary.

Three new work modes and three continuation forms expose this work in the same retained machine. `function_complete`, `function_branches_complete` and `invocations_complete` extend both directions of the general retained-configuration correspondence. The duplicate-type control traverses selection, head casting, raw argument delivery, call and publication and retains both answer occurrences. Its saved-stack variant retains prior work; a separate wrong-arity control completes with its diagnostic. The negative `duplicated_raw_type_occurrence_not_dropped` excludes deleting a type copy. Nonempty-frame controls cover head failure and binding changes through function assembly. These are source/AM results for the declared finite core; the full scoped-worker comparison, grounded hosts, effects and collection still require their separate joins.

Minimal bodies and saved contexts

`MinimalRel` separates ordinary data from an explicit invocation. Its evaluation rules use `eval` and `evalc`; continuation and matching use `chain` and `unify`. The latter includes the declared diagnostic for malformed arity. Structural rules use `cons-atom` and `decons-atom`. Other rules execute a `function`, enter full `metta` evaluation or read `context-space`.

`InvocationRel` consumes complete equation frontiers, retains one work item per stored copy, and completes an authored function wrapper once. Ordinary producer data stays raw. The typed return gate subsequently decides its use through the existing entry policy.

`BodyRel` runs minimal work and reads its reached frame before checking for another instruction. Five licensed boundaries share one ordered `BoundaryBatchRel`: local chain substitution, body resumption, function completion, producer completion and typed return. A local chain operand is substituted into the template without a caller-binding write. The substitution erasure law `substituteBinder_erase` agrees with the existing independent `Minimal.substituteName`. Contextual unification also projects to the independent `CoreStepRel`; the bridge is `completed_contextual_unify_step`.

The source families index the context space, so recursive children may use another space. A control `context` (S', C', K') retains that child's actual work and stack. `contextStep` lifts one child transition and `contextReturn` returns its completed batch. The outer space, stack and published prefix

remain saved. Neither rule calls a recursive evaluator. The additional six mutual completeness proofs and `local_source_back` extend the same retained-configuration correspondence to these modes.

`immediate_function_exact` proves that a raw function return transfers an arbitrary payload once, preserving open syntax and nested readiness; `immediate_function_cannot_reinterpret` is its negative machine control. The binding-changing control begins at $x = y$, unifies $y = T$ while yielding B, substitutes the chain-local x into `(return \$x)`, and returns B while the caller still reads $x = T$. The boundary's exactness law excludes writing that local selection into the caller frame. A separate control keeps the saved caller as the `NoReturn` diagnostic subject.

`evalc_saves_child` exhibits the actual paused child configuration; `evalc_resumes_child` continues from it and reads the child context, preserving arbitrary outer continuations and earlier publications. `metta_raw_child_retains_stack` exercises complete full-evaluation child delivery under an `Atom` demand. These results concern read-only contexts. Grounded hosts, `call-native`, captured snapshots for `collapse-bind`/`superpose-bind`, world effects and their native realizations remain separate coverage obligations.

Evaluation status is an entry policy

An occurrence which requires interpretation differs from a value ready for delivery, even when both print as the same expression. `Entry.Rel` separates delivery, type casting and expression interpretation. Its classification is total and functional without assuming that interpretation terminates. The `reuse` policy delivers held expressions; `reinterpret` enters them again. Fresh entries have the same classification under both policies. The theorem `erasing_status_cannot_classify` excludes an atom-only dispatcher for these two occurrence states.

Publication has a separate named policy: mark successful expressions, mark only unchanged successful expressions, or retain unmarked results. `publish_readout` preserves the entire occurrence list and binding frames, and `publish_priority` transports success-priority classification. This is not observational equivalence of later reentry: the status deliberately determines whether later computation occurs. The entry view specifies neither a storage bit nor a nested-value representation. The contextual values below retain child statuses. `AtomRel.delivery_exact` connects delivery after an actual frame read to the recursive source family under every declared policy.

Substitution must retain contextual values

`Entry.Value` records the status of every expression occurrence. Its finite environments retain whole values at variable lookup. Erasing the statuses commutes with simultaneous substitution (`erase_substitute`) and with the existing finite-table readout (`erase_substitute_environment`). One certificate for that table earns `selected_materialization` in the existing binding frame. `selected_held_variable_entry` additionally delivers a stored held expression under every demand, retaining its nested statuses.

Constraint readout alone does not determine readiness. Two nonempty environments can have the same certified erased table and different statuses for the same variable (`constraint_certificate_does_not_determine_status`). A contextual realization must retain that additional information. Scope renaming preserves statuses; inverse renaming restores the entire value (`rename_cancel`).

The constraint record itself can change without choosing a new annotated readout. `Frame.represent` retains the exact environment and constructs its new certificate from complete constraint-model agreement. `represent_read` preserves the whole annotated value under every policy; the identity and composition laws make successive record changes coherent. Full support agreement preserves the invocation's protected names as a set, without requiring equal record order. A reversed equality edge and a variable-valued assignment supply a nonidentity example. The separate `constraint_agree`

ment_alone_does_not_preserve_readiness control rules out replacing the retained environment by an arbitrary model-equivalent one.

Rebuilding exposes a distinct policy choice. The preserve policy keeps data ready through substitution and satisfies ordinary environment fusion (preserve_compose, preserve_fuse). The reopenChanged policy reactivates a parent whenever its substituted syntax changes. Swapping x with y twice in a held expression restores the printed syntax but leaves the expression active. Fusing the swaps first leaves it held (reopen_composition_separation). Thus syntax-level substitution fusion does not imply contextual-value fusion for that policy. This statement concerns the declared rebuilding rule; a dialect comparison requires its own source contract.

Retaining the pending substitution sequence preserves intermediate changes. substituteAll_append proves the prefix/residual law for either policy, using the reached value at the cut. These operations manipulate semantic values and environments; they neither evaluate language programs nor select a backend representation.

Ordinary unification transfers these contextual values through assignments. erase_bind projects each update to the existing finite-presentation operation. UnifyRel.erases and UnifyRel.lift prove both directions: annotations neither add nor remove successful ordinary matches. The matcher compares syntax; readiness is not another equality constraint. Each sibling uses the environment reached by its predecessor.

UnifyRel.of_source_candidate constructs a contextual trace and certified reached readout for every surviving source candidate, including open operands and incoming aliases. contextual_unify_step composes this readout with the existing minimal-instruction rule and materializes its continuation from the actual reached values. For a completed equation frontier, UnifyRel.equation_frontier pairs every physical work item with such a trace, retaining source order and separate occurrences.

The open-alias control starts with $x = y$, then binds y to a held value containing a free z and a nested held child. Repeated uses of x and y retain that whole value. Filling z later gives the same syntax under both rebuilding policies, with readiness following the chosen policy. Distinct symbols remain incompatible. These are source binding and materialization laws used by the completed recursive source family and contextual control machine. The core's completed source/control correspondence is proved above; the further construct coverage and concrete realization joins remain.

Complete equation selection precedes body evaluation

EquationFrontierRel supplies independent finite coverage for one equation query. Every physical row is classified as a non-equation, an equation with no viable fresh instance, or one selected principal constraint theory. Equal equations stored at different positions remain separate occurrences. Different proofs or equivalent records of one unifier do not add alternatives. This is the equality-grounded matching profile; custom grounded matchers that branch require their own occurrence contract.

The existence theorem equationFrontier_exists classifies every row of any finite space, including with alias-bearing callers. It does not run an evaluator or assume termination of an equation body. A negative row certificate covers every legal alpha variant, rather than merely one chosen spelling. row_miss_of_instance earns this universal coverage from the absence of a unifier for one legal instance. The general occurrence_alpha_transport_logical proves the required change of instance by logical conjunction, without the older restriction excluding variable-valued caller assignments.

The frontier laws prove both source soundness and coverage, distinct physical origins, and invariant occurrence counts. The stronger frontier_covers_readout also preserves the whole binding solution theory and RHS observation under one permutation fixing every caller-visible name. In particular, a name occurring only inside the value of a caller alias is protected. A completed empty

frontier is equivalent to absence of every source query candidate. For an ordinary call this earns unchanged query restoration, including its complete binding frame.

The actual binding-machine matcher and merge each return at most one candidate in this profile. This bound alone says nothing about exhaustion: it also holds at depth zero. The retained-machine controls compare a four-row source frontier containing two equal equation copies with its actual query, reject the empty truncated helper result as source completion, and prove that the checked machine preserves the query as pending work. General query refinement and recursively completed child evaluation remain separate obligations; selecting a raw RHS does not evaluate it.

Unification has a stronger completed primitive join. `covered_unifyReadouts_iff_source` proves that the actual covered match/reconciliation search has a surviving result exactly when `UnifyCandidateRel` admits a satisfiable conjunction. The caller may contain bare aliases. The reverse direction constructs a finite semantic size bound, earns actual matching and merging witnesses, and transports them back to the covered depth using stability. It assumes neither a desired result record nor whole-evaluator correctness. Consequently `covered_unify_empty_iff` earns the source failure judgment from completed search. The retained primitive derivation now uses this theorem instead of requiring a separate source-absence premise, and also covers rejection after a structural match succeeds. Its soundness transports failure through either admitted binding observer; its execution law keeps surrounding callers and earlier emissions. The positive alias, post-merge failure and truncated-search controls distinguish these cases.

The whole query has a completed negative connection as well. `source_occurrence_reaches_covered_query` transports any viable source occurrence to the retained fresh instance and constructs a surviving actual query item. It uses checked query and caller-merge coverage, without closedness or alias normalization assumptions. Conversely, each actual prepared item constructs an independent physical source occurrence and its readout. The theorem `covered_query_empty_iff_source_frontier` combines these directions: under the checked local query, merge and readout bounds, an empty actual query is equivalent to an independently covered empty source frontier. The frontier accounts for all stored rows and legal fresh variants; the runtime's empty list is not its definition. A structural hit rejected by the caller supplies the post-merge negative control. This settles this query's absence boundary, not nonempty recursive answer multiplicity or body termination.

A continuation slot is distinct from its binding environment

The distinction has an executable counterexample:

```
!(function (chain (unify $x A B refused) $x (return $x)))
```

Rust HE returns `B`. An earlier CeTTa image returned `A`, although renaming the chain destination to `$y` produced `B`. The child unification constrains `$x` to `A` and selects the branch `B`; these are different pieces of information. Chain substitutes the selected value into its saved template and carries the child's bindings separately. Applying those bindings to the template first erases the slot. For $\beta(x) = A$, the two operations need not commute:

$$[x := B](\beta(x)) = A \qquad \neq \qquad \beta([x := B](x)) = B.$$

The repaired instruction path retains literal instruction operands and performs the selected substitution before entering the continuation with its reached environment. This also protects a nested instruction's future destination from premature substitution. It does not turn minimal chain's structural substitution into capture-avoiding lambda substitution: replacing a slot can also replace the same name inside a nested instruction's syntax. Each language profile must state the operation it actually provides.

Raw instruction shape checks expose another boundary. A variable constrained to an expression is still a variable at a raw `cons-atom` or `decons-atom` operand; substitution is licensed at the

instruction's specified boundary, rather than inserted before every shape test. For `unify`, success requires a surviving match and merge. A match whose merge conflicts with the incoming bindings selects the failure branch with the original environment, rather than silently losing that occurrence.

The runtime qualification compares these cases with Rust HE across the declared HE profiles and both canonical and fallback execution paths. Nested collection additionally preserves captured bindings, repeated equal occurrences and one committed effect across every scheduler pause. The implementations can enumerate the same completed multiset in different orders; exact resumption continues to preserve each implementation's own chronology. These are local instruction and runtime results, not the complete native/source simulation.

A raw rewrite result still has a return continuation

The root rewrite relation and the public evaluator observe different boundaries. For example:

```
(= (freeze $x) NotReducible)
!(freeze (+ 1 2))
```

Rust HE returns `(freeze 3)`. The argument has already been evaluated; the returned protocol symbol asks the owning invocation to retain that assembled call. An earlier CeTTa image instead published `NotReducible`; an intermediate repair restored the unevaluated `(freeze (+ 1 2))`. Neither observation implements that return rule. The independent source relation also lacked this stage on its successful equation and emitted-grounded branches. A checked counterexample exposed the source defect independently of the native implementation.

Let a be the assembled invocation, v its raw result, D its selected return demand and β its reached bindings. The ordinary return judgment has three cases:

$$\mathrm{return}_D(a, v, \beta) = \begin{cases} [(a, \beta)] & v = \mathsf{NotReducible}, \\ [(v, \beta)] & v \text{ is Empty or an error}, \\ \mathrm{eval}_D(v, \beta) & \text{otherwise.} \end{cases}$$

This is an intermediate judgment; the public observation still performs its own filtering. The repaired source rules and retained worker both represent this continuation explicitly. The invocation, bindings and demand survive a scheduler pause. Restoring the invocation does not evaluate its arguments again. In particular, a raw successor alone is insufficient evidence for replacing a public call: its ordinary return evidence is also required. The smaller atomic-contraction fragment licenses non-protocol successors; the staged machine continues to cover protocol replies.

Downstream component realization examples

The following examples concern admitted formal component models. They are neither definitions of HE nor additional helper-by-helper acceptance rows. A claim about current native source additionally needs its revision and translation/ABI assumptions; compiled binaries require separate tests.

The selected native return clauses also have checked partial projections. Their source squares cover refusal, stopped replies and the actual typed callback request, retaining the assembled invocation and reached frame. The ordinary callback's variant-only operand and its fully materialized source value remain separate readbacks. A direct corollary applies when those inputs already agree; general open-input transport is still a proof obligation. The generic retained adapter starts after the C child push. A constructed native-memory join now supplies that entry from selected parsed field updates, operand reads, scalar initialization and intrusive push/pop. The paused sequencer's advanced counter determines the occurrence, while the active return frame's copied row supplies its

environment. The retained residual contains the actual stack address, child address and complete storage. Selected owned-data copy paths now derive identical scalar, binding-row, variant and goal-map images from the actual stores and array transfers. Repeated payload cells survive, while equal contents in separately allocated arrays retain distinct physical identities. The retained binding path copies through the actual temporary and aggregate assignment; the resulting child-request equality also preserves decoding refusal. These are data projections at explicit successful helper boundaries. Allocation, retention, index rebuilding, cache refresh, materialization, payload decoding and ABI layout remain separate contracts; the result does not establish the entire native caller body or a loader from arbitrary source syntax.

One materializer boundary now executes the complete parsed variant-only helper together with its actual presence and storage helpers. For a live inline outcome with null variant storage, it returns the stored atom and preserves arbitrary disjoint framed memory. The proof supplies neither a materializer-result oracle nor a helper-noninterference assumption. Controls separate null input, answer-reference inflation, dead or malformed storage, and a retained unfinished cut. The full parsed helper is also checked on its zero-binding and populated-cache paths, under the actual automatic storage entry and exit contract. A populated cache need not equal the stored atom; the full helper returns that cache, whereas the variant-only helper ignores it. The controls refute unconditional agreement between these two readbacks and show that a pause before the cache write retains that write without replaying it after resumption. Nonzero binding materialization, allocation and ABI interpretation remain separate joins.

The physical statement profile separates by-value record admission from logical cell extent. An extent supplies an array stride; it does not imply that a pointer or integer local is a record. The explicit finite record inventory selects aggregate copying and automatic record storage. One checked profile consequently supports pointer-array addressing and reads, scalar pointer assignment and complete two-cell record copying together. A pointer supplied as that record's source is refused. Byte extents and native layout agreement remain independent representation obligations.

The finite physical Atom reader also distinguishes numbered symbols from managed symbols whose stored number is zero. Managed symbols retain their exact spelling bytes and length, including embedded NUL and invalid UTF-8; conversion to a source String requires a checked UTF-8 boundary. Actual forwarding, terminal-tag inspection and owned byte reads compose with the existing graph reader. Repeated child pointers preserve repeated positions, while equal contents at distinct addresses preserve distinct storage nodes. The sorted frame-directory reader separately executes the parsed probe loop: capacity, half-load and full-epoch invariants earn completion without assuming a successful lookup. These are physical helper correspondences, with table publication, native layout and borrowed lifetime still explicit.

The current binding reader composes those representations at a logical checkpoint. `CurrentBindings.prepare?` traverses the represented current slots and the complete pending tuple using one checked map from full native IDs to generational references. It constructs a `ReadoutCertificate`; `prepare?_joint_models` and `prepare?_pending_absorbed` use that same certificate for all future operands. Repeated uses therefore retain their correlation. `prepare?_activation_slot` derives the selected activation's row read from successful whole preparation and the actual slot fields, rather than assuming a decoder's desired result. Unknown, known but unbound, and unreadable or stale references remain different outcomes. `readout?_joint_source_readout` compares this actual decoded packet with an independently certified source presentation of the same constraints. The source table need not choose the reader's alias representatives. `readout?_joint_source_readout_after_merge` retains that joint meaning after an independently derived source merge and under its next certified readout. Its extension law follows from the merge derivation. Actual reader controls keep a repeated open residual and its returned occurrence correlated, while refuting literal agreement with the source's opposite representative. These are observation and future-source-constraint laws, rather than proofs of all native binding mutations or readiness transport. The reader additionally constructs `logicalBindings` by reusing `aliasConstraints`: a variable-valued slot contributes an

equality constraint instead of a rigid assignment. This view preserves the complete solution theory and name support, and earns the nonvariable-assignment shape required by the finite binding-comparison interface. Successful preparation earns its normal-table certificate and joint packet observation; an independent future source merge preserves that observation. The view does not alter physical slots. Its shape condition implies neither groundness nor invariance of an executable cast under replacement of its input record. A checked open-alias example still admits two distinct future ground valuations. `readout?_unify_exists_iff` and `readout?_unify_noMatch_iff` apply the certified unification law to the actual decoded operand pair. `readout?_unify_transport` constructs the counterpart source operation and both output certificates, preserving the saved packet through the new constraint. Its positive control executes the memory reader, then consumes that decoded pair while retaining repeated saved operands and an independently oriented source alias. The operation is a source unification over the earned logical store; no physical slot mutation is inferred.

The framed receipt now has its own operation square. `receiveDirectory_coordinates` preserves the ordered lexical coordinates while advancing the directory's write clock. Consequently the same source naming decodes the new slot inventory. For a payload p decoded as a, `prepare?_receiveDirectory` constructs that updated snapshot from the actual receipt and Atom read:

$$\begin{array}{ccc} D & \xrightarrow{\text{receive}(r,p)} & D' \\ \downarrow \text{decode} & & \downarrow \text{decode} \\ B & \xrightarrow{r=a} & B'. \end{array}$$

The Atom graph remains the captured graph; the finite-reader check on the updated constraints is explicit. Every other row, repeated occurrence and activation boundary is retained, together with the updated write metadata.

If the selected coordinate was open, `open_slot_write_models` proves that the new model theory is exactly the old theory conjoined with the decoded equation. `receiveDirectory_open_renewal` constructs the actual shared-provider reply and the next readout certificate. Its later table absorbs the earlier table on every operand, preserving the entire correlated packet under the new constraints. The executed alias control writes the previously open target and reads both saved occurrences through its new value. A bound overwrite is different: the corresponding executed control remains readable but refutes preservation of the old theory. These are laws of the admitted framed component and its source decoder. They do not establish the encompassing native C helper, ownership acquisition, or annotated reentry.

> **Boundary.** This component covers finite acyclic ordinary Atom graphs in the maintained native interface. Borrowed directory completeness, lifetime, field ABI and the numbered-symbol API remain explicit contracts. A reader refusal does not establish source failure. It is not yet a proof of the surrounding RESULT/CALLER dispatcher or its binding-application transition.

`CurrentReturn.entered_current_return_readback` then connects a successfully prepared tuple to an entered native child. It constructs the partial payload reader from the represented storage, composes the owned frame, stack and retained binding-copy loaders, and retains the saved call, return demand, selected physical row and older continuation. Its ordinary return projection agrees with the independent `CallReturnRel` under the inherited protected scope. A freshly written current variable also earns selected-frame materialization through its actual activation slot. An open payload is read through the same certified table as the remaining packet; the materialization law does not require the payload to be closed or already normal. The existing ground-value control is its closed specialization.

> **Boundary.** The entry theorem admits frame-only HE bindings and the same finite Atom profile. It does not establish the dispatcher's branch choice, the CALLER binding fold, transition to typed evaluation, or final relay publication.

The actual open-addressed handle directory now also executes its helper after loading each reached epoch. The by-value reference check, automatic storage, full-epoch comparison and lexical cleanup compose into the complete parsed `find_ref` body. Its retained runner preserves the exact unfinished suffix and does not repeat a completed helper effect. A separate physical symbol-table proof executes the actual span lookup, retaining the distinct seven-tenths load invariant, native-width index arithmetic and byte comparison. The shared source reader admits the actual const character-pointer view by an independent C11 type judgment; it preserves nullable pointer identity and the operand's reached effects without inventing storage ownership. Earlier pinned seven-owner checks audited 7,141 kernel declarations and 3,946 theorem declarations; 55 generated implementations were checked separately. Those historical counts do not qualify the current importing closure. These source/helper joins do not yet close nonzero binding materialization or the complete activation and caller-return paths.

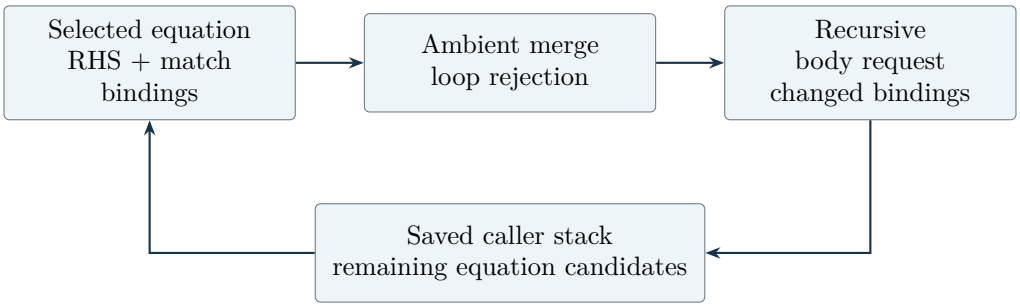
The current function-entry projection reads the body, checks its expression kind and records the selected native profile flag before deciding whether to push an instruction child. The checked branch preserves the parent stack, original bindings and private outcome target. Compatibility stepping, minimal-instruction stepping and a raw return are separate admissions to that branch. An ordinary evaluator call belongs to the other branch; lack of a projection for that call does not establish native failure. This selected entry theorem and the existing function-return square retain their independent body-execution and materialization obligations.

Provenance is essential here. A raw invocation refusal, a held Atom operand and a value explicitly completed by a native producer are different events, even when their syntax is the same symbol. Globally interpreting all completed values again would lose that distinction. A concrete Rust-compatibility defect exposed this boundary: `get-state` previously kept a stored callable unevaluated and omitted its effects. The repaired compatibility path first uses ordinary application admission, then emits its raw host result through the retained return continuation. Incompatible codomains are rejected at the owning call before the stored child runs. `get-atoms` retains its declared Atom demand, so stored callable data remains data. Explicit minimal `eval` uses the raw operation without adding ordinary return. Other profiles retain their declared completed-host policy. These producer-specific controls do not establish unrestricted HE conformance.

The same native qualification exposed two additional boundary errors. First, `case` must retain the source occurrence's bindings while matching and entering a branch. An Empty occurrence contributes neither a value nor its bindings; its empty fallback uses the incoming environment. Second, disabling an unbounded execution accelerator must not silently turn a requested lazy selection into eager evaluation. The repaired paths agree on both selected answers and committed effects. Pause/resume controls exercise the saved invocation and each allowed retained prefix.

Existing owners. `Mettapedia/Languages/MeTTa/HE/Spec/Eval.lean` and `Spec/Eval/Minimal.lean`; `Mettapedia/Languages/MeTTa/HE/Eval.lean`, `EvalSpec.lean`, `Correctness.lean`, `RecursiveTasks.lean`, `RecursiveMachine.lean`, `FirstSuccessSourceExecution.lean`, and `CeTTaRuntimeContracts.lean` (`CallReturnControl`); CeTTa's `eval_he_boundary.inc`, `eval_he_canonical.inc` and `eval_he_execution.inc`.

> **Boundary.** The source return judgments and their maintained evaluator refinement are kernel-checked. The native results additionally have executable Rust, profile, GC and retained-execution controls. These are separate evidence boundaries; the full native C simulation is still an obligation.



Local retained equation execution. The body returns through the saved continuations; it does not replace its changed bindings by the pre-call frame. This local realization law is separate from the general source/contextual correspondence and the remaining recursive worker simulation.

Native retained execution supports pause, continuation, inspection and cancellation. Checked readback components preserve contextual bindings and distinguish pending computation from written syntax, under explicit representation assumptions.

The general correspondence still needs arbitrary nesting, scope, quotation, typed raw arguments, shared choices, branch worlds, demands, ownership, effects and residuals. Flattened answer equality cannot establish this: losing a binding world may change a later demand.

Existing owners. `Mettapedia/Languages/MeTTa/HE/RetainedArgumentSpecification.lean`, `Mettapedia/Languages/MeTTa/HE/ScopedEquationQuery.lean`, `Mettapedia/Languages/MeTTa/HE/FirstSuccessSourceExecution.lean`, `Mettapedia/Languages/MeTTa/HE/ReflectedStepSharing.lean`, `Mettapedia/Languages/MeTTa/Bridges/GSLT/CeTTaNeedHistory.lean`, `Mettapedia/Languages/MeTTa/Bridges/GSLT/CeTTaFrameWriteback.lean`.

> **Boundary.** The model proofs and runtime controls are distinct evidence. They do not constitute a complete theorem about compiled CeTTa execution, nor freeze the eventual Prime candidate calculus.

PeTTa: ordered control through the common interface

The existing PeTTa evaluator has an explicit configuration: expression or delivery control, scoped bindings, caller and ordered search frames, the named-space store, input and output. Its transitions determine which alternative resumes next and where a selected case commits. The independent `DeclarativeSpec.Runs` judgment records completed answers or a fault with the reached effect state. No fairness policy is inferred from ordering the alternatives.

`Eval.CommonMachine` packages this actual step function as an `AnswerMachineCore` and uses the existing event-machine adapter. It lives with the downstream resumption laws; the base source evaluator does not depend on this adapter. `finished_iff_source` proves both directions between independent source execution and a common terminal observation; `loaded_iff_source` includes the actual loader. The terminal observation distinguishes a completed answer list from a source fault. A common `finished` outcome here means that this observation is available, not that the source program succeeded. `final_iff_observed` rules out an unobserved missing-step state.

The adapter publishes logical transition occurrences and retains the complete configuration at exhaustion. Its `resume` law appends only the saved suffix; controls preserve an effect committed before a pause and keep a selected empty case body from falling through to another case.

> **Boundary.** These laws cover the existing declared closed/saturated source profile. They do not identify it with CeTTa's scalar-delivery `CettaEagerDFS` profile, extend it to every open call, or prove a native SWI-Prolog or C implementation refinement. Those are separate comparison and coverage obligations.

The older PrologBridge dispatches a Pattern evaluator through an oracle; that interface does not discharge open-query execution or its fresh clause, DFS and cut obligations. Existing first-order LP substitution and unification results supply reusable background for that connection.

Existing owners. Mettapedia/Languages/MeTTa/PeTTa/Eval.lean, Resumption.lean, DeclarativeSpec.lean, CettaEagerDFS.lean; Mettapedia/Machines/ProductiveCoreAdapters.lean.

Complete ordered frontiers for open calls

The relational source component retains finite-term equations in each branch. An open call refines these equations jointly with one fresh instance of each selected clause. Its ClauseChoice.Decision judgment independently justifies selection, head mismatch, or absence of a common solution. Frontier scans every physical equation position and retains accepted positions in order. No executable matcher appears in those source judgments.

choices_iff_frontier proves exact equality of the computed list with a complete source frontier. Equal equations remain distinct occurrences; choices_ordered preserves their source order. Completed empty search requires a refusal at every position, by choices_empty_iff. scope_equation_frontier_iff joins this source criterion to the retained scope scheduler in both directions. Clause work precedes the saved alternatives and preserves their caller constraints, origin paths and reached world. The law holds for both declared destination policies, so changing their local evaluation-order choice does not require a new clause-selection proof.

A recursive source equation separates clause coverage from search fairness:

```
(= (dfs-loop) (dfs-loop))
!(superpose ((dfs-loop) (quote Later)))
```

authored_choice_diverges proves that every finite budget leaves a saved residual with no published answer. Its proof starts at the source superpose entry. The general dfs_loop_every_budget invariant retains arbitrary older alternatives, collection scopes and the reached world. The later value is independently publishable when it is at the head, by ready_answer_publishes. Thus a complete list of eligible clauses alone cannot justify eventual delivery under DFS. The scheduling condition belongs in the adequacy statement.

Executable controls also cover an aliased open call returning three ordered occurrences, a pause retaining its two remaining clauses, a completed incompatible call, and collection retaining duplicate rows while leaving the caller's variable open. These controls complement the general kernel proofs.

> **Boundary.** This is exact open-call entry in the finite relational component. Whole-program source adequacy, the declared cut boundaries, dynamic syntax and the complete SWI/native-profile comparison retain their own obligations.

Existing owners. Mettapedia/Languages/MeTTa/PeTTa/CettaEagerDFS.lean, CettaDFSStreaming.lean.

Licensed bodies and generative collection

The existing relational component now has independent local source judgments. Source.Prepared specifies body expansion for literals, quotation, bindings, fixed-head calls and constructors. Its finite demand and scheduling policies create pending work; they never supply an evaluated child. prepare_iff_source proves exact correspondence with the expander, retaining incoming constraints and the surrounding goal suffix.

Source.Step independently classifies expansion, accepted and refused refinement, completed return, inconsistent read states, argument-return splits, printing and retained operation requests. step_iff

`_source` proves both directions at a licensed active goal. This condition is substantive: `collapse`, `superpose` and `empty` own their scope boundaries. `scoped_form_has_no_local_step` prevents their treatment as ordinary body work. `source_step_scope` and `scope_has_local_source` join the local judgment to the retained dispatcher, with the actual world, context, origins and older alternatives. Serving a request is a further operational obligation.

Generative operations expose an opaque supply carrier through `FreshOperations`. Their signatures exchange the supply, syntax packets and protected names. The scoped driver owns caller advancement and world threading. `CollectionReturn` specifies completion at an empty child frontier, requiring the independent ordered-copy contract. It resumes the saved caller with the copied data and its original constraints. The reached world is preserved. `collection_return_iff` proves the maintained finite-supply operation implements this source boundary in both directions after token erasure. `owned_copies_valid` proves its fresh-copy contract; `shared_residual_copies_refused` rejects reuse of one residual variable across distinct answer occurrences. Partial child search cannot satisfy the completion rule.

> **Boundary.** The named-parser instance earns its post-reader inventory from the successful read and owned copy. Source occurrence binding is specified below. These local source rules compose into the scoped finite execution below; none verifies the compiled byte reader.

Existing owners. `Mettapedia/Languages/MeTTa/PeTTa/CettaEagerDFS.lean`.

Anonymous occurrences bind at the source boundary

The lexical reader recognizes `$_` as an anonymous marker. The scoped reader first reserves every authored variable name, then expands each marker to a fresh identity. A named variable retains its identity, including when it occurs repeatedly. Strings containing the same characters remain data. `Anonymous.Expansion` independently specifies this constructor-preserving binding traversal. `expand_sound` proves the actual traversal satisfies it, and `Expansion.card` counts exactly one new name per anonymous occurrence. The monotonically extended finite support prevents reuse of an earlier identity. `expand_of_named` proves named-only syntax is unchanged. The allocator reuses the existing finite-support fresh-string construction.

`readExpression_expansion` joins the lexical result to this source relation and its allocation count. `parseAtom_no_marker` proves dynamic parsing enters the same scoped boundary. The native-shaped Boolean view commutes with occurrence expansion and preserves the full variable inventory, so its existing reader-view correspondence remains exact. `loadParsed_named_result` joins that occurrence-bound read to the existing owned-copy allocator, with no caller-supplied inventory condition. `scope_parse_named_source` recovers the earned source result, token and exact continuation from the actual retained parse transition under arbitrary pending alternatives and collection callers. The source operation's `owned_named_parse` law likewise derives this result from its successful reply; it does not assume a separate raw parse or naming proof.

Controls match `(foo $_ $_)` against `(foo 1 2)`, while the repeated named variable rejects that pair. Other controls protect a later authored `$__`, quoted marker text and distinct top-level occurrences. The retained parsed-pair controls return unequal anonymous variables and keep a binding to one from constraining the other; the existing repeated-name controls still share. The machine's continuation grammar and source/execution correspondence do not change. These are laws of the source reader and its declared operation models; compiled parser agreement remains separate evidence.

Existing owners. `Mettapedia/Languages/MeTTa/PeTTa/SpaceSemantics.lean`,
`Mettapedia/Languages/MeTTa/PeTTa/CettaReaderView.lean`,
`Mettapedia/Languages/MeTTa/PeTTa/StdLib.lean`,
`Mettapedia/Languages/MeTTa/PeTTa/CettaDFSStreaming.lean`.

Complete scoped source correspondence

`Source.Invocation` distinguishes local primitive results, parse results, complete equation frontiers, faults and unserved requests. An equation call installs its fresh body work before the older alternatives. No constructor assumes a completed recursive evaluator reply. `ActiveStep` combines these cases with the independent ordinary rules and literal scope entries; `ScopedStep` adds exhausted root search, exhausted child collection and retained associated-work or allocation requests. Its source configurations use the existing scope carrier and one shared world.

`scoped_step_iff_source` proves exact correspondence with the entire scope dispatcher for every properly nested configuration. The rule family is parameterized by local generative operations; the theorem instantiates the maintained owned supply. Both declared destination policies are covered.

The common productive-block owner supplies `Derivation`, a finite execution judgment over independently supplied local rules. It distinguishes zero-budget residuals, silent steps, ordered publication, finishing, faults, requests and explicitly absent transitions. `Derivation.sound` composes local soundness; `Machine.derive` composes local reflection and the continuing-state invariant. `Machine.run_iff_derivation` joins them. `Derivation.resume` appends a suffix at the exact saved residual, carrying earlier publications once and in their original order.

The actual PeTTa instance is `Scoped.source_run_iff_derivation`. Every finite run of the existing scoped streaming adapter corresponds to a source derivation, in both directions, with identical answer occurrences, reached bindings, world and retained scopes. `completed_iff_source` is its completed-search specialization. `completed_results_unique` compares completed derivations even at different budgets; their ordered publications and entire reached configurations agree. The divergent-first-choice control also has no completed source derivation at any budget. A saved later answer cannot certify completion of that search.

Commit delimiters and first-success search

Commitment belongs to the source control state. Each branch records the boundary of the older alternatives outside its current predicate or meta-call. This logical delimiter is independent of a native stack address. Equation selection still earns the complete ordered frontier; `predicateEntry` then installs the callee's delimiter and saves a return marker restoring the caller's delimiter before its older goals run. The entry map is injective, so this bookkeeping cannot merge source occurrences.

`once` opens a child delimiter, evaluates its child, commits after the first success, and restores the caller. It does not collect or independently copy the child's answers. A child failure supplies no caller answer; an unreachable later success is not promoted past a divergent first branch. `cut` commits at the current predicate delimiter and returns its ordinary boolean result. These scopes follow PeTTa's translation to Prolog control; the corresponding host rules are described in the SWI-Prolog manual.[1]

For example, the following query retains the outer alternative and produces the ordered answers `A`, `Outer`:

```
!(superpose ((once (superpose ((quote A) (quote B))))
             (quote Outer)))
```

Pruning the whole frontier at `once` would lose `Outer`.

The independent source commit rule names a split of the pending frontier into a younger prefix and the retained outer suffix. `scope_commit_split` proves the actual scope step keeps that suffix exactly, together with the current bindings, source origins, reached world and parent scopes. `scope_commit_invalid` retains a malformed delimiter as a control request; it cannot become completed

[1] https://www.swi-prolog.org/pldoc/man?predicate=once/1 and https://www.swi-prolog.org/pldoc/man?predicate=!/0.

failure. The existing local and scoped correspondences, finite source derivations, completion and resumption laws extend through the new rules for both destination policies. The recursive DFS invariant now accounts for the growing saved return-marker stack.

The lifetime adapter uses that actual delimiter step. Let $C(s)$ inventory the contexts in all retained frontiers and saved collection callers, and let P be the externally pinned owners. The retired set is

$$D = C(s) \setminus (C(s') \cup P).$$

`commit_context_support` proves that $C(s)$ is the union of the younger-prefix contexts with $C(s')$. A discarded occurrence therefore does not retire a context still used by another occurrence, a parent or a caller. `commit_lifecycle_square` constructs the independent source cut and applies the existing `OwnedLifecycle.closeScope` to D. Every complete cell reachable through a surviving owner's roots is preserved, including shared descendants and cycles; the reached semantic world is unchanged. Checked controls retain duplicate outer occurrences, a collection caller, an external pin and prior output. Cancelling every context named in the discarded prefix instead destroys a still-needed shared graph. This local interface law assumes the supplied ownership roots; it neither discovers all native strong roots nor authorizes cancellation of an external service.

The general `commit_first_answer` law starts after the first child answer reaches its commit boundary. Three logical inspections discard the named younger prefix, restore the saved delimiter and record that answer, preserving its bindings, origins, reached world, allocator, outer alternatives and parent scopes. A root answer is published immediately; a nested collection retains it internally. No termination or exhaustion premise is imposed on the discarded siblings. `commit_first_answer_source` constructs the corresponding independent finite source derivation. These laws neither assert that a first answer exists nor grant permission to cancel an external service.

The controls cover nested commitment, surviving outer alternatives, later predicate clauses, failure after cut, quoted cut, collection-local cut, variable correlation and first success before a divergent later branch. An actual saved checkpoint already contains its printed effect; its resumed suffix produces the answer without repeating that effect. `protected_cut_source` and `local_cut_source` additionally construct completed source derivations checked in Lean. Their public observations differ, proving that the delimiter cannot be erased from the logical state. Return markers add declared logical transitions; no physical runtime cost claim follows from those counts.

> **Boundary.** These theorems cover the declared finite destination profiles and their local operation instances, at arbitrary scope nesting. They do not establish full SWI-PeTTa conformance, the remaining dynamic or conditional-control admission, global world/freshness and authority invariants, or native implementation refinement. Unknown operations remain explicit requests. The native saved-state view carries the logical delimiter; decoding it from native control remains a realization obligation. Fixed DFS order is not a fairness or effect-commutation theorem.

Existing owners. `Mettapedia/GSLT/Distinction/ProductiveBlocks.lean`, `Mettapedia/Languages/MeTTa/PeTTa/CettaEagerDFS.lean`, `Mettapedia/Languages/MeTTa/PeTTa/CettaDFSStreaming.lean`.

A shared clause-entry boundary

Native lowering can reuse a structural matching law without identifying the whole dialects. `ConstructorMatch.petta_match_he_pending` starts with an actual PeTTa match of a constructor pattern against ground data and constructs an HE query-matcher occurrence. Its `petta_match_he_pending_exact` strengthening constructs exactly one such occurrence at every larger matching depth. This is the equality-grounded matcher profile; it does not bound an extensible custom matcher or combine distinct stored equation copies. One resulting HE binding record agrees with every source lookup and materializes the complete pending tuple. The records may store assignments in different

orders. Nonlinear patterns remain admitted; PeTTa's special `cons` list view is excluded at every pattern position. The converse `he_match_petta_pending` reflects an actual HE matcher occurrence while preserving that same record's whole-tuple readout. `petta_mismatch_iff_he_empty` identifies a PeTTa mismatch with absence of HE matches at every depth. An empty zero-depth prefix alone supplies no such evidence. These are clause-entry/readout laws, not recursive program equivalence or a completed program-search theorem.

The comparison also has an independent equation-model boundary. `petta_match_iff_solution` characterizes constructor-match success by satisfiability of the pattern equation on ground query data. `query_occurrence_he_pending_exact` obtains that model from an independent query occurrence's actual result conjunction and constructs the exact HE matcher frontier together with the PeTTa tuple readout. `no_query_occurrence_of_petta_mismatch` excludes a rejected chosen fresh instance.

The comparison now also covers complete finite source selection with open queries and alias-bearing callers. The source-level `petta_accepted_iff_he_candidate` constructs HE matching and logical conjunction from a PeTTa constraint model, and constructs a PeTTa model in the reverse direction. `he_candidate_petta_solutions` equates their full solution theories, including every future observation of the caller store. No variable-representative equality is assumed.

The continuation boundary uses the reached theory rather than restarting from the caller's initial record. `petta_reached_refinement` constructs the next HE candidate from an actual PeTTa source refinement and preserves the complete solution relation for the following sibling. Its refusal law reflects absence of any compatible HE candidate. A control first binds an open caller variable to `A`; the next sibling's alias cannot then match `B`. Reusing the initial caller instead would admit that match. `source_equate_he_pending` connects the law to the actual destination equation step. A head's certified packet materialization remains valid after the next refinement, and one environment reads the entire packet of caller variables, saved values and pending siblings. The result is joint observation equality, not equality of independent marginal values or residual spellings.

PeTTa's allocator is injective on the finite names used by a clause; HE's source alpha-variant asks for a globally injective name function. A finite injective map extends to a permutation without changing the pattern or body. `protected_petta_clause_he_alpha` therefore earns HE's unchanged source freshness judgment from the actual PeTTa allocator after protecting the complete finite caller scope. A second constructed permutation relates any two legal variants while fixing that caller. It proves that refusal at one variant excludes every other legal spelling.

`petta_choices_he_frontier` lifts these row decisions over the actual complete ordered selector. Its incoming stores are related by their complete solution theories, allowing different constraint records after an earlier refinement. `reached_head_then_frontier` connects an actual open-alias source refinement to the following complete query using that earned relation. Its paired work items retain the same source positions, fresh bodies, older caller goals, destinations, reserved names and commit delimiters. `petta_choices_he_occurrences` proves that every completed HE source frontier has the same occurrence bag. In particular, equal stored equations at distinct positions remain distinct copies. `protected_petta_choices_he_pending` preserves the bag of source positions paired with these joint packet observations over the actual complete frontier, protecting the pending packet before freshening. Its duplicate-clause control retains both occurrences with a nonempty aliased caller and arbitrary pending operands. This composes clause selection with local constraint continuation; it does not yet prove arbitrary open recursive language equivalence.

A PeTTa private result destination need not exist in the HE store. `private_destination_pendin g_observations` hides a fresh destination by extending the source model only at that name, while preserving one joint value/caller packet through the existing valuation-congruence laws. Its incoming relation already permits private destinations from earlier returns. `petta_joint_refinement` constructs an HE matching candidate from equality of joint operand/continuation observations; complete-store agreement and identical operand syntax are unnecessary. Success and refusal are re-

flected at this boundary. `source_private_destination_pending` constructs the two actual source equation steps that introduce a fresh destination and match through it, while HE matches the value directly. Reusing a public variable as that destination has a negative observation witness. This is a local source composition law consumed by the completed recursive source-tree comparison below.

`joint_pending_prepend_fresh` constructs model extensions for fresh clause syntax on both sides of the scoped relation. Pattern and body share one interpretation of their variables. The scoped success and refusal laws then give `petta_selected_joint_row`: an actual PeTTa selection constructs the HE source row at the same stored occurrence, with joint observations of its reached body and caller. `protected_petta_selected_joint_row` earns the freshness premises from the existing allocator after protecting both packets and the incoming binding support. Clause variables are private; matching may still constrain public caller variables. `protected_petta_joint_choices_he_frontier` folds those proved local laws over the actual complete PeTTa clause selection and constructs the complete HE source frontier. Its scoped relation preserves the joint reached body/caller observations and the multiset of source-position occurrences, including duplicate clauses. A control combines an open aliased caller, a past private result destination, duplicate matching clauses and a refused different head. The two frontier comparisons reuse the same generic fold; there is no second matching semantics. These local laws feed the recursive source-tree comparison below.

The existing scoped PeTTa machine also has a source-level continuation attachment law. A saved goal suffix is protected before fresh allocation; attaching it shifts each saved argument-return depth by its length. `ProtectedContinuation.preReturn_derivation_attach` transports an arbitrary finite ordinary recursive prefix through this attachment. Complete equation frontiers, including rejected alternatives and duplicate source paths, retain their reached constraints. Only the current child's work gains the suffix; parent scopes and collected answers remain unchanged. `preReturn_return_continuation_iff` joins this prefix to an actual returned head and an independent suffix derivation. The reached binding equations, world, context, origins and pending frontier initialize that suffix. `preReturn_continuation_run` executes the actual machine from the reached cut, preserving the reached world instead of repeating the prefix. This source profile admits local silent steps, refusals and complete ordinary equation invocations. Local printed output is retained; external primitive replies, child collections, cuts and terminal answer publication are outside the prefix transport. Its terminal inversion is the independent source law `Source.Step.returned_iff`; no continuation adapter supplies a second definition of source return.

Open child values must be observed together with their unevaluated siblings and caller packet. `Destination.solutionTuples` is the canonical all-model source observer. `resolve_read_preserved_by_extension` proves that a materialized open value retains its meaning in any later model extending the reached equations. The head is not frozen to a separately chosen value before the remaining siblings run.

`OpenContinuation.OpenSpine.transport` uses this boundary for a common tuple/typed-argument spine induction. Each head has an independent HE child derivation and an actual retained scoped PeTTa path. The proof reconstructs the existing HE `TupleRel` or `ArgumentsRel`; it reads the whole evaluated prefix in each final tail frame. Thus later siblings can refine an earlier open variable while its correlation with the caller remains intact. The saved continuation contains the final equation or invocation goal: `assembled_equate` consumes its reached equation cut, and `call_frontier` constructs the complete protected clause frontier with physical positions and duplicate occurrences.

The earned recursive closure is `OpenRecursive.Evaluation.transport`. Five mutually inductive source judgments describe evaluation, arguments, argument alternatives, invocation and clause bodies. Every defined call carries a complete `ClauseChoice.Frontier`; no child-completion callback occurs in the induction. Their forward theorems construct the existing HE `AtomRel`, `TupleRel`, `CallsRel` and `EquationBodiesRel` derivations. `Evaluation.paths` reconstructs a source branch path to every outcome. A `List.Forall2` outcome relation gives exact ordered equality of the joint observation packets, hence also a multiset equality which preserves duplicate occurrences.

Fresh allocation is split at its actual boundary. A `PrivateDestination` is fresh for the reached equation store and its input syntax. Its saved continuation intentionally mentions the destination, because that is the child's return channel. `Evaluation.Allocated` separately protects the evaluated prefix and pending caller packet at every recursive cut. This split is required for nonempty argument lists and is exercised by the positive recursive control.

There is also a genuine source-policy boundary at an open variable entry. `active_binding_separates_dispatch` uses the pure rule `(= (f) A)` and a certified active binding `x = (f)`. Both source stores agree on every joint observation packet, but HE reads then evaluates `$x` to `A`, whereas PeTTa's variable rule returns `(f)` as data. The program satisfies the existing scoped code admission. Consequently, model agreement cannot supply arbitrary open-child evaluation correspondence. `AtomRel.input_read_eq` gives the valid HE transport law: the complete reached read, including readiness, must agree. A larger shared open fragment therefore needs an explicit compatible evaluation profile as well as earned recursive occurrence coverage; changing binding representatives alone does not resolve this separator.

Passive captures supply the structural variable-entry profile. `PureConstructor.Value` admits constructor data with an explicit scope of free names; its empty-scope instance is the existing closed-value profile. One completion induction, `Value.he_completes_unassigned`, serves both. `PassiveEnvironment` requires passive, error-marker-free stored values. Its substitution, binding and unifier preservation laws follow actual materialization and both binding orientations. Call-expression matching requires passive child symbols and arguments, so a defining outer head need not itself be a passive application. Certified normal readout then earns bound-variable evaluation. `OpenContinuation.passive_variable_child` constructs the actual two scoped source steps after any evaluated prefix, preserving the joint packet and every retained caller component. A positive control stores an open held constructor; the active-call separator is excluded for every name scope. The existing `CodeRel` admits scoped passive captures and restricts substitution to such captures; its read and scope extension laws preserve admission across materialized siblings. Active applications retain a literal ordinary head. A computed variable head can become a defining call under HE's initial read while PeTTa retains its raw non-symbol-head data dispatch. The comparison excludes that syntax through `CodeRel.no_variable_head`, without changing either source semantics. Open clause entry also needs passive patterns: unification may bind a caller variable to a pattern subterm. `ConstructorPatterns` requires ordinary heads and passive, error-free argument patterns. Its frontier law inherits the actual source row and transports that admission along its alpha-variant. `ConstructorPatterns.clause_entry` then earns both a passive reached environment and admitted materialized body, in the caller scope extended by the fresh pattern names. It uses actual unification and no child-completion premise. The existing closed body-admission proof factors through the common `CodeProgram.work_body_code` law. The recursive duplicate-clause control satisfies the admission; a pattern containing an applied defining head is rejected. The recursive source-tree induction consumes these closure laws at every selected clause.

$$\begin{array}{ccc} \text{PeTTa ordered clause work} & \longrightarrow & \text{HE source frontier} \\ \downarrow_{\text{source positions}} & & \downarrow_{\text{source positions}} \\ \text{occurrence bag} & = & \text{occurrence bag.} \end{array}$$

The principal positive control evaluates open `(fork \$x)` under `x = Z`. It enters both identical physical clauses, recursively evaluates `strip`, and assembles `Box Z`. The theorem constructs two result occurrences and their branch paths, proves exact ordered packet equality, admits the joint observation `[Box Z,Z]`, and rejects `[Box Z,S Z]`. Thus duplicate multiplicity and caller correlation are both non-vacuous. The active-bound-call and active-pattern examples remain the negative admission boundaries. Deterministic `CollectionScope` DFS queue linearization is a subsequent PeTTa machine theorem; it is not folded into the common-fragment source induction.

Exhausted calls give a small separation witness. With the sole equation `(= (f A) OK)`, both implementations return `OK` for `(f A)`. HE retains `(f B)` when no clause matches the defined head; PeTTa returns no answer. Both retain `(unregistered B)` when its head is entirely undefined.

Thus a common eager fragment must justify the coverage of each reached defined call, or explicitly translate this difference. An answer-bag quotient cannot erase a residual value versus no value.

Existing owners.
`Mettapedia/Languages/MeTTa/Translation/HEPeTTaNativeLoweringContracts.lean`;
`Mettapedia/Languages/MeTTa/Translation/HEPeTTaOpenRecursive.lean`;
`Translation/he_petta_defined_miss.metta`.

Recursive source completion and tuple-protocol separation

`PureConstructor.Evaluation` is a finite, proof-relevant source tree for ordinary eager constructor programs. Its mutually defined argument, invocation and body families account for complete structural clause scans, physical row positions and recursive child completions. Every argument answer finishes its continuation before the next answer is explored. The tree is independent of machine transitions, runner fuel and evaluator callbacks. Typed demand, primitive/control heads, the special `cons` list view and exhausted calls to a defined head are outside this candidate shared fragment.

The checked `Evaluation.petta_path` constructs an actual path of the maintained PeTTa machine for every such tree in an annotation-free program. `Evaluation.petta_source` connects that path to the independent completed `PeTTa.DeclarativeSpec.Runs` judgment, preserving the entire ordered answer list and the initial pure world. All five mutual families have path theorems; the recursive bodies are not left as assumed evaluator replies. `Evaluation.petta_completed_iff` characterizes every completed machine observation from a load with such a source tree, including its whole store and I/O. It does not claim source-tree admission for excluded programs. The control `duplicate_recursive_petta` proves the exact two-answer list for two identical stored `fork` clauses, each invoking a recursive `strip` on a constructor natural, at every natural-number input depth.

`Evaluation.values` derives the passive return contract from the whole source tree. `Evaluation.answer_read_exact` then preserves any annotated representation of an answer through every certified HE frame and every declared rebuilding policy. The generic source law `Frame.read_variable_free` proves that result without examining binding storage. `ConstructorHead.he_frontier_empty` also constructs complete HE absence evidence against the actual equation table for an undefined constructor head, with arbitrary retained caller constraints.

These connections do not establish unrestricted recursive HE equivalence. Two source controls locate a precise obstruction. Active `(Empty A)` reaches HE's empty marker, whereas explicitly held syntax is reused under the ready policy. `no_atom_only_ready_evaluator` proves that erasing readiness cannot define a complete evaluation state for that policy.

The return base case now has a constructed HE connection. `Value.he_completes` evaluates arbitrarily nested passive data with unchanged syntax and the same certified frame, allowing every nested occurrence status and every declared local policy. Its output annotations come from HE's actual publication constructors. The explicit `NoErrorSymbol` admission excludes the scalar-tag collision described below; it is checked on the materialized value, not merely its raw template. `Value.he_return` joins that phase to the full return boundary.

`passive_equation_call` then constructs a complete HE call from its physical frontier when the binding constraints determine passive body instances. Contextual unifiers and certified frames are constructed from the source candidates. Each stored work occurrence produces one result, retaining its reached binding record; no complete evaluator reply is an assumption. The duplicate-call control proves this for every certified caller, including nonempty alias-bearing callers. It does not assert that arbitrary recursive equation bodies are passive.

Fresh-body materialization is also representation-independent. `ReadoutCertificate.forced_read` proves that a value forced by all models of a constraint record is the literal read of its retained certified table. `query_occurrence_template_readout` derives the read for an actual selected source

occurrence from a closed query, one source pattern instance and coverage of body variables by the pattern:

$$\begin{array}{ccc} \text{raw body} & \xrightarrow{\text{source substitution}} & \text{materialized code} \\ \downarrow\alpha & & \| \\ \text{fresh body} & \xrightarrow{\text{certified frame read}} & \text{materialized code.} \end{array}$$

`frontier_body_read` applies that square to the actual work occurrence and reached contextual frame. The materialized code may itself contain further calls. `Evaluation.substituted` constructs the corresponding finite source tree with its input substitution moved into syntax, preserving its complete answers. A positive control keeps two uses of one fresh variable correlated through any qualifying table; a negative control shows why an escaping body variable is not determined by its pattern.

Complete structural clause scans now construct that HE frontier directly. `fresh_pattern_model` combines any satisfiable caller model with a fresh rule model, preserving the caller on its entire constraint support. On closed ground query data, `fresh_candidate_iff_source_match` equates viable HE candidates with the independent structural source match. The shared match specification owns the substitution-stability and ground-roundtrip laws used here; the inversion development imports them rather than duplicating them. `ClauseScan.he_frontier` then constructs every selected physical work item and excludes every omitted row under all legal fresh spellings. Its pairing retains the original clause position, raw body and actual source substitution image. No evaluator answer or implementation record is supplied to this construction.

The explicit `RangeRestricted` admission covers body variables by their own raw pattern. `ClauseInstance.body_model` proves that every model of the reached work item reads its fresh body as that source substitution does. The code so obtained may contain further calls. The control `duplicate_recursive_frontier` retains positions two and three and the exact two recursive tail calls at every constructor-natural input depth, with arbitrary certified caller constraints.

`fresh_candidate_caller_theory` and `ClauseInstance.caller_observations` also preserve the existing `BindingTheoryEquivAt` observer on the full visible query, live continuation and incoming constraint scope. Every caller model extends to a model of the selected private constraints with the same protected values; every reached model restricts to a caller model. Equality of binding records or representative names is unnecessary. An open-assignment counterexample shows why this unchanged-caller law requires a closed query: a public query variable may legitimately acquire a new binding.

For passive instantiated bodies and passive argument syntax, the composition reaches the full evaluation boundary. `values_he_tuple` reuses the nested data proof as a complete annotated tuple packet. `ClauseScan.he_passive_evaluation` joins that traversal to clause selection, return, reached-frame readout, success priority and publication, under every declared local policy. `ClauseScan.passive_completed_pair` constructs completed HE and PeTTa source evaluations with the same entire answer list from the same admitted source work. The operational bridge's `ClauseScan.passive_machine_pair` composes these derivations with the existing correspondence laws, constructing a path of the retained HE machine and a sufficient-fuel PeTTa run with those same answer occurrences. Every published result additionally preserves the protected caller-binding observer. It introduces no additional machine or evaluator. The captured-variable control uses its argument twice in each of two equal stored clauses, obtaining two equal data results for every certified HE caller. The raw-variable counterexample shows that omitting fresh instantiation can instead conflict with a satisfiable caller. The recursive construction now extends beyond that passive-body result. `CodeRel` distinguishes active source syntax from passive annotated data and records the scope of variable leaves. Its renaming and substitution laws retain every nested readiness mark under either rebuilding policy. `CodeProgram` admits each raw rule body in its pattern-variable scope; it is a syntactic condition, not an evaluator postcondition or a termination assumption. Actual structural matches earn their safe captures and variable coverage. `ClauseInstance.body_code` then derives closed executable code for the fresh body in the reached certified frame, with the same

materialized syntax as the source substitution. No recursive result is a premise of this transport.

`RecursiveTransport.he_evaluation` constructs a complete HE batch from every admitted finite source tree. Its five-way induction composes expressions, arguments, argument alternatives, invocations and recursive bodies. It retains the entire ordered answer occurrence list and each actual result frame under every declared local policy. Tuple assembly reuses HE's existing tuple and call judgments; it does not introduce another evaluator. `Evaluation.recursive_machine_pair` joins this construction to the maintained HE and PeTTa machines, preserving the pure world and complete answer list. The `nested_recursive_completed_pair` control proves four equal answers when two separate argument positions each run a recursive binary choice, at every constructor-natural input depth and for arbitrary certified HE callers. `held_recursive_call_not_admitted` supplies the opposite readiness control.

The reverse direction now has two general base laws. The independent `EquationFrontierRel.source_related` pairs every two complete scans of the same stored presentation by physical position, raw pattern and raw body. `ClauseScan.he_frontier_all` transports the structural source scan to every such HE frontier, without requiring equal private names or constraint records. `RecursiveTransport.passive_atom_exact` and `passive_tuple_exact` characterize every completed phase on admitted passive data: there is one value with the same erased syntax and the same caller constraint record. They retain the actual reached annotated value and certified frame; they do not identify alternate environment layouts. The source type owner supplies complete-query uniqueness and the exact undefined-demand cast law. `AtomRel.undefined_operator_phases` exposes tuple/call/publication children for either presentation of undefined operator selection, under every declared local policy.

`RecursiveTransport.he_evaluation_exact` proves the universal reverse answer law. For any admitted finite source tree and closed annotated input, every completed HE phase has that source tree's entire ordered answer list. The five mutual proofs invert the actual tuple, alternative, invocation and body phases, pairing every complete physical frontier with the independent source scan. They retain arbitrary certified callers and all declared local policies; a recursive evaluator's desired answer is never a premise. `Evaluation.recursive_machine_observation_iff` composes this result with the maintained HE machine's completed-path correspondence and PeTTa's independent completed-observation theorem. A complete HE answer observation exists exactly when the PeTTa machine completes with that same ordered list and unchanged pure world. `nested_recursive_completion_exact` characterizes every completion of the four-answer control at every input depth, and `nested_recursive_cannot_drop_choice` excludes a two-answer completion.

The ordered equality above is a presentation law for one fixed equation traversal. The public completed HE observation is its multiset quotient, retaining duplicate occurrences. `recursive_machine_bag_iff` compares that bag with the bag view of PeTTa's completed ordered execution; it does not erase PeTTa's operational prefixes. The stronger storage law is separate. `Evaluation.perm_program` transports an independent finite source tree across any permutation of the stored equation copies, including nested alternatives and recursive bodies. Clause positions are recomputed, while each selected substitution/body pair is retained with its multiplicity. `recursive_machine_storage_permutation` composes this construction with both directions of machine adequacy. It thus preserves completed HE bags without assuming termination of the reordered computation. The control `[A,B,A]` versus `[B,A,A]` distinguishes list order from bag equality and rejects dropping the duplicate. These laws apply to the declared pure, annotation-free fragment. First-applicable type selection and effects retain their declared enumeration and causal order; a bag observer alone cannot justify reordering them.

The recursive proofs also compose the caller-binding observer. `ConservativeExtension`, in the existing binding-observation owner, retains all incoming constraint support and compares models on that support together with the protected continuation names. Its transitivity law keeps earlier support visible to later fresh instances, including tautological constraints. `ClauseInstance.caller_extension` earns the invariant from actual selected source work. The five inverse laws carry it through argument alternatives, recursive bodies, return and publication, without requiring equal

environment layouts or private representatives. `he_evaluation_caller_observations` projects the complete `BindingTheoryEquivAt` observer on the visible query, live continuation and incoming constraint scope. Every completed machine path satisfies the same law through `Evaluation.recursive_machine_caller_observations`; `recursive_machine_pair` constructs both runs with that observer. The all-depth `nested_recursive_caller_observations` control includes arbitrary certified callers. `visible_assignment_refused` proves that assigning a previously unconstrained protected name fails this invariant. These answer and binding-observation laws establish the declared recursive comparison fragment; they do not extend its admission to either full dialect.

The second obstruction remains even with no equations, bindings or effects:

$$\texttt{(Box Error A)} \quad \begin{cases} \longmapsto \texttt{(Error A)} & \text{HE completed tuple rules,} \\ \longmapsto \texttt{(Box Error A)} & \text{PeTTa completed source.} \end{cases}$$

HE's published `interpret\_tuple` rule tests the recursively assembled tail for the `Error` shape before prepending its head. It consequently propagates the successful tail `(Error A)` as a failure and discards `Box`. `he_propagates_assembled_error_tail` proves this derivation for every declared marking policy; `petta_retains_error_tail` proves the other observation independently. Installed Rust HE 0.2.10 and SWI-PeTTa reproduce the respective observations on the tiny discriminator. This is a source-protocol separation, not a claim about a backend storage defect.

A stronger shared fragment must therefore constrain such control-shaped data or translate the protocol. A distinct internal success packet carrying tuple elements would remove this collision; propagating only tagged failures would deliberately change the displayed HE observation. That is a local semantic proposal, not an established compatibility result or a change to the faithful source rules. The constructed recursive join uses this explicit admission; it does not establish equivalence for the separating program or for either unrestricted dialect.

Existing owners. `Mettapedia/Languages/MeTTa/Translation/HEPeTTaSound.lean`, `Mettapedia/Languages/MeTTa/HE/Spec/Eval.lean`, `Mettapedia/Languages/MeTTa/PeTTa/Eval.lean`; `Translation/he_petta_marker_head.metta`, `Translation/he_petta_recursive_duplicates.metta`.

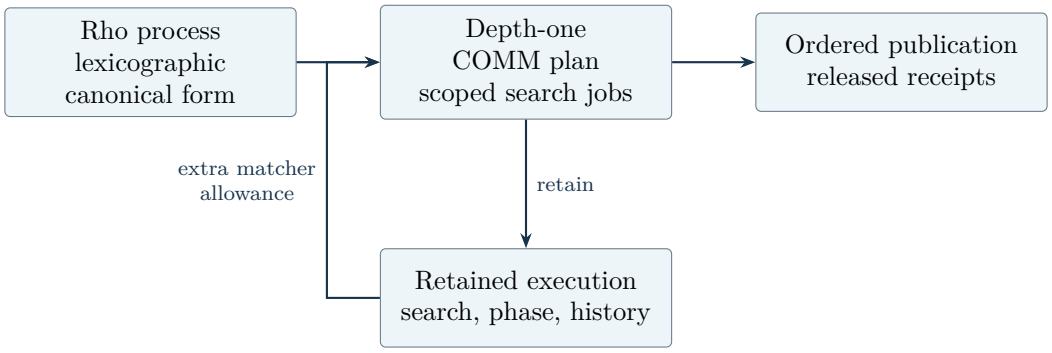
8 GSLT and rho: retain searches, publish occurrences

An authored language also reaches the common execution interface through the existing GSLT-to-MeTTa bridge. Under its explicit first-order, provider and premise admission conditions, `observes_contextual_iff_gslt_step` relates the actual PeTTa source judgment to the independent GSLT step. The corollary `common_machine_observes_contextual_iff_gslt_step` composes that result with PeTTa's common-machine adequacy: a completed run returns the encoded reduct frontier, and membership in that frontier is equivalent to a licensed source step. The evaluator budget and the recursive-premise depth remain separate parameters. The ordered frontier retains repeated witnesses; the source step's membership observation alone does not count them.

This connection does not identify every syntactic context with an evaluation context. Authored operational premises license reduction; equations use their independently admitted congruence contexts. The emitted runner and graph entrypoints additionally have exact PeTTa return theorems. Their current reachability-completeness clauses require no declared collection or scope-extrusion laws. The presented-semantics development supplies forward soundness with those laws, under `ExtrusionsFlatten`; the broader operational coherence and completeness result is separate. Visited-state deduplication in that explorer preserves reachability, not the number of derivation paths or answer occurrences.

Existing owners.
`Mettapedia/Languages/MeTTa/PeTTa/Bridges/MeTTaIL/ContextualRuleExecution.lean`,
`Mettapedia/Languages/MeTTa/PeTTa/Bridges/MeTTaIL/CompleteEmissionExecution.lean`,
`Mettapedia/OSLF/MeTTaIL/Complete/Presented.lean`.

The scoped GSLT scheduler advances actual premise, matcher and query frames. It retains pending tasks, proved results and unresolved obligations. PeTTa's authored program driver packages these searches into resumable jobs. Its aggregate and publication are deliberately distinct: a later job may already have results while an earlier incomplete job still controls their public order.



The actual `rho:run:resume-search` equation advances the pending `ProgramObservation` and returns the full updated `RhoSearch`. The evaluator-return and completion theorems cover that quoted equation, including unchanged ambient state and empty effect queues. `Search.resume_valid` retains earned validity, and publication grows by prefix.

A released receipt supplies a source-valid rho step modulo structural congruence. The existing `step_sound`, `step_quiescent` and saturation results distinguish a step, quiescence and insufficient search allowance. Empty publication establishes quiescence only when the search is complete. Stopping with no answer so far is not evidence that there is no answer.

The shared trajectory law is instantiated by scheduled states and actual PeTTa jobs/programs:

$$\mathsf{resume}(s, m+n) = \mathsf{resume}(\mathsf{resume}(s, m), n).$$

The units differ: scheduled fuel counts ticks; PeTTa adds allowance per job. Publication cursors emit only a newly released suffix and keep absolute occurrence ordinals, including equal-valued duplicate answers.

Existing owners. `Mettapedia/OSLF/MeTTaIL/StructuralDescentScheduledExecution.lean`, `Mettapedia/Languages/MeTTa/PeTTa/Bridges/MeTTaIL/GeneralProgramResumptionExecution.lean`, `Mettapedia/Languages/MeTTa/PeTTa/Bridges/MeTTaIL/RhoRunnerExecution.lean`.

Retained `Execution` adds successive completed COMM commitment and cancellation to the search state. A commit selects one completed publication, records its before/receipt/after triple, and begins a new search for the successor process. Its quoted `rho:run:commit-candidate` equation checks the entire supplied receipt wire against that local selection. A foreign candidate or an immediate stale candidate does not extend the history. Resume preserves the committed prefix; cancellation preserves history and drops live work. `Execution.advance_add` instantiates the shared resumable trajectory. Its fuel counts commit attempts, including no-ops; matcher allowance is resumed separately. A two-communication control keeps both commits in chronology.

Every execution reached by any sequence of retained operations is sound (`Retained.Reached.sound`): its history is a chronological sequence of committed depth-one COMMs from the canonical start (`Retained.Reached.chain`), its live search runs at that history's endpoint, and a done phase there is exact quiescence. A commit placed at the wrong state, or with a result that is not the receipt's, is not a chronicle. Under the common owned host (`Retained.owned_publication`) committed COMMs reach the world exactly once, copies of used tickets are refused, and the delivered history is that chronological COMM sequence.

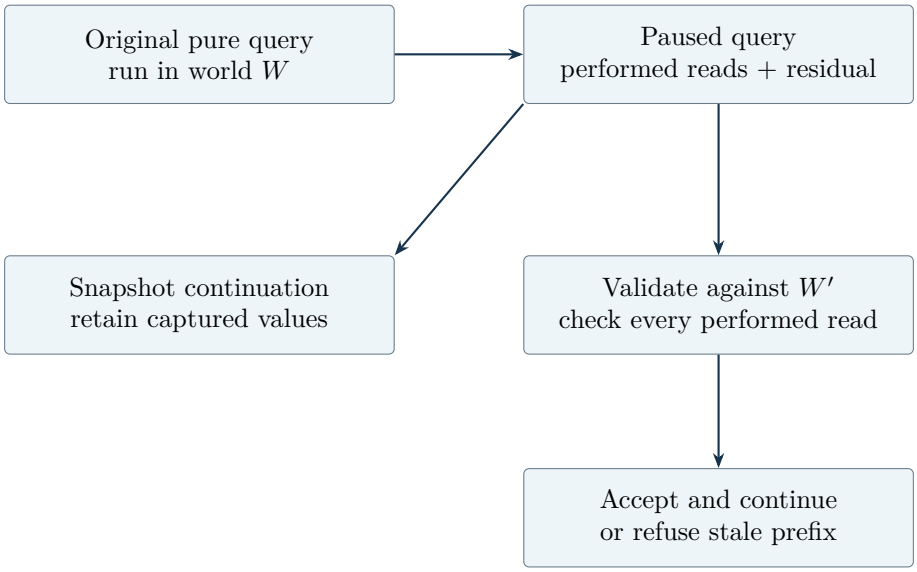
Selection need not always use the first published receipt. `commitPublished` validates membership of the complete supplied receipt, including its matcher, source job, bindings and premise history. Two ground communications in disjoint multiset components construct a retained diamond (`disjoint_retained_diamond`): each edge is an actual completed search and licensed commit. Both schedules reach the same process; their chronological histories remain distinct, with equality only under the declared causal observer.

The local dependency interface names input, output and fresh allocation occurrences. Distinct keys retain distinct resources even when their process values coincide. Fresh, live and spent states prevent reusing a consumed key. The actual read program produces the candidate's dependencies; disjoint read/write footprints prove that an unrelated commit preserves its validation. `replay_after_history_refused` excludes reusing a consumed occurrence through any later accepted history. Cancellation closes the owned publication channel, and stale reads publish no world change even with a current ticket. The quoted `rho:run:commit-published` and cell-store commands have source execution and store-refinement proofs, as well as matching native controls. An old whole-process search receipt is never relabelled for the successor: its search is rebuilt there, while the local dependency certificate can survive independent updates.

> **Boundary.** A commit's independent rho-step licence requires valid search and a well-typed process. The quoted bridge uses the pinned depth-one COMM plan and prepared encoded values. The owned host is a sequential model of authority; durable or distributed exclusive delivery is a host obligation, and fairness over all alternatives is not proved here.

9 World queries: what can safely survive a change?

A pure adaptive query can choose its next read from a previous result. Its observation therefore contains both its answer and the ordered reads it actually performed. A read certificate records complete values, including absence and complete ordered match lists.



Suppose the query reads all matches at a key and gets $[a, a]$. Replacing that list by $[a, a, b]$ changes the read even though both results are nonempty. Replacing it by $[a]$ loses an occurrence. Reordering it can change a first answer. An empty result must also be recorded: a later insertion invalidates the earlier absence. An update to an unread key can remain compatible.

`run_eq_of_agrees` proves that agreement on the complete performed certificate preserves the query's answer and read path. `Retained.resumeValidated` checks the performed prefix before executing any pending read. `Retained.validated_fresh` connects an accepted earned prefix to fresh execution of the original query in the new world.

Snapshot continuation is a different policy. A residual may already have closed over a captured answer, and retains it even when the current store changes. Failed validation does not retroactively turn that old residual into a fresh query. Starting the original pure query again is possible; replaying committed effects to refresh knowledge needs a separate effect contract.

Writes computed from reads follow the same rule. An owner runs a transaction, a pure query whose answer is a list of writes, on the snapshot it saw and proposes the writes with the complete read certificate. The ledger commits the proposal only when every recorded read agrees with the current store. An accepted proposal writes what its transaction would write now (`commit_fresh`), so accepted commitments are serializable (`commitAll_serial`); a proposal is refused exactly when one of its reads changed (`commit_refused_iff`). Two increments of one counter from the same snapshot commit once; rerunning the refused one on a fresh snapshot commits it; writing both blindly loses an update. The ledger is a retained state for the owned host, which publishes each accepted write list once (`ledger_published_once`).

Existing owners. `Mettapedia/Machines/ReadOnlyQuery.lean`, `Mettapedia/Machines/ReadCertifiedQueryReuse.lean`, `Mettapedia/Machines/ValidatedCommit.lean`, `Mettapedia/Coalgebra/CoherentPrefixTower.lean`.

> **Boundary.** This is a pure read interface. Query identity, program revisions, ownership and service authority are additional dependencies; an unchanged read list alone does not validate a changed program or authorize consumption.

Context identity is not a space snapshot

Two fresh mutable spaces can have equal current contents and still denote different places to write. `ModuleStore.equal_contents_forbid_content_codec` proves that no payload computed only from those equal contents can recover both identities. `annotation_write_lookup` and `annotation_write_other` give the positive distinction: a write changes the selected context's lookup and preserves another context's lookup. The maintained `he_context_identity_discriminator.metta` allocates two empty spaces, writes to one, and asserts that the other remains empty. Rust HE passes this control; it is an abstraction discriminator, not evidence of an engine defect.

The existing `Contextual.Control` is parameterized by its context-label carrier. Its work, results and continuation grammar are unchanged. `WorldConfiguration` carries one shared world, the root context identity, that control, the caller stack and prior publications. `snapshot` reads all currently retained operational context views from the reached world. It is a phase projection, not an encoding of mutable identity. Its footprint congruence law and `disjoint_context_write_snapshot` preserve a current read-only phase under writes outside the declared direct-key footprint; module dependency views need their own complete footprint.

`WorldLocalStep` reuses the old ordinary primitive transitions at the current context's view through `UnscopedControl`. Context entry stores labels, and its nested rule propagates the child's reached world back to the caller. The three context instructions use the context-label payload. `managedContextControl` prevents the old contents-valued `context-space` path from bypassing that boundary. A `call-native` rule installs a completed local host result and its reached world. Its contract takes one invocation and a logical activation, without the caller control or a recursive evaluator. It specifies reply installation; service execution, reentrancy and cost require their separate refinement laws. `WorldStep` adds the existing ordered root-publication discipline, and `worldSpan` instantiates the same relation-level machine interface.

The read-only instance uses the supplied snapshot service and disables native completion, as the old completed core does. Local transitions and whole paths lift and project in both directions. `world_readOnly_source_iff_observed` then preserves the entire existing independent completed source profile, including nested contexts, bindings and retained continuations. A nontrivial model native-write instance separately changes an actual context store under a saved child. `nested_native_write_reaches_caller` and `native_return_keeps_reached_world` keep its reached store, activation, inner and outer continuations and prior publications. No native plugin implementation is inferred from that model instance.

The independent `Completed.Stateful` judgments now thread the reached world through each completed source phase, including all recursive children, captured equation alternatives, function returns and context changes. `WorldEnvironment` supplies current context views, casting and local host contracts; it contains no evaluator. The existing entry, type selection, logical unification, annotated readout, marking and success-priority judgments are reused. Remaining alternatives are the frontier captured before their children execute; later queries use their actual reached world.

In the stateful execution namespace, `source_iff_finished` and `source_iff_observed` relate every retained configuration's independent source completion to its machine run, with exactly the same answer history and final world. The proof follows source trees forwards and reconstructs them backwards through the actual control zipper. A completed source suffix therefore reconstructs the earlier completion across any retained prefix. `readOnly_source_iff` identifies the read-only instance with the entire previous completed source meaning. These are ordered exploration judgments; no permutation law is asserted for noncommuting effects.

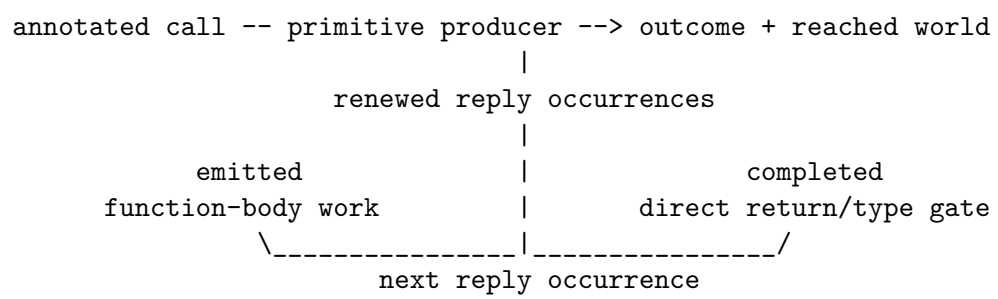
The model writer control performs two sequential writes. Its source proof constructs a complete machine run; a separate concrete cut pauses after the first write and resumes with only the second retained in the chain frame. The final store keeps both writes, the cut is not complete, and a fully published root has no successor that could reissue an operation. These laws concern the logical completed-operation model, not a native plugin or an executed host-service protocol.

Grounded producers retain their value representation

`GroundedResult` is parameterized by its value carrier, with bare atoms as its existing default. Its representation map preserves outcome classes, binding records and occurrence order. The stateful producer receives an annotated call and returns annotated values. It receives no caller stack or expected return type. `readOnlyGrounded` supplies a concrete fresh-syntax adapter from the old dispatch, and erasing its result recovers the exact old primitive outcome. This adapter does not reconstruct previously held marks. `native_identity_needs_annotations` proves why: a relation whose only input is erased call syntax cannot specify exact identity-returned readiness for both active and held arguments with the same syntax. Implementations may represent this information in the value or in separate evaluator metadata.

Native binding renewal packages the entire new constraint record as one syntactic equation between two tuples. The existing contextual unifier earns the reached annotated table and its readout certificate. `Frame.NativeRenewRel.of_logical_merge` constructs that table for every surviving finite logical merge. Raw producer values keep their annotations; materialized values use that reached frame. `NativeRowsRel` visits every physical reply copy, rejecting only finite-model-incompatible bindings. Its length theorem preserves every compatible duplicate occurrence. `NativeRowsRel.represent_before_iff` transports the complete returned batch across `Frame.represent`, with identical annotated outputs and the same rejected rows. Its control retains two equal successful copies around an inconsistent middle reply. This is postprocessing of an already returned primitive batch; it does not assume that the host callback itself is independent of its input representation.

The primitive outcome is installed once in its reached world. Each successful row then has its own saved native return boundary:



An emitted function completes its body before the full caller returns; raw invocation leaves its ordinary returned data raw. A producer-completed value uses a pure return/type gate and does not activate function-shaped data there. Refusal restores the saved full call or emits the raw refusal sentinel; runtime errors and empty producer batches retain their distinct outcomes. The extended `source_iff_finished` and `source_iff_observed` prove both directions through these rules. The same prefix and resumption laws continue to apply. Controls execute both raw and full provenance cases with one primitive-counter advance, preserve annotated completed values, forbid duplicate loss, and exclude activation at the completed raw return boundary.

9.1 Binding collections over logical snapshots

Binding collection saves a value together with its complete logical frame. `SnapshotServices` supplies two local relations: capture a frame in the reached world and restore it from a payload. Neither relation receives an evaluator or continuation. The separate laws require roundtrip lookup, functional restoration and preservation of older snapshots when a new one is captured. They prescribe no native handle, allocation or memory layout.

`CaptureRowsRel` traverses the completed minimal child batch, omitting only final `Empty` occurrences. It retains errors and duplicate copies, the value's annotations and the full frame. Each capture uses the world reached by the previous one. The retained task stores its remaining rows

and reversed encoded prefix. `RestoreRowsRel` visits each saved pair and conjoins its captured bindings with the current caller's constraints. `Frame.SnapshotMergeRel.of_logical_merge` constructs the actual annotated table and certified readout for every compatible finite merge. Incompatible bindings reject that occurrence; a missing snapshot lookup does not justify completed empty output. `RestoreRowsRel.represent_caller_iff` preserves the entire traversal when the caller's constraint record is represented differently while its annotated environment is retained. The relation `RestoreRepresentation` identifies these actual pending configurations and otherwise retains identical states. Its `step` law matches one actual world transition with one actual transition in either direction; `steps` retains related residuals at every finite pause. `completed_paths_iff` preserves the entire finished configuration, including the world and publication history. The relation explicitly refuses a caller change that preserves only the erased table while changing readiness. `restore_representation_square` constructs both corresponding retained machine paths to one identical configuration, under arbitrary caller continuations and prior publication history. The machine control retains a saved result, two equal restored occurrences and a nonempty caller stack; the reached cut is still unfinished.

The snapshot laws prove exact occurrence encoding at the reached world and an exact restore roundtrip into the empty caller. Restoration into an arbitrary caller is total on valid captured pairs; its length theorem keeps every compatible occurrence, including duplicates. The source theorem `MinimalRel.collection_roundtrip` composes the actual program

```
(chain (collapse-bind p) $saved (superpose-bind $saved))
```

with the complete child derivation. Its output is precisely the child's non-`Empty` rows, with their original frames and annotations.

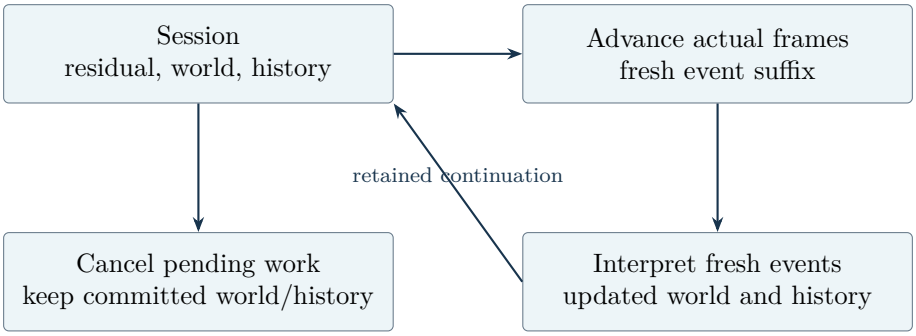
The retained AM has one collecting continuation and explicit per-row capture/restore work. The existing two-way completed-source correspondence extends through those transitions. A concrete append-only mathematical snapshot model satisfies all three service laws. In that model, `collection_run` executes an arbitrary row-producing child once; `collection_pause_after_child` and `collection_resume_after_child` exhibit the actual saved cut and its remaining execution. Snapshot capture preserves the child's reached program world. The paused collection is not complete, and a completed run cannot replay the child. These are logical execution results, not native snapshot encoding or authority proofs.

> **Boundary.** The stateful completion correspondence is checked for the declared typed/minimal profile including ordinary grounded production and binding collections. Binding renewal and snapshot merge use the profile's explicit finite-model and certified-readout assumptions. Remaining dialect coverage, the actual retained host-service protocol, codec/authority validity and native realization remain obligations. Equal snapshots do not identify future mutable worlds or confer permission.

Existing owners. `Mettapedia/Languages/MeTTa/HE/ModuleSpace.lean`, `Mettapedia/Languages/MeTTa/HE/Spec/Eval.lean`, `Mettapedia/Languages/MeTTa/HE/RecursiveCorrectness.lean`, `Mettapedia/Languages/MeTTa/HE/RecursiveTasks.lean`, `Mettapedia/Languages/MeTTa/HE/RecursiveMachine.lean`, `Mettapedia/Languages/MeTTa/HE/RecursiveExecution.lean`, `artifacts/conformance/he_io_cases/he_context_identity_discriminator.metta`.

10 Effects, services and the limit of a copied continuation

An effect session retains an actual driver residual, the current interpreted world, and its committed event history. Continuing advances the residual and interprets only the new event suffix. Splitting a budget preserves all three components, including when the handler performs a non-idempotent increment.



The independent recursive `Runs` relation supplies the meaning of children and caller continuations. Explicit frames realize that meaning with the exact ordered trace. `Session.resume_add` gives the complete sliced state law. `Session.cancel` discards pending execution and preserves the committed world/history; later resumes stutter.

Pairwise commuting actions permit a permutation theorem for the final world. The ordered trace still records a different schedule. A negative control shows a reversed noncommuting pair changes the world. Commutation must be proved for the actual actions and interference conditions.

The general event machine also has resumable service handling. A request can run another machine while retaining the caller and service continuations; faults, stuck endpoints and divergence remain distinct from replies. This is related to the handler-as-computation approach of interaction trees [5]; the present interface uses explicit states and finite budgets.

First-order configurations in the shared scheduler

`ProductiveCoreAdapters.EventCursor` connects an event machine to the existing indexed cursor protocol. Its retained client control is an event history paired with the machine's existing outcome; a provider inspection executes exactly the declared machine transition. The adapter introduces no host closure or additional program syntax. `advance_run` preserves the whole finite run and its declared transition account, and `advance_exhausted` identifies every live cut with its actual paused packet. An unbounded execution remains live at every scheduling allowance. Terminal client bookkeeping cannot answer a machine request or turn it into successful completion.

The PeTTa `Scoped.Scheduler` instance retains the actual allocator, choice delimiters, bindings, collection scopes, callers and reached world. `visit_source_iff` and `resumed_source_iff` compare the real cursor execution with independent source derivations in both directions. `resume_source_cut` continues the exact source residual and appends only its later publications. The source delimiter counterexample remains observable through this adapter. These laws use the saved world; changing that world, consuming publication authority or licensing external cancellation still requires the corresponding interaction contract. Transition accounts do not bound time spent inside a primitive.

The checked HE worker uses the same first-order adapter with its complete configuration and an explicit type-lookup policy. Its scheduling split retains the reached cursor and account. A separate depth grant replenishes only the saved evaluation or leaf operation, preserving prior publications and the selected lookup policy. It leaves primitive-service requests unchanged. The controls begin at a real zero-depth request: scoped lookup then keeps two polymorphic calls independent, whereas

literal lookup retains its type conflict. This worker/protocol connection is distinct from proving the remaining general scoped source correspondence and from granting external effect authority.

The downstream `RecursiveService.CheckedAdmission` instance checks currentness before advancing this read-only worker. It uses the existing ordered dependency store and indexed authority: the captured read stamp must validate in the current store, and the current right must admit the required frame-preserving update. Along the store's actual update history, validation preserves the complete ordered space query, including absence and duplicates. An invariant of actual checked transitions identifies that query with the saved worker's space. Equal counters in unrelated stores do not establish this fact.

`resume_reached_cut` and `resume_reached_depth` preserve the actual suffix, old publications and transition account under the same type policy. A denied entrance retains the complete paid cursor. Explicit depth grants are checked at this same boundary; an external primitive request remains unserved. `scoped_source_resume_completed` additionally joins the independently proved covered source family with closed type annotations. Its result is the exact source result history and terminal configuration. Schematic annotations retain their separate correspondence obligation. This is admission under a supplied current world and right, not authority consumption, external cancellation, atomic publication or a native concurrency theorem.

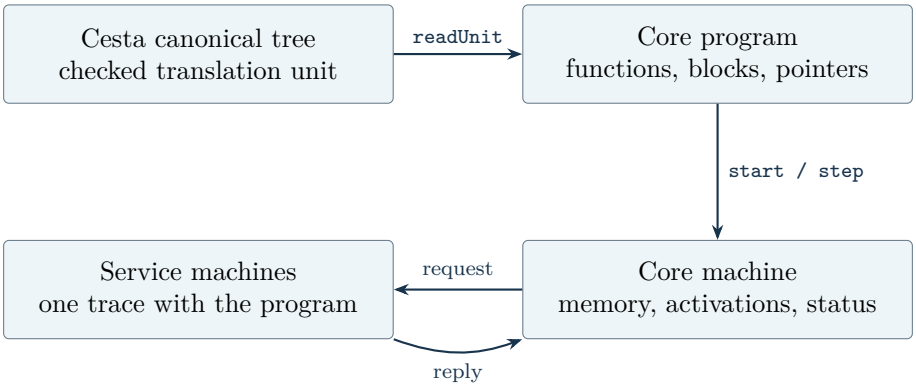
Owned interaction

The pure session and cursor laws do not enforce exclusive ownership: copying an earlier cursor can redeliver, and restarting a program can repeat an increment. The owned host adds consuming authority. A retained computation begins under a ticket; submitting its derived actions consumes the ticket, applies the actions once to the shared world and issues the successor ticket. A used ticket is refused (`Owned.used_ticket_refused`), so copies of a computation act at most once (`Owned.copies_act_once`); a cancelled computation refuses further work, private forks never touch the shared world, and every reached host state accounts its world as the initial world under the delivered actions. For commuting actions the final world does not depend on the interleaving (`interpret_commute_past`, `interpret_interleaving`); a noncommuting pair is a negative control. External crash recovery, durable exactly-once delivery and the disposition of non-droppable obligations need a service protocol beyond this sequential host.

Existing owners. `Mettapedia/Machines/OwnedPullDriver.lean`,
`Mettapedia/Machines/ExecutionHistoryReadout.lean`,
`Mettapedia/Machines/ResourceOwnership.lean`,
`Mettapedia/Machines/ProductiveCoreAdapters.lean`,
`Mettapedia/Languages/MeTTa/HE/FirstSuccessRecursiveExecution.lean`,
`Mettapedia/Languages/MeTTa/PeTTa/CettaDFSStreaming.lean`,
`Mettapedia/GSLT/Distinction/ProductiveBlocks.lean`.

11 The fixed C-language machine

The fixed core is a source language over the existing C11 semantic modules: typed integer operations under a selected integer model and signed-conversion policy, byte memory with effective types, automatic object scopes and array pointers. Names resolve through the lexical scopes of the current activation. Pure expressions read integer elements, compute, take addresses and move pointers within live arrays. Statements assign through places, set pointer variables, call functions recursively with parameter objects, return values, branch, loop, break and continue, open blocks that reserve declared objects and retire every owned lifetime on every exit, allocate and release dynamic objects, and suspend on external requests.



The executable step is sound and complete for an independent step relation whose premises are the memory, scope and expression judgments (`step_iff`), up to the two choices C11 leaves open: the identity of each new object and whether an allocation succeeds. Undefined behaviors stop with their reason: arithmetic overflow or division by zero, an access to a dead, indeterminate, out-of-bounds or misaligned element, a trap representation, a pointer formed outside its array, an invalid or repeated release, or a used missing return. A constraint violation and a failed host service are reported separately.

The common `FixedCore.machine` selects the canonical fresh-identity, successful-allocation policy. The choice-complete `FixedCore.Scheduled` instance instead retains a C configuration and a finite tape of permitted choices. Every finite independent source path constructs such a tape; `source_path_iff` proves preservation and reflection with the complete represented endpoint. One choice is consumed per C step, so repeated visits to the same C state may make different choices. Each step publishes its actual configuration pair. Exhausted prefixes retain exactly the unused tape and the corresponding source path; an empty tape requests replenishment without reporting C failure. A supplied invalid identity is distinct from source undefined behavior. External service requests retain their own request kind, and replies preserve the tape. `finished_source_iff` relates completed results to finite source paths, while `resume_with_supply` appends future choices without replaying committed steps.

The Cesta reader is a static checker for the core fragment. Names resolve through the C block scopes, operands are integers or pointers to a declared integer type, and calls are checked against the signatures of the program's functions and of the host's services. Interleaved declarations become nested blocks; `E1[E2]`, `&*E` and `&E1[E2]` follow their C11 equivalences; returned values and service arguments are converted to the declared types; the end of `int main` returns `0`. Every reading function is equivalent to an independent reading relation (`readUnit_iff`). Constructs outside the fragment are refused, not approximated. Sources parsed by the Cesta binary run on the core to the values GCC and Clang compute. On the undefined controls, AddressSanitizer and UndefinedBehaviorSanitizer report the use after scope, use after release, double release and overflow under both compilers; only Clang reports the pointer formed past its array.

Requests are served by service machines through the common callback protocol (`served_call`); running services as machines is a compilation stage from atomic answers (`servedStage`). With the retained HE evaluator as the service, a C loop calling a MeTTa function returns what GCC computes for the same loop over a C definition, and the trace contains exactly the evaluator's result occurrences (`metta_call_publishes_evaluator_meaning`). Several results, or an unreadable one, stop the program with the service's fault: a reply never selects one result from several.

The scoped service also has an independent source connection. `checked_service_source_completed` constructs completion from an admitted `AtomDerivation` and its local operation coverage; it does not assume an already completed evaluator run. The transfer profile has closed type annotations, while equation instances and caller bindings may remain open. Its events are exactly the source occurrences, and its reply is determined by the complete batch. Publication of the final occurrence alone is still an exhausted configuration, rather than a completed reply.

`scoped_metta_call_source` lifts this law through the actual C call and request codec. `scoped_metta_resume_source` keeps the waiting C caller attached to the saved HE configuration and executes only its remaining events before returning one reply. The saved prefix and the suffix partition the original source batch. A loop control returns nine through this scoped service; the two-answer control reports the service fault instead of selecting an alternative. These are formal machine and callback laws, not verification of a compiled CeTTa binary or of unrestricted schematic types.

The located layer remains the place for side effects inside expressions. Its expression scheduling makes full-expression boundaries and unsequenced conflicts explicit, and `StatementExecution.execute_iff` connects its executable transition to independent rules.

The common fixed/located fragment now has a constructed translation. An executable lowering agrees with an independent admission relation for arbitrary finite pure expressions and for assignment, conditionals, loops, break, continue and return. Pure expression schedules preserve the same scalar and semantic memory. An assignment additionally executes its actual conversion, store and returned-value occurrences. Both legal store/result orders reach the same boundary; neither may bypass an unfinished operand.

`statement_step_simulation` and `control_step_simulation` replace one fixed-core control step by its actual located expression and boundary steps. Their finite-history composition covers arbitrary admitted successful runs. `Block.finished_realization` then executes the final verdict on the existing budgeted runner and retains the complete receipt list and unused schedule suffix. The common memory relation preserves object allocation and lifetime headers as well as contents; reads outside their admitted live catalogue are refused. A counter loop reaches zero with the committed values $[2, 1, 0]$ under both store/result policies.

Different legal evaluation orders can encounter different first faults when both pure operands fail. The cross-layer observer therefore distinguishes a defined scalar result from failure without identifying those diagnostics. Each located execution retains its own exact fault and chronological prefix. The whole-run reverse construction now discovers the owning source control through every partial expression and assignment history. Its `ReachedControl.step` either constructs a genuine source step or strictly decreases finite pending-expression work. Thus arbitrary source loops remain possible while an infinite stretch of source-free administration is excluded. `terminal_observation_iff` proves preservation and reflection of complete success or refusal under the admitted scheduler. The schedule license checks the actual next ready occurrence; an unready operation is malformed scheduling, not source undefined behavior. `ReachedControl.run` additionally retains the earned source prefix and exact receipts at every budget exhaustion and selection request. No supplied decomposition into source blocks is required.

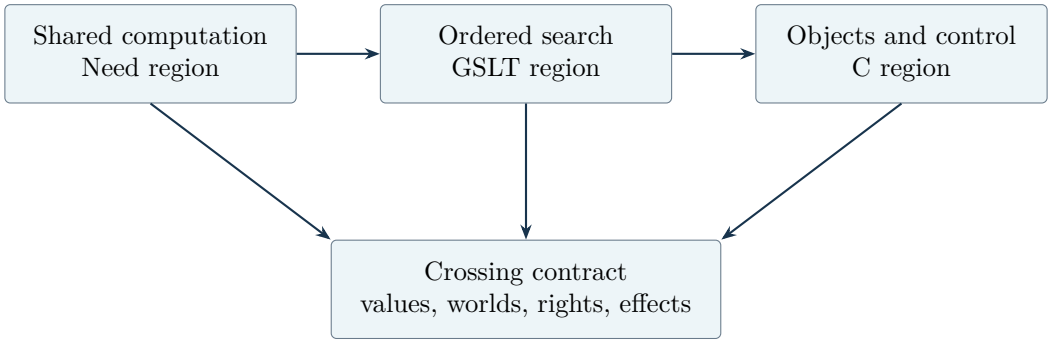
Existing owners. `Mettapedia/Languages/C11/FixedCore.lean`,
`Mettapedia/Languages/C11/FixedCoreStatements.lean`,
`Mettapedia/Languages/C11/CestaFixedCore.lean`,
`Mettapedia/Languages/C11/FixedCoreServices.lean`,

`Mettapedia/Languages/MeTTa/HE/RecursiveService.lean`,
`Mettapedia/Languages/C11/LocatedStatements.lean`,
`Mettapedia/Languages/C11/StatementExecution.lean`,
`Mettapedia/Languages/C11/ExpressionEventHistory.lean`,
`Mettapedia/Languages/C11/StatementMachineRefinement.lean`.

Boundary. Object layouts are the selected ABI's, an explicit parameter; pointer variables hold an abstract provenance-carrying value; external services answer each request at most once. Named extensions outside the core include calls inside expressions, pointer-typed objects in byte memory, structures, floating types, `for`, `do`, `switch` and `goto`, array initializers, function pointers and threads. The common-fragment translation does not additionally cover calls, allocation instructions or side-effecting operands. These remain available only at their separately specified machine boundaries.

12 Regional machines and explicit crossings

A useful machine comparison fixes three things: a region of admitted programs, the observations to preserve, and the resource objective. Removing an observation can permit more representations, but does not by itself establish a faster implementation. Operational accounts and measured native costs are different claims.



A crossing states what is translated, which observer is preserved, what dependency/authority evidence travels, and how return or refusal behaves. Pairwise agreement is insufficient for arbitrary networks: composed paths must satisfy the applicable compatibility laws. Existing gluing results cover equation models over a shared binding algebra, equivalence-valued view systems with trivial cycle transport, and compatible causal histories with explicit read/write and observation hazards. None of these identifies arbitrary guest logics or supplies their complete native execution. A general sheaf account of regional machines still needs its actual restriction and gluing maps. Trivial cycle transport concerns changes of view that claim to preserve the same object. An effectful computation cycle can change the world and emit events; it is not such an identity transport.

Observation, demand and retention are separate controls

An observer chooses what distinctions a user sees. Demand chooses how much work is requested. Retention chooses what remains available for inspection or continuation. `once` and `select k` bound requested successful occurrences; they are not general permission to erase faults, reorder effects or discard owned pending work. Zooming out can license a coarser readout; zooming in later is possible only if the required information was retained.

Machines do not impose one universal logic

Conversion within one shared logic and transport between distinct logical guests have different contracts. A selected machine family retains each kernel's own state and question type. It does not make HOL, dependent type theory and set theory one calculus. A stable guest can remain stable while Prime's candidate develops, provided the crossings state their assumptions.

ZINC [6] is useful historical comparison for an ML implementation. Here ZAM denotes a *zipper abstract machine*, which is a different idea and is discussed in Section 15. No general WAM optimality theorem is claimed. New regional machines can be derived or compared through these interfaces, then judged by correspondence and an explicit cost model.

> **Boundary.** The integration target is faithful higher-order MeTTa execution. CAM and the fixed C core supply substantial instances and reusable laws; they do not displace that language-facing purpose.

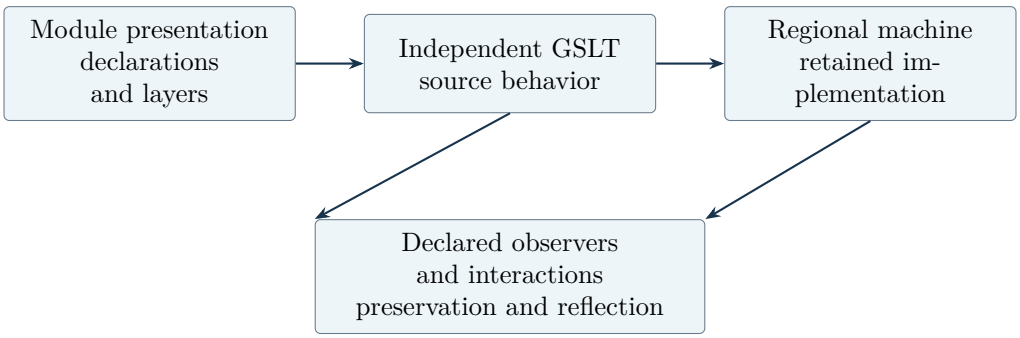
12.1 Modules, presentations and necessary machine state

A module assembles declarations and their selected policy layers. A GSLT states their operational meaning. A regional machine realizes that meaning with its own representation and control. These are distinct objects: joining declarations neither selects an evaluation strategy nor proves that two machines can interact safely.

Universality and the absence of a globally best optimizer do not determine how many machine families an implementation must expose. One interpreter can itself dispatch among several strategies. Regional specialization is justified by a stated program region, correspondence and resource objective, rather than inferred from undecidability alone.

A region can support a portfolio of implementations, including difficult higher-order regions; it need not have one fixed algorithm. Vampire provides an operational precedent for scheduling configurable proof strategies from workload features [26]. The existing bounded-search portfolio retains private batches, distinguishes cutoff from exhaustion and publishes one strategy's batch. Combining strategies' rediscovered answers would require a separate occurrence correspondence. Fair scheduling alone does not certify exhaustion after a finite budget.

Constraint-solving backends are also candidates for finite collection matching. A checked encoding must relate decoded solutions to the declared source matches, preserving their occurrence identities. Auxiliary Boolean assignments need not correspond one-to-one with those matches, so ordinary model counting does not automatically count semantic answers. SAT witnesses and checked unsatisfiability certificates can justify different conclusions; the encoding and coverage obligations remain [27]. Specialized AC representations can also be effective for important restricted rule classes [28]. The general complexity bound does not choose the best implementation for a workload.



An admitted execution region restricts a backend's qualification, not the module format. A first-order backend can refuse a presentation containing binders while retaining that presentation for a suitable scoped interpreter. A general interpretation path and its specialized realizations should share an independent source observation; the existence of that path does not by itself establish coverage of every binder or premise form.

The presentation and its surface syntax also have different responsibilities. The existing five-field `LanguageDef` carries sorts, constructors, equations and rewrites; `ExtendedLanguageDef` attaches independently authored dependent layers. A proof calculus is one such layer. Several concrete syntaxes can elaborate to this shared structure. Upstream syntax is an interchange route, not a requirement that every local extension fit a historical grammar. Improvements to notation must preserve binding, declaration origins and the laws of the selected layers.

Four current boundaries must not be conflated: the closed-theory `moduleTheory!` Lean quotation; the structured `module/1` envelope; textual `.module` languages; and the live runner's emitted MeTTa program. The quotation and envelope already have structural elaboration laws. The runner's `ReadAfterRender` theorem concerns MeTTa program text, not the upstream module grammar. Exact program-text rendering excludes the anonymous variable marker as a literal variable name,

since the reader allocates a fresh occurrence there; the underscore symbol remains ordinary data. A canonical portable surface now establishes

$$\mathsf{readBytes}(\mathsf{renderBytes}(P)) = \mathsf{some}(P),$$

with a separate elaboration theorem into the existing assembled presentation. The character reader uses the existing lexer/parser/atom-reader pipeline; the byte reader validates UTF-8. The complete finite `LanguageDef` codec retains binders, collection metadata, equations and scoped premises. Dependent layers carry an explicit indexed codec; proof calculi, export names and products are concrete instances. This does not claim that every Lean function or relation has a finite serialization.

The surface is an inert `(module:document payload)` declaration. Origin-bearing entries survive the text boundary before the existing import contract decides sharing or collision. Unknown envelope payloads remain data; a declared layer rejects an incompatible schema. The canonical syntax does not implement a historical upstream grammar. Preserving comments, formatting and import expressions still requires a separate source representation: an assembled presentation cannot reconstruct information it has discarded.

A finite rule list need not imply first-order syntax: its rules can have binders and higher-order premises. An effectively enumerable rule family additionally needs a finite generator description and evidence for each selected instance. An arbitrary semantic relation in Lean's abstract `GSLT` record need not be effectively enumerable or serializable. Keeping that semantic generality is compatible with requiring explicit presentation and realization evidence at an executable boundary.

Even a simple specialization must carry the typing boundary. A variable-root rule $x : \mathsf{Nat} \rightsquigarrow \mathsf{zero}$ structurally matches a Boolean term if its type is forgotten. Checked invocation-sort selection must exclude that candidate before execution, retaining the original indices of later rules. Comparing two untyped implementations would miss this error. The same contract must fix when query inputs become available. In the checked provider profile considered here, an output of the current query does not retroactively satisfy a required input to that query.

The question of necessary state is relative to the admitted observations and future interactions. Write

$$s \approx_{O,\mathcal{C}} t \quad \Longleftrightarrow \quad \forall C \in \mathcal{C}, \quad O(C[s]) = O(C[t]).$$

The observation may include outcomes, residuals, effects or probability laws. An encoding supporting every such interaction cannot identify states outside this equivalence. Turning it into a machine quotient additionally requires stability under the actual transitions and contexts. This identifies necessary distinctions, not a fastest implementation or a computable finite quotient.

The contextual observed-coalgebra owner constructs a quotient, its transition square and exact factorization for compatible natural consumers. That owner uses labelled set-valued futures: order, multiplicity and probability require their additional representations and laws. Its quotient is not already the complete MeTTa machine. The ordered-machine bridge also reads an ordered successor list as a GSLT membership relation; that modal view alone forgets list order and repeated successors. Occurrence simulations transport these ordinary modal formulas without a positional alignment. The converse fails for occurrence preservation: a one-successor machine and a machine with two equal successor occurrences satisfy the same formulas, while their occurrence bags have different cardinality. This limitation is proved in the native-type bridge.

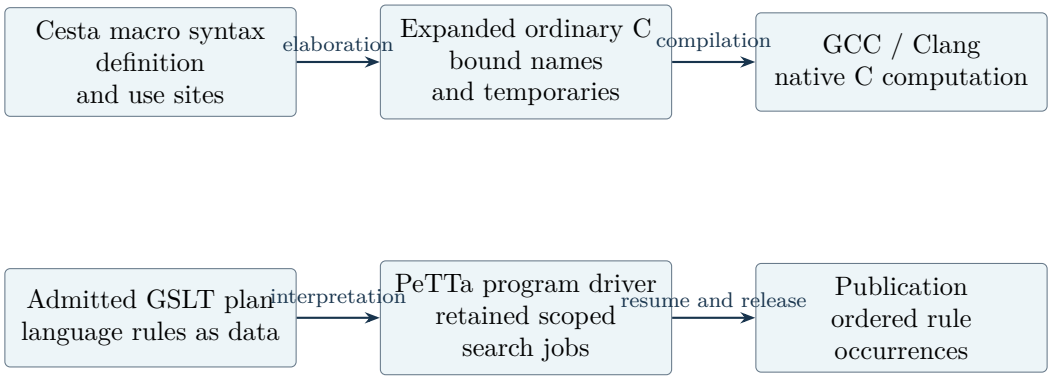
Admitted region	Useful machine structure	Distinction to preserve
Functional terms	CEK, Krivine, CAM or Need	Strategy, captured environment, shared demand
Rule search	Retained matcher and frontier	Occurrence identities, selected traversal, residual work
Reflective processes	Communication/reaction machine	Names, quotation, ports, permitted scheduling
Certificate checking	Decoder, proof state and snapshot	Declared theory, references, checked evidence
Probabilistic programs	Kernel-valued machine and history	Draw identity, scores, captured random values

Algebraic operational semantics connects syntax and behavior through a distributive law when the required rule-format and categorical hypotheses hold [30]. Polynomial syntax alone does not provide that law. Likewise, module-join compatibility is distinct from preservation of conversion or effects. The existing first-match priority counterexample fails substitution stability; ordinary relational rewriting has its own substitution theorem.

Existing owners. `Mettapedia/GSLT/LanguageDef/ModuleAlgebra/Semantics.lean`, `Mettapedia/GSLT/LanguageDef/ExtendedLanguageDef.lean`, `Mettapedia/GSLT/LanguageDef/ModuleAlgebra/UpstreamTheoryQuotation.lean`, `Mettapedia/GSLT/LanguageDef/ModuleFormat/Authoring.lean`, `Mettapedia/GSLT/LanguageDef/ModuleFormat/Text.lean`, `Mettapedia/GSLT/LanguageDef/ModuleFormat/Portable.lean`, `Mettapedia/Languages/MeTTa/PeTTa/Bridges/MeTTaIL/GeneralLiveRunnerEmission.lean`, `Mettapedia/Machines/BoundedSearchPortfolio.lean`, `Mettapedia/GSLT/Logic/ContextualObservedCoalgebraQuotient.lean`, `Mettapedia/OSLF/Framework/OrderedSystemNativeTypes.lean`, `Mettapedia/GSLT/Dedukti/LeanMinusPriority.lean`.

12.2 Staged lowering and runtime interaction

There are two different ways for one language machine to use another. *Staged lowering* transforms the source before execution. *Runtime interaction* keeps both computations alive across requests and returns. Both need semantic preservation, but they retain different states. Operational semantics based on explicit language boundaries is established work [7]; hygienic macro binding also has its own semantic models [8].



Cesta's current macro engine instantiates templates, resolves explicit binding origins, introduces fresh locals, and emits C. Argument metadata distinguishes syntax, values and modifiable places, with evaluation contracts. For the checked narrowing library, source and target types are syntax; the input value and output place are evaluated once by the emitted C. Macro expansion does not execute the input expression. The native compiler supplies active-header name inventory and checks the final C. This is a staged language layer; it is not an extra interpreter running beside the resulting native program.

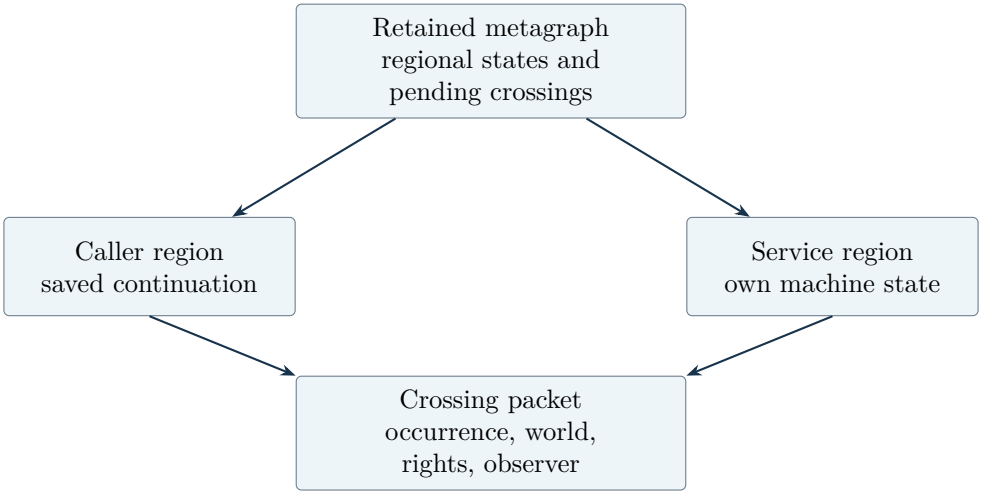
The current GSLT driver instead retains a licensed request and actual search state for every job. Requests carry the presentation/environment, permissions, binding context, expected type, source and physical rule/site identity. Receipts retain matcher choices, captured and completed assignments, premise history, and local and reconstructed targets. The quoted authored equations are connected to PeTTa evaluation. The general target-step theorem requires an equation-free, checked-stratified plan and sufficient search allowance. Its exact canonical reduct-list theorem has a narrower root query-profile scope. A presentation emitter and a general whole-run loop still need their complete integrated proofs.

Treating an expander as an abstract machine can clarify its own binding and phase semantics. It does not require paying for that machine at program runtime: justified lowering can remove it. A live heterogeneous computation needs runtime crossing rules in addition to staged source transformations.

Existing owners. `Mettapedia/GSLT/LanguageDef/MacroTemplateNames.lean`, `Mettapedia/GSLT/LanguageDef/MacroDefinitionPorts.lean`, `Mettapedia/GSLT/LanguageDef/MacroCheckedNarrowing.lean`, `Mettapedia/Languages/MeTTa/PeTTa/Bridges/MeTTaIL/GeneralProgramDriverExecution.lean`.

12.3 A computational-metagraph machine as a candidate

A candidate Meta-MeTTa machine would retain regional computations at nodes and pending crossings at edges. Admission, transfer, advancement, reply, refusal, cancellation and dependency validation would be explicit steps. This heterogeneous machine remains an integration target.



The state needs continuations, ordered occurrences, dependency evidence, consuming authority and committed history. Each crossing must realize an independent semantics preserving the selected observations.

`MachineCore.familyCore` retains heterogeneous state/question types but cannot change the selection. `Machine.handle` constructs interaction with retained caller and service states. `ScheduledFrontier` preserves occurrence slots across differing completion times, under independence hypotheses. Publication, read-validation and effect-session laws still need the common ownership join.

`GSLT.Poly.BindingSignature` represents binding terms as a context-indexed free monad. The indexed polynomial client in `GSLT.Poly.SortedTraversal` retains argument occurrences and reply slots. Neither polynomial structure nor a slot determines a child's interpretation or force rights.

In the MM2 evidence-material theory, two actual matcher paths conclude the same fact but deliver different origin lists. Their consuming continuation cannot factor through conclusion support. An ordered material batch or receipt retains enough information; the recovery theorem uses a classical partial inverse, rather than an executable native decoder.

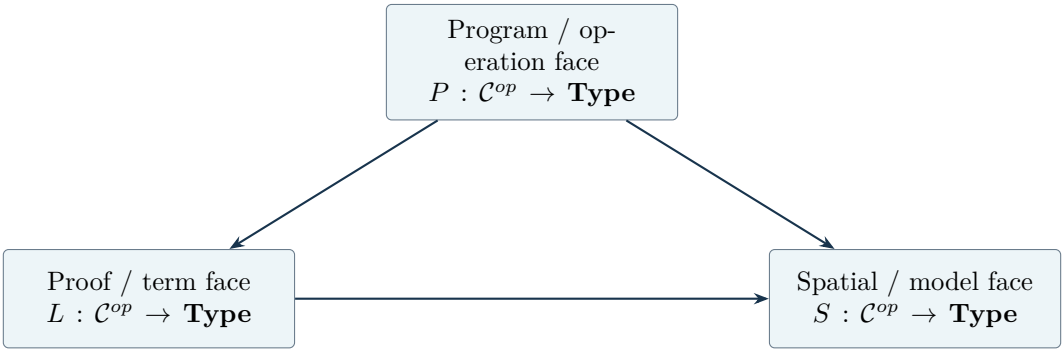
Native MM2 controls separately run a finite staged consumer on three input occurrences, two with equal payloads. Shared choice gives six deliveries; separate choice coordinates give twelve. Sequence fields recover their order. Retained cursors preserve the answers and three counter updates across reloads. Restarting repeats an update; changing the captured environment changes later answers. `head 1` alone still permits sibling writes. This does not establish a general MeTTa correspondence or external-effects protocol.

Producer identity supports sharing; delivery identity supports occurrence observation. Indexed set storage can retain both views. The richer encoding then carries the surrounding machine's cursor, world and commitment protocol.

Existing owners. `Mettapedia/Machines/MachineSubstrate.lean`, `Mettapedia/GSLT/Distinction/ProductiveBlocks.lean`, `Mettapedia/GSLT/Distinction/OrderedSimulationSpans.lean`, `Mettapedia/GSLT/Distinction/CausalGluing.lean`, `Mettapedia/GSLT/Scope/Holonomy.lean`, `Mettapedia/GSLT/LanguageDef/ModuleAlgebra/SemanticGluing.lean`, `Mettapedia/GSLT/Poly/SortedTraversal.lean`, `Mettapedia/GSLT/Logic/ProgrammableSpaceMM2EvidenceMaterial.lean`.

12.4 The trinity as a fibred web of presentations

The language-facing trinity names intensional mathematics, extensional mathematics and a GSLT-IL operation presentation. These are semantic roles, not necessarily separate kernels. The contextual computational-trinity framework compares programs, proofs or typed terms, and generalized elements in spaces. Each face is a presheaf over contexts, so substitution is part of the presentation rather than an untracked surrounding operation.



Comparison means a commuting triangle of natural maps. Exactness additionally requires invertible maps. The abstract construction does not instantiate the whole native DTT/HOL/set/GSLT triangle.

Actual categorical owners already have different variances. Theory models reindex by reducts along theory arrows, forming a Grothendieck fibration. Typed open code has forward transport and cocartesian lifts. Finite retained execution paths form another category-valued index. These constructions do not make every theory arrow an executable, resource-safe or conservative machine crossing. The complete step, residual and observer comparison must be added for the actual route. The additive world-model doctrine likewise cannot silently justify deleting a record or normalizing a Bayesian update.

The STT/DTT and selected set-family bridges are scoped interpretations. Representing a HOL formula does not prove it; a native inhabitant is not automatically a source HOL derivation. Logical borrowing needs the selected conservativity contract. Adding a stronger conversion bubble also needs its own substitution and join laws. Two component normalizers do not necessarily decide their combined conversion.

Hyperset presentations retain graph occurrences; material equality quotients by bisimulation. A dependent consumer therefore needs family and section descent before those distinctions are erased. Internal coalgebraic membership and the host's well-founded membership remain distinct. Initial/final comparisons also keep their bounds and existence hypotheses: identifying an initial algebra with a final well-founded coalgebra does not prove that every set functor has either [29]. Unrestricted powerset has no set-sized fixed point. Qualified bounded constructions state their size conditions separately.

Existing owners. `Mettapedia/Computability/ComputationalTrinity.lean`, `Mettapedia/GSLT/LanguageDef/TheoryGraphModels.lean`, `Mettapedia/Languages/MeTTa/PrimeCandidates/ContextualDataFibration.lean`, `Mettapedia/GSLT/Core/OperationalPathFibration.lean`, `Mettapedia/TypeTheory/MaterialSets/Hypersets/FamilyDescent.lean`, `Mettapedia/Logic/WorldModel/Indexed/Doctrine.lean`.

13 Proof machines: scoped, licensed composition

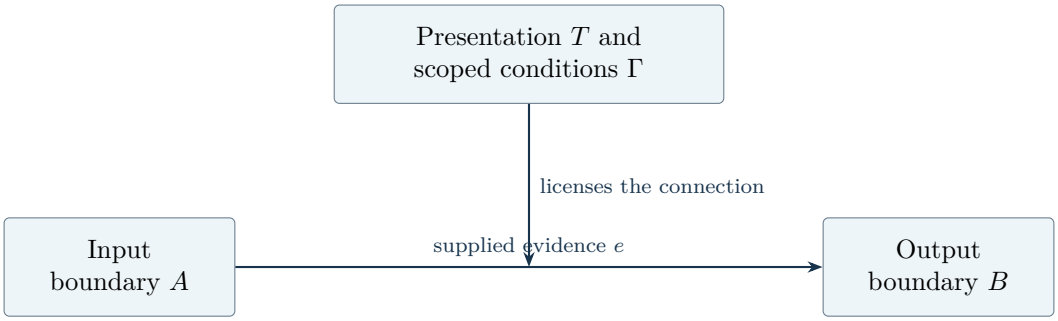
What follows from what, under which conditions?

A checking machine makes a proposed answer to this question inspectable. The local rules specify the permitted connections; evidence identifies the connection being proposed; composition connects its output to the next input. The interface shared by proof checking and rewrite checking is therefore *scoped, licensed composition*. This identifies a common organization, while leaving each presentation responsible for its own judgments, equality, binding and resource rules.

Write

$$T; \Gamma \vdash e : A \longrightarrow B$$

for a supplied piece of evidence e connecting boundaries A and B under presentation T and scoped conditions Γ. The notation is schematic: the boundaries may be premise contexts and conclusions, rewrite states, or typed interfaces carrying a world and resource obligations. An independent judgment determines which such connections are licensed. The machine realizes its checking discipline.



A boundary states what must fit. The selected theory determines whether evidence is reusable, whether an input must be consumed, and which conversions are admissible.

This perspective is continuous with the categorical treatment of proofs and programs as composable processes [15], and with rewriting logic's treatment of computation by proof terms [16]. Variable binding requires its own substitution algebra, including weakening, capture avoidance and metasubstitution [17]. These ingredients explain how a small checker can serve many presentations without choosing one object logic as their common meaning.

There are distinct categorical levels here. A rule with several premises is naturally a generator of a multicategory; substitution composes proofs into its premise slots. A sequence of unary rewrites instead gives paths in a free category. Exchange, contraction and weakening require the structural laws of the selected calculus. The category of checking machines describes realizations of these proof structures, with operational observations and costs. It is not the proof category itself. This agrees with the nLab's accounts of deductive systems, multicategories and syntactic categories [32]. Institutions additionally organize signatures, models and satisfaction under change of notation; they do not by themselves provide an executable certificate checker.

Storage navigation is a further realization level. A cursor, stack, stream, tree focus or dependency schedule may organize the same proof evidence. None is a primitive of every proof calculus. A focused position can be an observation of a machine—an account of what it is currently attending to—but a comparison must state whether it preserves that focus or hides it. Forgetting navigation need not forget the proof's premise dependencies. The dependency direction is from source rules to checking realizations and then to storage adapters, rather than from a particular backend to logic. For the authored-calculus classifying category, closed source derivations are equivalent to arrows from the empty premise context to a singleton goal; the two executable maps preserve the actual proof term. VibeITP and Metamath connect their source judgments at this level, before choosing graph replay or retained source-address checking.

13.1 Local evidence, exact checking and coverage

For one admitted effective presentation, let $\mathsf{Cert}_T(\Gamma; A, B)$ be the supplied certificate data and let $\mathsf{Valid}_T(\Gamma; A, e, B)$ be the independent validity judgment. The local checker earns

$$\mathsf{check}_T(\Gamma, A, e, B) = \mathsf{true} \quad \Longleftrightarrow \quad \mathsf{Valid}_T(\Gamma; A, e, B).$$

The separate coverage theorem relates this judgment to the intended source relation:

$$A \ \mathsf{Step}_{T,\Gamma} \ B \quad \Longleftrightarrow \quad \exists e.\ \mathsf{Valid}_T(\Gamma; A, e, B).$$

The first theorem qualifies the submitted evidence. The second establishes which source steps have a representation in that evidence language. Neither asserts that a search procedure will find the evidence within a given budget. Existential coverage alone also supplies no polynomial-time certificate translator.

For an ordinary calculus with implication elimination, a local rule is

$$\frac{T; \Gamma \vdash p : A \qquad T; \Gamma \vdash q : A \Rightarrow B}{T; \Gamma \vdash \mathsf{mp}(p, q) : B}.$$

Its checker resolves the two premise citations, checks the rule instance and compares the claimed conclusion. A resource-sensitive rule instead states how its inputs are split or shared. Local computation can instantiate a rule; evidence must still establish its side conditions. A negative premise such as "no equation applies" requires a checked exhaustion argument or a justified index certificate.

Declaration identity belongs to this boundary. Content addressing can locate a rule, while exact comparison or stated digest assumptions establish which rule was found. A repeated declaration can have several occurrences despite identical content. Certificate references must retain the occurrences required by the observer. The host additionally identifies the theory it intended to authorize: a valid proof under a packet's supplied theory does not identify that theory with an externally requested one.

13.2 Proof terms, slots and retained ambiguity

A proof term records enough information to reconstruct the licensed use. For an authored rewrite this can be a rule occurrence, a typed site, matching choices, premise references and equation evidence. Deterministic fields can be computed during replay. The current `GeneralRewriteRecipeExpansion` distinguishes a compact invocation from its expanded firing receipt and proves sound expansion and reconstruction of valid expanded firings.

Consider a two-input rule $r : (A, A) \to B$ and evidence p, q of A. The terms $r(p, q)$ and $r(q, p)$ record different slot assignments. Their conclusions may coincide; identifying the derivations requires an explicit equation or observer. Unique scoped names for slots, together with a substitution, can encode the assignment. They do not remove choices caused by collection matching, equations or alternative premise derivations.

Ambiguity may be useful compression. A packed proof structure can retain a family of alternatives, just as a packed parse forest retains multiple parses. Its boundary must say whether it certifies one selected derivation, all represented derivations, or a complete collection of derivation occurrences. Finding one valid branch licenses an existential claim. Certifying every represented branch requires their local validity; claiming that the family contains every source answer additionally requires coverage and multiplicity laws. Backend traversal order grants none of those permissions.

The following quantities are distinct:

stored nodes, reference uses, derivations distinguished by the selected observer.

A shared node can have many uses. Two positional replay choices can yield the same firing observation, while two identical declaration copies can generate distinct answer occurrences. The quotient must name what it forgets and prove that the remaining consumer descends through it.

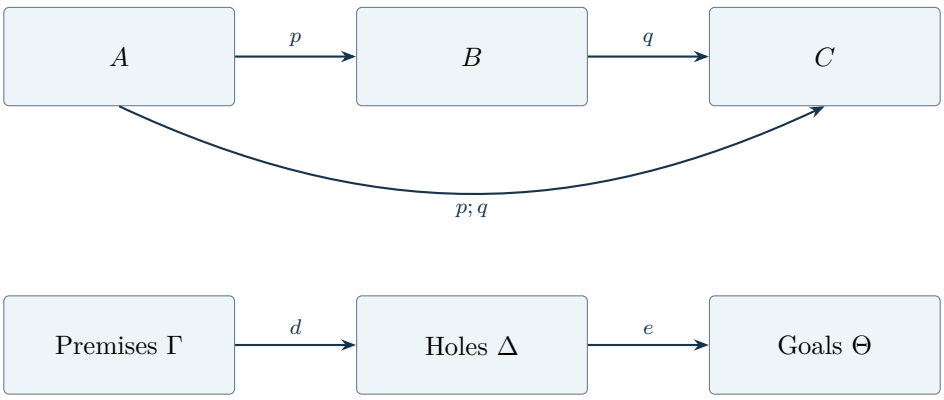
Sharing can change certificate size exponentially. A fixed example in `ProofSharingBounds` has an axiom q_0 and the rules $q_n, q_n \vdash q_{n+1}$. Every accepted proof tree of q_n contains exactly $2^{n+1} - 1$ nodes. A chronological DAG instead stores $n + 1$ nodes and two references at each nonbase node. Its checked input size is at most $4(n+1)^2$ even with unary rule and reference indices. The resulting theorem excludes every polynomial-size translation from this calculus's DAGs to its trees. This is a lower bound for the target representation, rather than a claim about the behavior of one unfolding algorithm. It does not separate Frege from extended Frege.

Textual parsing is another boundary. A structural certificate reader is not the lexer and grammar of its source language. Source ambiguity must survive in the packed parse result until an explicit policy selects, ranks or rejects it. Binding lowering then preserves the selected scopes. The portable presentation roundtrip supplies a proved textual surface for its supported document schema; it does not identify every upstream module grammar with that surface.

13.3 The algebra of checked composition

Five operations expose the common algebra:

Operation	Evidence being composed	Required law
Sequence	$A \xrightarrow{p} B \xrightarrow{q} C$	Exact joining boundary.
Hole filling	Premise outputs into an open derivation	Substitution respects scope and the selected structural rules.
Context	A local rewrite inside $C[-]$	The context is licensed and replacement preserves its typing and binding interface.
Sharing	Several references to stored evidence	Every reference use remains justified and counted where required.
Parallel	Two orders of independent actions	Checked residual edges close the selected square.



Temporal composition and premise substitution are distinct constructions. A particular checker realization relates them by adequacy.

The common executable law is

$$\mathsf{replay}(s, p +\!\!+ q) = \mathsf{replay}(s, p) \gg\!= (\lambda t.\, \mathsf{replay}(t, q)).$$

It holds for success and refusal in `DerivationGraph.replayFrom_append_eq`; the authored rewrite instance is `GeneralRewriteRecipeComposition.Trace.replay_append`. Its categorical readout is

composition of *partial state transformations*. A bare Boolean verdict forgets the intermediate state needed for composition. On successful paths, identity and concatenation give a category of checked executions.

Open derivations have another category: `CertificateGSLTClassifyingCategory` takes premise contexts as objects and vectors of open derivations as morphisms. Composition substitutes those vectors into premise holes. Its existing Cartesian structure licenses reuse and discard. A multi-input or linear presentation needs its corresponding multicategorical or monoidal structure; these structural permissions are not consequences of sequencing alone.

The authored checker already realizes the other three operations. `GeneralRewriteRecipeContext` transports local receipts through admitted typed contexts, including binders. `GeneralRewriteRecipeDAG` admits local recipes and reuses the existing shared-certificate checker for references and joins. `GeneralRewriteRecipeParallel` checks four original residual firings, independent addresses, stable site labels and literal endpoint closure; the accepted square constructs an existing Mazurkiewicz swap. It does not infer the missing residuals from two disjoint sites. Changed provider worlds or observable effects require further independence evidence.

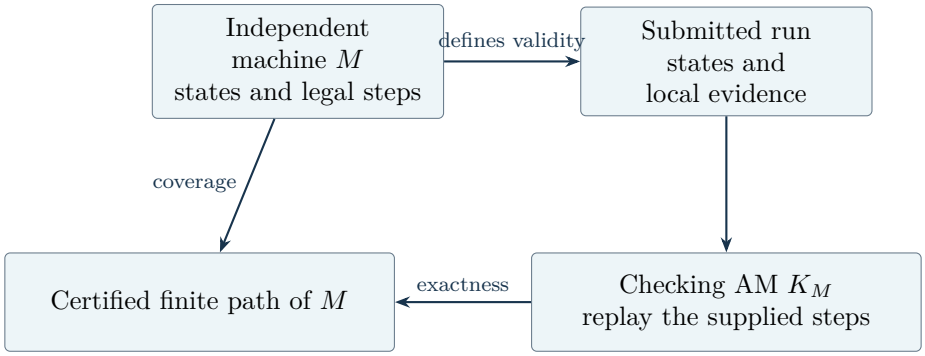
13.4 Checking whether another machine was stepped correctly

Let M have states S and an independently stated transition relation $s \longrightarrow_M t$. Suppose local certificates c have earned the exactness and coverage contracts above. A submitted finite execution is

$$s_0 \xrightarrow{c_1} s_1 \xrightarrow{c_2} \cdots \xrightarrow{c_n} s_n.$$

The checking machine threads the actual state s_i and validates c_{i+1} against the proposed s_{i+1}. Induction on the submitted sequence proves that acceptance is equivalent to validity of all its local certificates. Coverage then represents every finite M-execution. The induction uses the actual intermediate state, so scope, heap, continuation and world components cannot be reset between steps.

A checked instruction log is evidence of machine execution; a source evaluation derivation supplies evidence for the source judgment. Adequacy relates their claims. One possible result, completed evaluation and a complete result multiset are different judgments: the latter two require the corresponding terminal or exhaustive-branching evidence, including duplicate occurrences and pending work.



The checked machine and its checking machine are separate roles. A machine can play either role once the relevant evidence interface is constructed. An empty path alone proves neither program formation nor loading validity.

This applies to proof-kernel instructions as readily as to CAM instructions, rho communications or ISA transitions, when their local certificate contracts are available. The authored GSLT implementation supplies one concrete route: `GeneralRewriteRecipe.check_exact` checks its independent receipt judgment, and `exists_checked_iff_step` covers the original equation-saturated step relation. Composed recipes reach the existing finite may-reachability native types. That modal readout establishes the witnessed finite path; exhaustive search, termination and physical execution cost have their own obligations.

Checking a run of K_M is another instance of the same pattern. Such reflection requires an encoding of K_M and adequacy of its transition certificates. Encoding its states as numbers or graphs preserves only the distinctions proved for that encoding. It does not turn a certificate about a checker's run into unconditional soundness of every theory that checker might load.

13.5 Reconstruction, elaboration and reproducible hints

A partial reconstructor r can process output from an external prover, tactic or repair procedure. With destination checker c, publication is

$$\mathsf{reconstruct}_{c,r}(A, o) = r(A, o) \gg\!= (\lambda p.\ \mathsf{if}\ c(A, p)\ \mathsf{then}\ \mathsf{some}(p)\ \mathsf{else}\ \mathsf{none}).$$

It returns p exactly when r returned p and c accepted it. No soundness of the producer is assumed. Producer failure and checker refusal both publish nothing; the submitted-output coverage

theorem states exactly which accepted certificates the producer can return. Composing two such stages retains both checking boundaries. This is a general reconstruction interface, not a completed translation of an arbitrary ATP proof format.

An elaboration has another obligation: an accepted source certificate must produce accepted destination evidence. Existing local elaborations compose and lift to articles. Polynomial output size, actual translation time and logical soundness are separate properties. An output-size bound yields a size simulation; it does not manufacture an executed polynomial-time translator. A model interpretation likewise supplies no resource bound.

For reproducible checking, fix a deterministic strategy $f : S \to S$ and an explicit observation $v : S \to V$ with decidable equality on V. A checkpoint (k_i, z_i) requests k_i strategy ticks followed by the comparison $v(f^{k_i}(s_i)) = z_i$. The checker continues from its *computed* internal state, never from a replacement state supplied by the certificate. On success,

$$s_{\text{final}} = f^{\sum_i k_i}(s_0), \qquad \mathsf{tickCalls} = \sum_i k_i.$$

This remains true when v forgets private theory, ownership or environment data. A wrong checkpoint refuses at that boundary, retaining the work count of the attempted gap. Checkpoint sequences compose from the actual residual.

Fuel alone records the budget; checkpoints additionally cross-check selected residual observations; a full trace records one observation per tick. The formal materializer produces exactly n trace entries from budget n. All three replay the specified computation: checkpoints alone do not make the intervening computation cheaper. Tick calls, observation comparisons, stored observation sizes and native instructions require distinct accounts. Implementation heartbeats are not interchangeable with these strategy ticks.

The Lean4Lean instance uses its existing finite-catalog beta/delta/head algorithm. Its checkpoints include the expression and a terminal flag. Acceptance proves a path in the independent head-event relation and an endpoint with no remaining head event. Completeness is relative to this strategy reaching a common head-normal form. The raw self-application term Ω remains pending at every budget, yet every submitted finite normalization certificate is refused. This instance neither requires nor asserts termination of arbitrary Lean conversion, and it is not the complete Lean type-checking kernel.

The algorithmic and ideal judgments must also be distinguished. Lean4Lean implements a particular checking algorithm and separately studies the declarative type theory; verification of its implementation is partial work [20]. An exact contract for an algorithmic judgment does not decide every ideal equality. If a family of budgets enumerates exactly the same algorithmic successes, an ideal equality absent from that set remains absent at every budget. This is a theorem about the acceptance boundary, independent of implementation heartbeats.

Within a fixed formed-expression fibre, checked comparisons may instead be leaves of explicit reflexivity, symmetry and transitivity recipes. The checker computes their endpoints and refuses a mismatched intermediate join. It has exact authority for the equivalence closure $\mathrm{EqvGen}(R_{\text{alg}})$ of the independent leaf relation. Soundness into ideal equality follows from a sound leaf interpretation. Completeness for ideal equality additionally requires the generation theorem $R_{\text{ideal}} \subseteq \mathrm{EqvGen}(R_{\text{alg}})$; it is not supplied by fuel or by categorical composition alone. On raw syntax, ideal typed equality need not be reflexive: formation remains an explicit obligation before entering this fibre. The finite nontransitive control in the library illustrates this distinction without claiming to be a counterexample from Lean's full conversion implementation.

Lean4Less gives a concrete reason to separate these routes: it replaces selected implicit proof-irrelevance and K-like conversions with explicit equality proofs and transports, then checks the output with those conversion features disabled [21]. This is an elaboration between equality profiles, with separate declaration, universe and resource obligations. Destination rechecking can validate a

proposed output even when the producer is partial; neither a total translator nor a polynomial size bound follows from that check. The present boundary theorems do not complete either external system's metatheory.

The practical propositional connection reuses Lean's verified LRAT checker for RUP, RAT and deletion actions [27]. Its adapter preserves native clause satisfaction pointwise, including repeated literals and clauses. Accepted refutations imply an empty-clause derivation in the independently complete resolution calculus. That implication supplies no polynomial LRAT-to-resolution compiler. The public checker may stop at a successful empty-clause action, so its verdict does not validate an unused action suffix.

Finite intuitionistic and classical Hilbert certificates are checked directly by their formula instances and ordered premises. The existing standard Hilbert calculi in Foundation supply the independent source judgments; soundness and tree/DAG coverage are proved against them. The classical profile admits excluded middle, while the intuitionistic profile refuses that rule. A computable elaborator now returns terms in the existing `PropDeriv` family from accepted certificates, using the supplied children and a computed finite lookup rather than choosing an existential proof. The retained-term interface publishes a term exactly when the native checker accepts the requested formula.

Intuitionistic non-theorems have a complementary finite certificate: a partial-order table, a persistent atomic valuation, and a refuting world. The checker validates those tables and computes the Kripke semantics of the formula. Soundness follows through the independent source semantics; Foundation's finite-model completeness supplies a finite data certificate for every non-theorem. A two-world control refutes excluded middle, while malformed orders, nonpersistent valuations and out-of-range worlds refuse.

Proof and countermodel certificates are countably encoded as finite data. Enumerating those codes until one validates gives an executable terminating decision procedure for the original intuitionistic theorem judgment. Termination uses the two coverage theorems, not an ideal decision oracle. Submitted proofs or countermodels can be checked first as hints; incorrect hints are skipped and the same total search remains available. The least successful index is bounded by the code of any supplied valid certificate. This exhaustive construction proves decidability. Its search order has no practical complexity claim and is separate from the native replay kernel.

The classical extension checker adds literal hypotheses $p_i \leftrightarrow A_i$ with fresh names. It refuses a name appearing in the final claim, any earlier declaration, or its own right-hand side. Definitions are hypotheses, never additional substitution-closed axiom schemes. A valuation of the old vocabulary extends in declaration order to satisfy every definition while preserving the claim. Consequently the extension checker has exactly the ordinary classical theorems. Native tree and shared-DAG coverage are proved. The direct Frege-DAG embedding retains all formulas, parent references and node counts, and has the same replay verdict; no tree expansion is involved.

Literal elimination of definitions has a separately measured cost. The admitted chain $p_{i+1} \leftrightarrow (p_i \wedge p_i)$ has encoded size at most $5(n+1)(n+2)$, including unary name costs and the final reference. The specified last-definition-first unfolding algorithm returns a formula with exactly $2^{n+1}-1$ nodes. This bounds that literal materialization policy. It is not a lower bound on every equivalent formula, nor a separation between unrestricted Frege and extended Frege. An executed Cook–Reckhow polynomial bound remains a further obligation.

13.6 Retained checking, suspension and evidence release

A checking configuration retains what its remaining work needs:

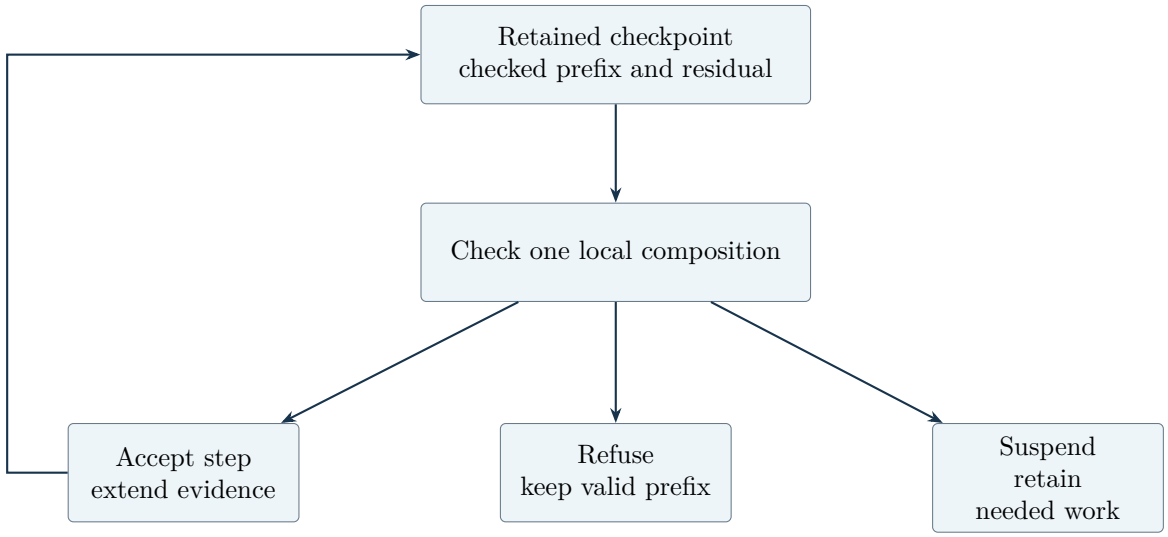
$$K = (T, \Gamma, \mathsf{cache}, \mathsf{boundary}, \mathsf{residual}, \mathsf{obligations}, \mathsf{status}).$$

The concrete representation may use references and shared tables. World revisions, provenance and consuming rights enter those components where the selected calculus requires them. Accepting a

local instruction extends the checked prefix; final acceptance additionally requires the requested root and all obligations required for that claim.

As one storage realization, the authored-calculus cursor instantiates this retained machine directly. Its completed successful runs are exactly licensed runs of the original source rows, including unused rows in the submitted schedule. VibeITP's instance preserves its independently specified static theorem judgment and auxiliary term-operation judgments. Metamath's projected snapshot connects common graph acceptance to successful original normal-label execution under its projection and empty-stack hypotheses. Neither connection replaces parser provenance or native storage refinement.

Portable calculus modules retain both the complete term-language base and the proof-calculus layer. The verified text reader, followed by the original calculus validator, recovers the exact selected package. Malformed text, another layer schema and failed validation cannot select that package. This is the portable module data surface, separate from the upstream theory-builder grammar. Reading declarations grants no theorem authority.



Missing evidence, exhausted checking budget and an unanswered external request are different suspension reasons. Refusing the supplied certificate establishes neither falsity nor unreachability of its claim.

The retained law is

$$\mathsf{advance}(m+n, K) = \mathsf{advance}(n, \mathsf{advance}(m, K)).$$

It compares complete checkpoints. MM0 assertion execution and the HOL Light source receiver already instantiate `ResumableTrajectory`. For MM0, the command tail and unification continuation remain live; final assertion validation is a separate observer. For HOL Light, `EventWork` retains the accepted theory/proof state, remaining JSON events and refusal status. A refused event stays at the residual's head. Accepted receipt prefixes grow without replay, and two separately identified reflexivity events with the same conclusion remain two occurrences. The budget counts source events, whose local costs need not coincide.

The authored recipe DAG and equation-application DAG now have retained workers over their original nodes and checkers. Each checks one submitted row per transition, retaining the last accepted table and the entire refused suffix. Their completed verdicts are exactly the existing certificate verdicts, and their actual runs finish within the number of rows plus one finishing transition. The rho tree worker similarly inspects one original inference occurrence per transition, with exact recursive-checker agreement and a bound of the tree's node count plus one. Endpoint formation remains an additional rho request-boundary check. These bounds count calls of the local worker; they do not price every internal substitution or equality operation. For authored GSLTs, existence of an actual terminated checker run with an accepted root is equivalent to inhabiting the existing reaching native

type for an equation-respecting target predicate. The native-type readout therefore comes from the original finite rewriting semantics, rather than a command layer with an unspecified local step.

All these workers are objects of the existing machine category at their own observation interfaces. The MM0 instance keeps its command and unification microsteps, the HOL Light instance keeps its registered source-event receiver, and the Lean4Lean head-profile instance keeps its actual head strategy. Choosing a common observation interface and proving a bounded stage between two objects are additional obligations. Object construction alone supplies no translation of one native logic into another.

Retention is governed by future use, not by a requirement to keep every detail forever. For ordinary value readouts d in a declared consumer family $\mathcal{D}$, replacing a detailed record by $q(x)$ is licensed when

$$\forall d \in \mathcal{D}\ \exists \bar{d}. \qquad d = \bar{d} \circ q.$$

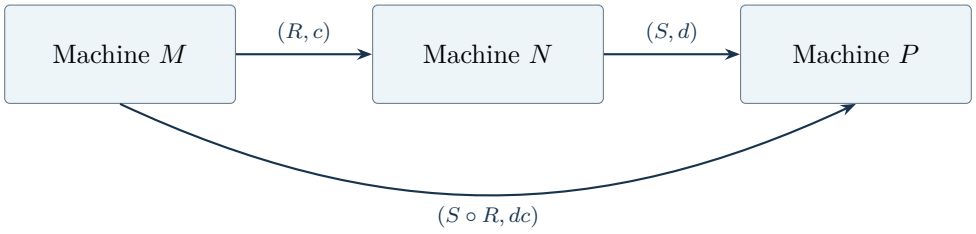
Dependent evidence additionally needs coherent family and section descent. An outstanding obligation must be discharged or transferred under its own authority contract. Once a recipient owns the needed evidence and every remaining local consumer factors through the retained summary, the sender may release detail that no longer serves those obligations. A newly admitted consumer cannot recover erased distinctions merely by requesting them.

13.7 A category of machines and realization stages

For fixed event, verdict and request types (E, V, Q), the existing `ObservedMachine` bundles an actual state carrier S and productive stepper $M : S \to \mathsf{Option}(\mathsf{Transition}(S, E, V, Q))$. Its morphisms are the existing `Stage` contracts: a relation between source and target states, a forward cost-bounded simulation and completion reflection. Identity uses the equality relation at bound one. For stages (R, c) and (S, d), composition has

$$(S \circ R)(x, z) \iff \exists y.\ R(x, y) \wedge S(y, z), \qquad \mathsf{cost} = dc.$$

`Stage.refl_comp`, `Stage.comp_refl` and `Stage.comp_assoc` supply the category laws. The `ObservedMachine.relabelFunctor` maps a common observation interface to another while retaining the underlying state carriers. Noninjective label maps can forget distinctions; each dependent consumer needs its own descent law before using that coarser view.



Composition retains the declared relational contract and resource bound. It neither selects a compiler implementation nor identifies all intermediate execution witnesses.

The controls make the distinction concrete: two stages with bound three require composite bound nine; the proposed sum six fails. A checker/interpreter round trip preserves its final observation but differs from the identity stage because its retained cost bound is three. The bound is a proved account for the chosen machine transitions, not a measurement of native time. The present `Stage` contract reflects completion; preserving stuckness versus silent divergence requires further endpoint laws. An existing negative control relates a stuck source to a spinning target under the weaker contract. The extra `MatchesStuck` obligation excludes that case.

There is also a general finite-trace relational envelope. Fix observable words w, with internal steps erased by a declared observer. For branching machines, take relations R satisfying

$$R(s,u) \ \wedge \ s \overset{w}{\Longrightarrow}_M t \quad \Longrightarrow \quad \exists v.\, u \overset{w}{\Longrightarrow}_N v \ \wedge \ R(t,v).$$

Equality is an identity relation. Relational composition preserves this law: choose the intermediate related state, match the source path in the middle machine, then match that middle path in the final machine. Relational unit and associativity laws therefore give a category at this extensional contract level. This mathematical schema states a finite may-preservation profile; backward, divergence, probability and resource obligations enrich it separately. The concrete kernel-checked category above currently uses deterministic steppers, whose schedules may realize branching source calculi only through their own adequacy theorem.

Vacuous relations are not executable realizations. Loading must relate every admitted source entry to an actual target entry; representation, request/reply and authority laws complete the applicable contract. Furthermore, forgetting the middle state can erase information required by a dependent result. `DependentComposition.Path` retains that state as data; its descent theorems identify when transport is independent of the chosen middle witness. Thus a category of endpoint contracts and a richer account of executions can coexist, with information release justified at the selected boundary.

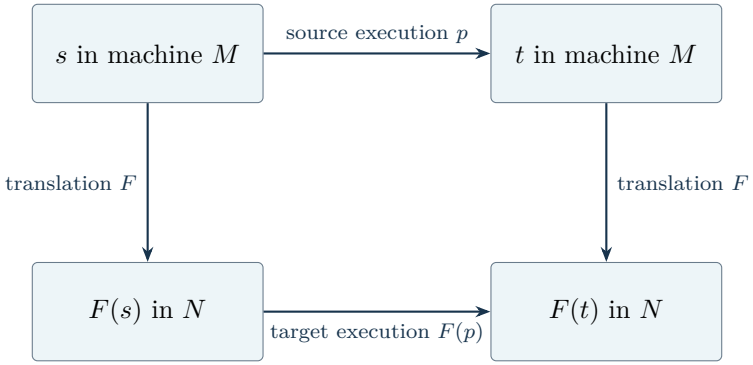
For the MeTTa realizations, the remaining obligations are stated at semantic operations: complete occurrence selection, solution-preserving binding reads and merges, retained call/return control, and world/dependency/publication updates. An implementation may realize one such operation by a size-dependent finite block. Storage layouts and individual C helpers lie downstream of this boundary. Their existing formal proofs are worked representation instances; they do not enlarge the logical machine or require a new source rule for each helper. Formal local simulation laws, source/ABI premises and tests of a compiled engine are distinct evidence. A tested binary link cannot substitute for a required unproved formal component law.

13.8 Fibres, crossings and a machine of machines

The functional GSLT construction is already literal category theory. `OperationalTranslation` preserves equations and source steps; `CoveredTranslation` additionally covers target steps at image states. The functor `executionIndex` sends each operational theory L to its category of finite execution paths. Its Grothendieck construction has objects (L, s) and arrows

$$(L,s) \longrightarrow (K,t) \quad = \quad (f : L \to K,\ p : F(f)(s) \longrightarrow t).$$

For $f : L \to K$, $g : K \to H$ and compatible paths p, q, the composite is $(g \circ f, F(g)(p); q)$. Identity and associativity follow from the base and fibre laws and the functor laws. This is a constructed total category for functional operational transport. Its underlying proposition-valued steps do not distinguish two rule occurrences with identical endpoints; receipts or labelled edges retain such distinctions when required.



A functional path map supplies a commuting square. Relational block simulations instead retain related endpoints and their actual matching paths; their coherence is an additional construction.

The executable `GSLTILNativeKernel` separates stepping inside a fibre from crossing an explicit route. The installed `GSLTILAuthoredFibre` uses the actual authored rewrite checker and its source-coverage theorem. It currently installs no translation routes. Its `at/via` dispatcher can host locally justified routes, but does not manufacture their interpretations, resource transport or conservativity.

A broader web can retain relational simulations and comparison cells instead of forcing all crossings into state functions. Horizontal paths then concern executions within a regional machine; vertical comparisons relate configurations of different machines; a cell supplies the matched execution evidence. These cells need their own composition and coherence laws. Substitution, logical interpretation, world update and authority transport retain different preservation contracts even when they occur in one diagram. This is the remaining generalization, not an implication of the Grothendieck name.

13.9 Regional kernels, semantic faces and open-ended reflection

The checking machine has an operational run; its accepted evidence inhabits an intensional derivation judgment; model interpretation supplies an extensional face. These comparisons form the computational trinity around the checker. Logical soundness goes from derivations to validity in the selected models. Logical completeness, where established for the particular fragment, goes in the other direction. Coverage of certificate representations is instead a comparison with derivations. These three claims have different hypotheses and do not identify the faces.

MM0 makes the distinction between its readable specification and binary proof program explicit [18]. Its verifier computes with stacks, hypotheses, terms and binder data, and declaration agreement binds that computation to the intended specification and preceding environment. HOL Light supplies kernel-authorized inference through the LCF interface [19]; its theorem cache and receipt graph have a separate retained execution account. Megalodon checks typed proof objects with normalization and delta expansion under its selected environment. OpenTheory articles give a textual proof stack machine [24]; CakeML includes a verified article checker and Candle's HOL Light implementation [23]. These are regional realizations, with distinct logical commitments and checking disciplines.

Lambdapi and HOL Light exercise different binder obligations. Lambdapi's restricted higher-order rewrite patterns test scoped matching under binders [22]. HOL Light's primitive replay tests term/type instantiation and theorem construction. Real HE and PeTTa evaluation kernels must likewise use their own independent judgments: a certified rewrite at one atom's root does not establish the dialect's nested evaluation, quoted/raw arguments, complete result multiplicity or retained effects.

Structural compilation of an admitted profile and checking a finite execution are complementary hosting routes. Partial recursive checkers admit finite bounded-evaluation witnesses [25]; actual hosting still requires the source encoding, target certificate construction and checker adequacy. A p-simulation further requires a polynomial-time proof translator under declared encodings, hence polynomial output size. A short operation name does not make a large substitution or conversion cheap. Shared-node size, expanded uses and executed work require separate accounts.

Small primitive inventories, weak logical assumptions and cheap checking are different objectives. The existing inference-basis theory supplies inclusion-minimal generating subsets for a finite inventory, including incomparable minima with no least generating basis. It supplies neither an efficient minimizer nor a universal least kernel. The common interface is licensed composition; individual presentations choose their generating rules and prove their interpretation.

An interactive prover additionally retains open goals, scoped metavariables, substitutions and delayed constraints. Its search may diverge or remain incomplete even when each closed certificate is

finitely checkable. Reflection makes these presentations and checking processes available as data. Goertzel's reflective metagraph account motivates this internal availability of rules and their transformations [12]. Weinbaum's open-ended intelligence motivates admitting new vocabularies and observers rather than requiring a final fixed inventory [31]. The architectural proposal is to combine that openness with explicit crossings: changing a theory exposes new commitments, replacing a machine preserves the declared observations, and neither silently changes what an earlier certificate proved.

Existing owners. `Mettapedia/Logic/DerivationGraph.lean`,
`Mettapedia/GSLT/LanguageDef/CertificateGSLTClassifyingCategory.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipe.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipeExpansion.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipeComposition.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipeContext.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipeDAG.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipeExecution.lean`,
`Mettapedia/OSLF/MeTTaIL/GeneralRewriteRecipeParallel.lean`,
`Mettapedia/GSLT/Distinction/SimulationComposition.lean`,
`Mettapedia/GSLT/Distinction/SimulationCompositionControls.lean`,
`Mettapedia/GSLT/Distinction/DependentComposition.lean`,
`Mettapedia/GSLT/Core/OperationalPathFibration.lean`,
`Mettapedia/GSLT/LanguageDef/GSLTILAuthoredFibre.lean`,
`Mettapedia/Languages/MM0/Formats/MMB/MachineCost.lean`,
`Mettapedia/Languages/MM0/MeTTa/Formats/MMB/ExecutedSoundness.lean`,
`Mettapedia/Languages/HOLLight/IncrementalExecution.lean`,
`Mettapedia/Languages/HOLLight/KernelAtlas.lean`,
`Mettapedia/Languages/MeTTa/EquationKernel/Execution.lean`,
`Mettapedia/Languages/ProcessCalculi/RhoCalculus/NativeExecution.lean`,
`Mettapedia/Languages/Lean/HintedHeadConversion.lean`,
`Mettapedia/Languages/Lean/Lean4LessCheckerBoundary.lean`,
`Mettapedia/Logic/EqualityRecipe.lean`,
`Mettapedia/Computability/FuelledConversionEvidence.lean`,
`Mettapedia/Machines/ExecutionHints.lean`,
`Mettapedia/GSLT/LanguageDef/KernelCheckingClasses.lean`,
`Mettapedia/GSLT/Logic/PropositionalLRAT.lean`, `Mettapedia/Logic/PropositionalExtension.lean`,
`Mettapedia/Logic/IntuitionisticDecision.lean`,
`Mettapedia/Logic/PropositionalHilbertChecking.lean`,
`Mettapedia/Logic/ProofSharingBounds.lean`, `Mettapedia/Coalgebra/CoherentPrefixTower.lean`,
`Mettapedia/Logic/InferenceBasis.lean`.

Boundary. Local authored checking, its composition operations, functional path transport, costed stage composition, checked reconstruction and the retained native checker instances are constructed at their stated contracts. Authored packet coverage uses the existing admission compiler and an empty dynamic provider; textual module parsing is a separate front end. The finite-trace relational envelope is a mathematical schema. A complete heterogeneous execution/certificate web, its route authentication, general relational coherence and native physical resource bounds remain further constructions.

14 Execution contracts: determinism is one dimension

A deterministic implementation can enumerate many answers, compute an entire distribution, or simulate concurrent processes under a selected schedule. A sampler can return exactly one answer while representing a nontrivial probability law. These properties therefore cannot be represented by one exclusive choice between deterministic, nondeterministic and probabilistic.

Dimension	Contract question	Distinguishing example
Transition	At most one next state?	An ordered search driver steps deterministically through many alternatives.
Cardinality	How many successful occurrences in a completed run?	Two equal answers count twice.
Completion	Is any work still pending?	One answer followed by a suspended branch is not a closed singleton.
Probability	What measure describes outcomes?	A fair and a biased coin have the same finite support.
Concurrency	Which enabled actions may be scheduled?	Two independent actions can commute without one global deterministic schedule.
Commitment	Which alternatives or effects may be discarded?	`once` selects a success; it does not prove there was only one.

Cardinality belongs to a calling mode and its world

The existing answer grades bound the number of occurrences in a completed answer collection: zero, exactly one, zero-or-one, one-or-more, or unrestricted. Relational choice adds occurrence counts; an exclusive conditional combines bounds by an interval hull. Sequential binding applies a bound separately to each reachable input occurrence with its bindings. Types, instantiation, termination, faults and effects remain additional obligations. A declaration of `det` expresses a contract; an optimization needs evidence for the actual calling mode and relevant provider revision.

The current weighted cardinality consumer accepts a complete cut only when it is empty or contains one returned occurrence. It preserves that occurrence's coefficient, including zero. A cut with a pending state is refused even if its current returned prefix has length one. The accepted result is stable under further resumption. This is a representation theorem about an already computed cut, not a procedure for deciding termination or safely pruning an unfinished search. Probing for another answer can perform effects, force otherwise unused work, or diverge.

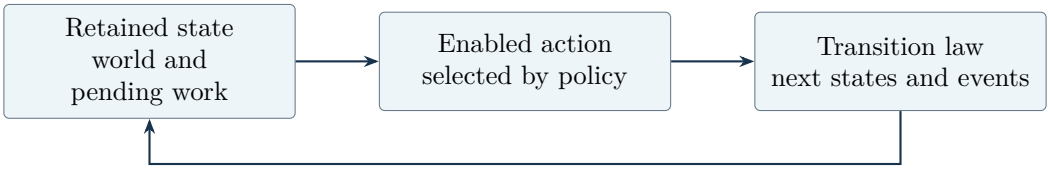
What a deterministic region can improve

The independently defined queued and direct executors already agree on ready operands, functional primitives, worlds, ordered effects, failures and instruction-boundary residuals. Their primitive invocation counts agree; their modeled dispatch counts differ. This licenses eliminating that queue within the admitted region. It establishes no native speedup and does not yet cover arbitrary expressions, binding merges or concurrent mutation.

Existing owners. `Mettapedia/Machines/AnswerCardinality.lean`, `Mettapedia/Machines/DeterministicRegion.lean`, `Mettapedia/Machines/OccurrenceMachine.lean`, `Mettapedia/GSLT/Dynamics/WeightedBranchingCardinality.lean`.

14.1 Probabilities, coefficients and schedules

A coefficient is not automatically a probability. The weighted resumption interfaces work with algebraic coefficients and preserve pending contributions and occurrence identity. A probabilistic interpretation additionally needs nonnegative mass and a specified normalization or subnormalization law. An importance weight, a scheduling priority, a cost and an ownership permission are different quantities even when they use similar algebraic notation.



A scheduler selects an enabled action; a probabilistic kernel specifies its stochastic transition. Relational alternatives need not carry a canonical probability law. A point mass recovers deterministic transition.

Combining nondeterministic choice with probability is a semantic choice: the available scheduler information and the order of the two effects matter. One cannot assume that an arbitrary combination of the familiar monads has the desired distributive laws [11]. A machine contract should state the admissible policies and what the observer compares: possibilities, ordered occurrences, trace distributions, or another declared quantity.

The existing finite probabilistic systems use subdistributions and account for missing mass as declared refusal. Their probabilistic bisimulation retains more information than the support relation. However, the system record does not itself prove that its kernel follows the underlying GSLT rewrite relation; that is an operational bridge still to construct. Erasing probabilities to possibility support cannot justify a probability-preserving implementation. For a continuous distribution, a singleton can have probability zero without being an impossible value; positive point mass is not a general support test.

Sampling, scoring and resumption

A retained draw needs its dynamic occurrence identity and original world. Resuming a committed draw must reuse its result. Calling a resampled suspension creates a new occurrence; merely sharing written syntax does not establish stochastic independence. Independence requires a law for the actual joint execution, often conditional on captured latent variables.

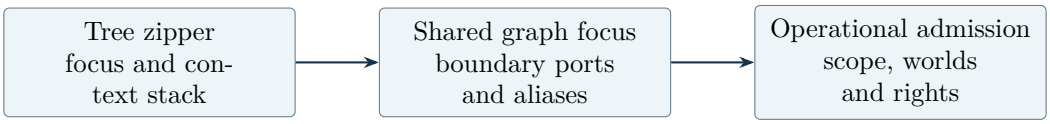
Scoring generally produces an unnormalized measure. Forming a posterior requires an evidence mass Z with $0 < Z < \infty$; zero or infinite evidence needs explicit treatment. Retention must preserve committed factors and avoid applying them again. Existing coefficient plumbing is useful here, but it does not supply this entire probability-and-authority contract.

> **Boundary.** No production port is licensed solely by cardinality or coefficient laws. The complete world, bindings, effect history, faults, ownership and residual demand must travel through the same constructed correspondence.

Existing owners. `Mettapedia/GSLT/Dynamics/WeightedResumption.lean`, `Mettapedia/GSLT/Dynamics/WeightedBranchingResumption.lean`, `Mettapedia/GSLT/Distinction/Probabilistic/System.lean`, `Mettapedia/GSLT/Distinction/Probabilistic/Semiring.lean`, `Mettapedia/ProbabilityTheory/IndependentFiniteSampling.lean`.

15 Focus, zippers and typed graph holes

For regular tree syntax, the derivative of a datatype describes its one-hole contexts [9]. A zipper retains a focused subtree and the context needed to rebuild its surroundings [10]. Indexed polynomials extend this organization with context and sort information; dissections distinguish already processed positions from remaining positions. These are representation and navigation tools. A legal position is not by itself permission to evaluate, force, sample or commit an effect.



The existing PathMap zipper interface transports query membership under iteration and store-agreement hypotheses. Concrete flat and trie instances prove particular traversal and rebuilding laws. This is useful ZAM structure; it is not a proof of the full native Rust machine, ordered query multiplicity, or its world and demand semantics. A membership theorem alone cannot establish that two equal rows produce two answer occurrences.

A shared hole needs an interface, not just a path

Replacing a shared subgraph affects every alias to it. A graph hole therefore needs the incoming and outgoing ports crossing its boundary, their scoped types, identity and sharing requirements, and the resources or obligations that move with them. A single tree path loses those other connections. Cycles require the corresponding consistency and update rules. Bag or set quotients also need occurrence identities when their physical ordering is not canonical.

Goertzel's reflective metagraph account makes programs and rewrite rules graph data [12]. Graph rewriting's preserved interface and gluing conditions are useful guidance: a replacement must account for the links it preserves and those it removes. This motivates a runtime contract; it does not already provide a typed effectful graph-hole machine.

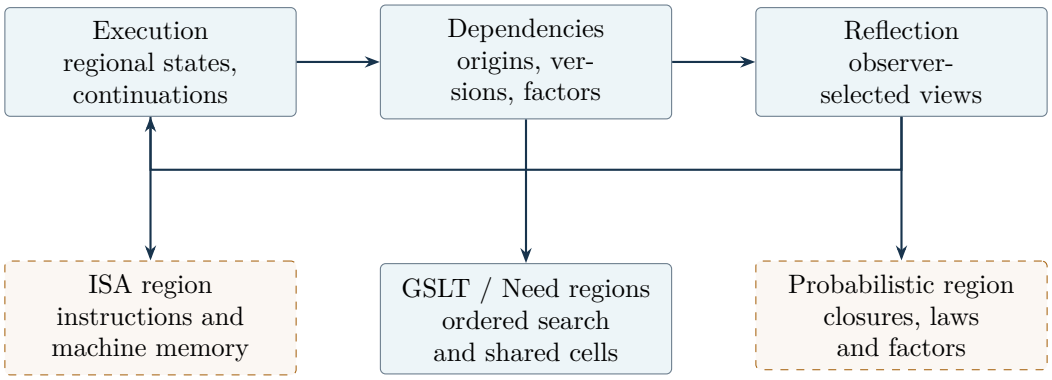
Different holes have different filling rules

An unknown source fragment, a paused computation, a stochastic latent variable, an external response and an unfulfilled proof obligation are not interchangeable. The existing pure higher-order hole evaluator retains environments and indeterminate operations. Its filling/resumption equivalence has a termination condition. Its controls show that divergent call-by-value fillers and effectful reordering can break an unrestricted equivalence. The tree multi-hole monad also deliberately permits capture; it is not a ready-made hygienic shared-graph interface. Filling must establish the applicable scope and execution contract.

Existing owners. `Mettapedia/GSLT/Poly/BindingSignature.lean`, `Mettapedia/GSLT/Poly/SortedTraversal.lean`, `Mettapedia/OSLF/PathMap/ZipperExecution.lean`, `Mettapedia/OSLF/PathMap/Trie/HuetZipper.lean`, `Mettapedia/OSLF/Programs/HoleEvaluation.lean`, `Mettapedia/OSLF/MeTTaIL/MultiHoleContext.lean`, `Mettapedia/GraphTheory/Representation/Metagraph.lean`.

15.1 A core MAM for heterogeneous retained computations

The strongest candidate here separates execution, dependency accounting and reflection. They refer to the same occurrence identities, but they need not use one physical representation. A machine can execute a compact packed region while exposing a metagraph view on demand. Reflection need not allocate a public syntax object for every primitive instruction.



Dashed regions mark unconstructed general interfaces. All regional calls use the crossing contract; the connecting lines do not assert an ISA or higher-order probability correspondence already exists.

The proposed retained state includes local computations, typed open boundaries, shared producer/use identities, ordered alternatives, world dependencies, committed draws and factors, consumed rights, and selected observations. Its transitions admit and focus a region, advance it, deliver a request or reply, fill an authorized hole, validate dependencies, pause, or dispose of cancelled obligations. Sampling and scoring come from an admitted regional semantics. Neither graph shape nor a type annotation silently authorizes them. Logical guests retain their own judgment systems; a common port protocol does not make their propositions interchangeable.

Qualification would lift local correspondences through admitted graph contexts, fillings and crossings. At publication, suspension and return, the chosen trace observations and related residuals must survive together. Probabilistic regions compare the resulting laws under declared scheduling policies, or a stated coupling bound; equal possible values are insufficient. These compatibility laws must be proved for the actual assembly. Listing types and authority metadata does not establish them.

An ISA region would load code with an instruction-validity contract and retain registers, memory, faults and external-service state. Its source translation must state ABI and memory assumptions. The existing register-code machine supplies a narrower lowering and execution connection; a proposed `lts:isa` interface is not established by that result. Cesta macro elaboration belongs to a staging phase, before the resulting C/native region.

A GSLT interpreter is a general rule-execution machine. Calling it UTM-tier requires an encoding of universal computation and a correspondence theorem. Even such a theorem would not preserve fairness, ordered occurrences, costs, probability laws or ownership automatically. Those are regional contracts, not consequences of computational universality.

> **Boundary.** This architecture is a research target assembled from existing owners. It is not an implemented universal MAM or a claim that one representation is optimal for every region. Its first consumer remains the actual MeTTa source/native and owned-interaction connection.

Existing owners. `Mettapedia/Machines/RegisterCode.lean`, `Mettapedia/Languages/TuringMachine/MathlibBridge.lean`, `Mettapedia/GSLT/Dynamics/ObservationSchedulerSufficiency.lean`, `Mettapedia/GSLT/Dynamics/ScheduleObservationFactorization.lean`.

16 Higher-order probabilistic holes

Consider a hole with type

$$\square_{\text{make}} : \text{Unit} \longrightarrow \mathcal{P}(\text{Unit} \longrightarrow \mathcal{P}\text{Bool}),$$

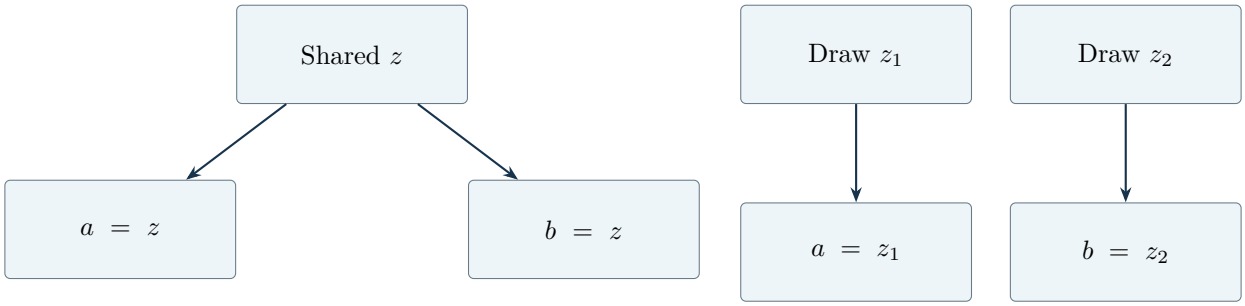
where $\mathcal{P}$ denotes a probability computation. A consumer obtains f from `make ()`, calls f twice, and returns the pair (a, b). There are two well-typed fillings:

Shared draw	Fresh draw per call
`cached () =` draw $z \sim \text{Bernoulli}(1/2)$ return $(\lambda().\ \text{return}\ z)$	`fresh () =` return a closure which, on every call, draws a fresh $z \sim \text{Bernoulli}(1/2)$ and returns z.

Both give a fair marginal for either individual call. Their joint laws differ:

Filling	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
Shared draw	$1/2$	0	0	$1/2$
Fresh draws	$1/4$	$1/4$	$1/4$	$1/4$

Thus $\Pr(a = b)$ is respectively 1 and $1/2$. Matching the hole's type or one call's marginal cannot license replacing one filling by the other. One shared draw has one producer with two use edges; two independent draws have two dynamic sampling occurrences, even if their source syntax is equal.



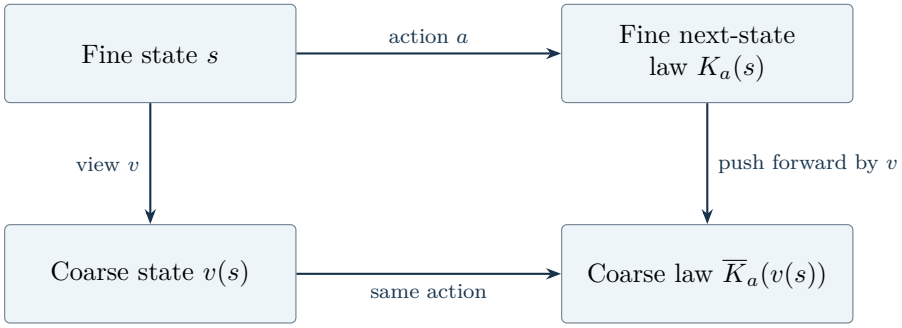
Two semantic routes remain open

An operational route places measures over encoded closures and retained machine states, with application specified by the evaluator. A denotational route supports function spaces directly. Ordinary measurable spaces are insufficient for the desired general higher-order structure; quasi-Borel spaces provide a compositional alternative [13], with domain-theoretic extensions for recursion and recursive types [14]. Neither route should be confused with the existing mixture-of-kernels or probabilistic-HOL-model modules, which do not implement a general probabilistic lambda calculus. The small example above is a mathematical discriminator, not a newly proved machine instance.

Existing owners. `Mettapedia/ProbabilityTheory/HigherOrderProbability/Basic.lean`, `Mettapedia/ProbabilityTheory/HigherOrderProbability/GiryMonad.lean`, `Mettapedia/ProbabilityTheory/IndependentFiniteSampling.lean`.

16.1 Zooming requires a law for future computation

Let $v : S \to C$ forget details of a fine state. Equal coarse readouts now are insufficient: different hidden states can react differently to the next command. An exact coarse machine needs a law relating its future transitions to those of the fine machine, for the declared action and observer family.



For finite probabilistic states, existing strong lumpability says that states in one view fibre have the same aggregated next-state law for every admitted action. It constructs a representative-independent coarse kernel and proves

$$v_*(\mu \text{ bind } K_a) = (v_*\mu) \text{ bind } \overline{K}_a.$$

Here v_* is pushforward. Without this condition, retain additional state or history, restrict actions/observers, or accept a weaker approximate contract. A policy depending on hidden details also needs its own factorization law.

Existing probabilistic-square results offer a finite controlled-chain route to approximation: local couplings bound expected discrepancy, and Lipschitz and discount hypotheses control its accumulated observational effect. They are not exact quotients or reversible maps. Numerical closeness also depends on the observer: a finite floating-point distribution and a continuous law can be maximally different in total variation despite close numerical samples. A metric suited to the permitted queries may be more appropriate.

Zooming in can expose retained fine state or reconstruct an authorized view. It cannot recover arbitrary forgotten choices from a non-injective quotient. For the shared-draw example, the first call's marginal hides the captured draw and whether subsequent calls reuse it; reconstructing by resampling can break those correlations. Similarly, replaying a native trace must not repeat committed effects. Observation, demand and retention therefore remain separate in-language controls.

The remaining research question

The missing joint theory is typed open-metagraph filling with scope, shared identity, effects, probability and ownership, together with observation maps that commute with continuation. Useful research would compare existing models, identify incompatibilities, and work one higher-order shared/fresh example through both exact finite execution and a continuous denotation. General recursion, conditioning and scheduling must keep their explicit conditions; universal complete inference is not a consequence of a universal machine.

Existing owners. `Mettapedia/ProbabilityTheory/FiniteLumpability.lean`, `Mettapedia/Cybernetics/ApproximateAdequacy/ProbabilisticSquares.lean`, `Mettapedia/GSLT/Dynamics/ObservationSchedulerSufficiency.lean`, `Mettapedia/GSLT/Dynamics/ScheduleObservationFactorization.lean`.

17 Qualification and the remaining joins

The principal inventory is maintained with the eight accepted obligations: common operational theory, historical CAM, fixed C, HE/PeTTa correspondence, shared and distinct dialect behavior, retention/interaction, component realization, and joined qualification. It indexes existing declarations by source contract, observer and assumptions. Historical proof and test receipts remain evidence for their recorded source states; they are not current qualification after a dependency changes.

What the principal theorems establish

Layer	Constructed connection and remaining obligation
Common theory	`DerivationSimulation.transport`, `trans` and `resume` compose constructed local blocks. `CostSimulation.toDerivationSimulation` requires `MatchesStuck`. `EventCursor.advance_simulation` and `resume_simulation` reach actual scheduled execution. Missing local instances are not supplied by those generic laws.
CAM	Historical compiler/readback, typed strict meaning, graph recursion and Need have their independent source correspondences. Graph/tree scheduled refinement and abstraction use nontrivial representation maps. Original coverage and current importing consumers still belong in final qualification.
Fixed C	`FixedCore.Scheduled.source_path_iff` preserves permitted choices and retained supply. Cesta reading, the stated fixed/located fragment and service calls have separate correspondences. A compiled CeTTa program is not thereby a program of the fixed core; the named extension ledger remains authoritative.
HE source/AM	`Contextual.source_iff_observed` and `Protocol.public_completion_iff_source` connect independent completed source judgments, actual contextual paths and joint caller/value answer bags. The stateful correspondence additionally retains its reached world. Declared construct coverage and external-service obligations remain explicit.
HE worker	Scoped casts, certified readout, argument spines and stack-generic invocation blocks are local realization results. General recursive, multi-result block adequacy with its preserved entry/freshness invariant remains open. A supplied child execution is not that theorem. The multi-row equation spine composes supplied producer and return runs; recursive coverage remains separate.
PeTTa	`Scoped.source_run_iff_derivation` and `completed_iff_source` cover finite scoped source execution; `commit_first_answer` reaches an answer without waiting for younger search to finish. Wider dialect admission, interaction and native component laws remain distinct obligations.
Shared fragment	`recursive_machine_bag_iff` and `recursive_machine_storage_permutation` preserve the admitted pure recursive fragment's multiplicities and protected caller observations. Its closed profile uses materialized inputs. For the open profile, `OpenRecursive.Evaluation.transport` constructs the HE derivation and source branch paths from one complete recursive source tree, preserving the ordered list and bag of joint caller/value observations. CollectionScope DFS linearization remains a PeTTa machine obligation.
Interaction	The finite weighted Need chain, scoped cut/lifetime and authorized append reached-cut laws are constructed instances. Complete cancellation, nested services, COMM, ownership and publication joins on actual dialect/declared Prime configurations remain required.
Realization	Store, control, selection, world/service and publication must discharge their local laws and compose into retained-run correspondence. Existing reader, open-slot, return and service instances do not imply correctness of every native mutation, dispatcher or compiled binary.

Readout equivalence does not erase operational distinctions

A binding observation is joint:

$$\mathsf{observedTuples}(B, \vec{a}) = \{\sigma(\vec{a}) \mid \sigma \models B\}.$$

This preserves repeated-variable correlation, while separate scalar observers can lose it. A `Readou tCertificate` connects one retained normal table to the complete constraint theory and supports future operands and independent constraint extension. Hygiene support is separate: a redundant equality can change the names that must be protected without changing the solution set.

Changing a constraint record through `Frame.represent` retains its exact annotated environment. Changing that environment to another model-equivalent representative is a stronger operation. Opposite open-variable representatives can give the same joint observations but different literal syntax at the next atom dispatch. The worker's canonical control simulation consequently uses `ScopedActivationRel` and earned entry/materialization admission; the general finite-model cast result uses joint observational adequacy instead. Neither theorem may be substituted silently for the other.

Readiness has the same boundary. Equal erased constraints do not determine whether an expression is active or held. The contextual machine retains that status under the selected policy, whereas the plain-Atom worker is a stripped representation with a narrower admission. Its private-sibling freshness law protects both substitution keys and variables occurring in stored values. A materialized open result instead earns stable entry from the certified normal read. General recursive realization must preserve the appropriate invariant at every reached call and return; it cannot assume all caller variables remain private or unassigned.

Current qualification boundary

The independent source ownership boundary is checked. The import-free `MeTTailCore.MeTTaSyn tax.Atom` leaf owns the actual shared declarations; HE mathematics and implementation studies depend on it in the same direction. Every LeaTTa package module is excluded from the protected HE source and AM closure. The boundary check covers 117 roots with zero violations and 30 regression controls. The current global subject-boundary gate also passes with zero findings. A fresh audit checks 326 earlier principal roots, and the E2 follow-up checks ten additional roots including `Evaluation.transport` and `open_recursive_duplicate_transport`. They use only `propext`, `Classical.choice` and `Quot.sound`. The changed proof owners contain no proof admissions or unchecked evaluation. This does not establish termination or implementation refinement of every executable helper in the wider dependency closure.

The current ordinary Lake join passes all 22 targets and 5,304 jobs with zero warnings or errors. It includes the scoped worker and service consumers, the open recursive comparison, the public HE/PeTTa translation aggregator, and the GSLT ordered-realizations driver and resumption consumers. These checks qualify the recorded source snapshot and the actual theorem premises. They do not discharge the unfinished worker, queue-linearization, interaction or component-composition obligations, and they do not constitute a renewed binary-conformance run.

Final qualification uses ordinary dependency-tracked builds of the named owners and impacted public consumers, strict warnings, proof-dependency audits and the relevant existing Rust HE, SWI-PeTTa and CeTTa gates. Binary evidence must name its revision, profile and expected differences. Source proofs, formal component models, source/ABI assumptions and tested executable facts remain separate. A discovered engine defect needs a minimal repro and an honest failed contract; it does not authorize adopting the defect as a language rule or expanding this tranche into engine debugging.

Finite closing obligations

1. Consolidate the existing owners around principal statements, explicit profiles and reusable lemmas. Preserve valuable downstream examples without counting their size as source/AM coverage.
2. Close the scoped worker's recursive multi-result block simulation and its preserved admission invariant. Rebuild its affected consumers after the active owner stabilizes.

3. Linearize the open recursive source tree through PeTTa's `CollectionScope` DFS queue. Prove that installed alternatives drain in source order with their occurrence origins, cut boundaries and exact retained outer frontier, then join that machine theorem to the completed open HE/PeTTa source comparison.
4. Finish the declared retained interaction and five-component realization joins. Loading, selection, readout, call/return, world/effect, retained-state, future-use and productive resumption laws must be constructed locally before whole-run composition.
5. Recheck original CAM/fixed-C coverage, run the final named-entrypoint and consumer joins, audit principal roots and execute the relevant full runtime gates. Synchronize this explanation with the resulting evidence.

No new reference interpreter, C-helper inventory or universal effect framework is needed to discharge those obligations. The stopping condition is their proved coverage and current joined verification, not line count, module count, a diagram or examples with supplied executions.

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